Materials Science Study Guide

A Learning Companion for University Students

Chapters 1–11 · Structures, Defects, Phase Diagrams & Mechanical Behavior

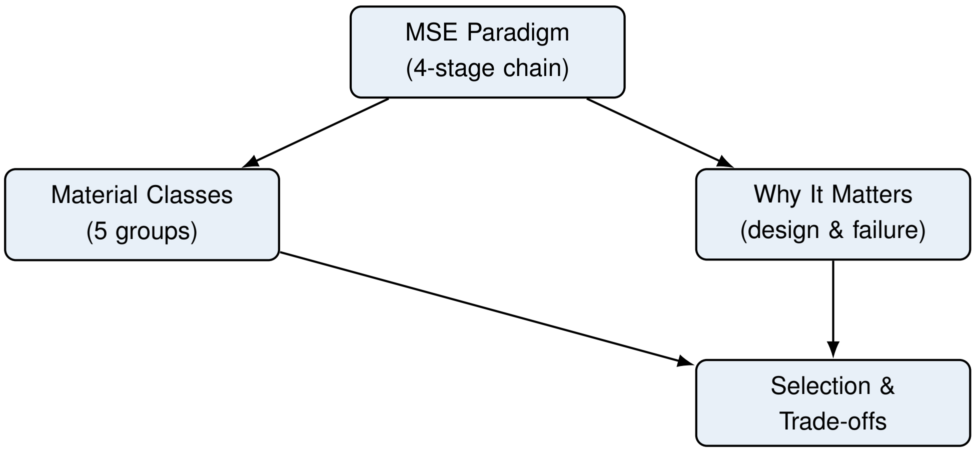
For learning and self-study. All figures drawn by the author.

Contents

Chapter 1

Introduction & the MSE Paradigm

1. Roadmap



This chapter builds the foundation: the framework that links *processing* to *performance*, the five material families, and why this knowledge prevents engineering disasters.

2. Learning Objectives

1. **State and trace** the MSE paradigm: processing → structure → properties → performance.
2. **Classify** any engineering material into the five families and justify a choice for a given application.
3. **Compute** theoretical density from unit-cell data (preview of Ch 3).
4. **Explain** why materials failures (Liberty ships, bridges) motivate the field.

3. Why This Matters

Materials selection decides cost, safety, and performance of every product. Roughly every major engineering disaster (Liberty ships, Silver Bridge, Challenger O-rings) was fundamentally a *materials* failure. Roughly 10–15% of exam points test classification and the MSE chain directly, and they underpin every later chapter.

4. Intuition First

Intuition First

Think of a material like a recipe. *Processing* is how you cook it; *structure* is the microscopic arrangement of ingredients (crumbs, layers, bubbles); *properties* are what you can measure (crispy, soft, strong); *performance* is whether it satisfies the eater. Change the cooking, and you change the crumb structure, which changes the crispiness. You can never change a property without first changing the structure.

5. Visual

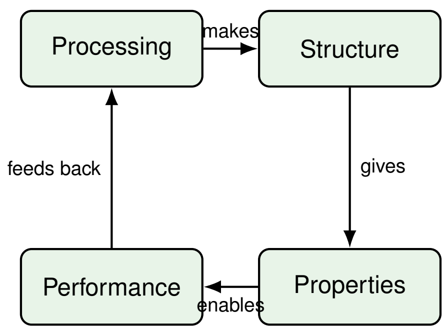


Figure 1.1. Figure 1.1 — The Materials Science & Engineering paradigm: a closed loop from processing to performance.

The MSE Paradigm

Four linked stages: **processing** creates a **structure**; structure determines **properties**; properties enable **performance** in service. The chain also runs backward as a feedback loop.

Structure across length scales. Structure is studied at nested scales: subatomic (electrons/bonds) → atomic (crystal lattice) → microscopic (grains, phases) → macroscopic (components). A change at any scale propagates outward.

6. Memory Box

Memory Box

- **MSE chain:** Processing → Structure → Properties → Performance.
- **5 families:** Metals, Ceramics, Polymers, Composites, Advanced materials.
- **Metals:** electron-sea bonding → conductive, ductile, stiff.
- **Ceramics:** ionic/covalent → hard, brittle, insulating, high T_m.
- **Polymers:** covalent chains → light, flexible, low T_{max}.
- **Composites:** $\geqslant 2$ phases → tailored, anisotropic.
- **Advanced:** semiconductors, biomaterials, nanomaterials, smart materials.

7. Compare Box

Metals vs Ceramics vs Polymers

Metals	**Ceramics**	**Polymers**
Electron-sea bond	Ionic/covalent bond	Covalent chains
Conductive	Insulating	Insulating
Ductile	Brittle	Flexible/viscoelastic
High T_m	Very high T_m	Low T_{max}
e.g. steel, Al	e.g. Al_2O_3, SiC	e.g. PE, nylon

8. Common Mistakes

Common Mistakes

- **"Processing changes properties directly." Why wrong:** it acts *through* structure; the intermediate step is what you are tested on.
- **Calling a composite an alloy. Why wrong:** an alloy is a single-phase solid solution of metals; a composite mixes *distinct* phases (e.g. CFRP = carbon fiber + polymer).
- **Assuming all ceramics insulate electricity. Why wrong:** some are ionic conductors or semiconductors (ZrO_2, SiC).
- **Confusing "advanced" with "stronger." Why wrong:** advanced means a *special function* (e.g. semiconductors), not necessarily higher strength.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$\rho = \frac{nA}{V_c N_A}$	Theoretical density from cell	g/cm^3	n atoms/cell, A at. wt (g/mol), V_c cell vol (cm^3), N_A	Perfect crystal, no voids	Ignores defects/-porosity	Screening mass-critical parts
$\rho = \frac{\sum m_i}{\sum V_i}$	Bulk/-measured density	g/cm^3	measured mass, volume	Homogeneous sample	Needs accurate volume	Quality/void check

10. Worked Examples

Basic

Example

Problem: Which material family fits a part needing high electrical conductivity, moderate strength, and ductility?
Thinking: Conductivity + ductility points to the electron-sea bond.
Solution: A **metal** (e.g. copper or aluminum). Ceramics/polymers insulate; composites are not intrinsically conductive.

Intermediate

Example

Problem: A bicycle frame needs high strength-to-weight, moderate stiffness, low cost. Choose among: alumina ceramic, 6061 aluminum, nylon, tool steel.
Thinking: Rank by σ/ρ and service temperature. Ceramic is brittle (rotating load → catastrophic); nylon too weak/soft; tool steel too heavy.
Solution: 6061 aluminum alloy — good specific strength, stiff enough, low cost.

Advanced

Example

Problem: Copper has $n = 4$ atoms/cell, $A = 63.55$ g/mol, FCC with $a = 0.3615$ nm. Compute theoretical ρ.
Thinking: $V_c = a^3$; convert $\text{nm}^3 \rightarrow \text{cm}^3$.
Solution:

$$V_c = (0.3615 \times 10^{-7}\ \text{cm})^3 = 4.724 \times 10^{-23}\ \text{cm}^3$$
$$\rho = \frac{4(63.55)}{(4.724 \times 10^{-23})(6.022 \times 10^{23})} = \frac{254.2}{28.45} = 8.94\ \text{g/cm}^3$$

Matches the handbook value (8.96), confirming an FCC assignment.

11. Engineering Insight

Engineering Insight

The paradigm is a *design tool*, not just theory. To improve a part you choose the lever:

- Want higher strength? Change **structure** via processing (grain refinement, alloying, heat treatment).
- Want a new function? Design a **new structure** (nanocomposite, metamaterial).

Reverse-engineering a competitor's part means reading its microstructure to infer its processing history.

12. Industrial Applications

Industrial Applications

- **Aerospace:** Ti–6Al–4V fan blades (light + creep-resistant at 500°C).
- **Electronics:** silicon (advanced/semiconductor) enables all ICs.
- **Transport:** CFRP in Boeing 787 cuts mass ~ 20% vs aluminum.
- **Medicine:** Ti and Co–Cr biomaterials for implants (biocompatible).

13. Summary

Materials are classified into five families whose properties follow from bonding and structure. The MSE paradigm — Processing → Structure → Properties → Performance — is the lens for every design and failure question. The single most useful equation here is $\rho = nA/(V_c N_A)$, which links atomic structure to a measurable bulk property.

14. Quick Review

Quick Review

1. The fourth stage of the MSE paradigm, after properties, is ____________.
2. Name the five material families: ______________.
3. True / False: An alloy is a type of composite. __________
4. A material with ionic bonds, high hardness, and brittleness is a ____________.
5. CFRP stands for ______________.

15. Practice Problems

⋆ Which is **not** a primary classification of engineering materials?

(a) Metals

(b) Ceramics

(c) Semiconductors

(d) Polymers

(p. 144)

⋆ In the MSE paradigm, which stage directly determines properties?

(a) Processing
(b) Structure
(c) Performance
(d) Cost

(p. 144)

⋆⋆ A jet-engine turbine blade needs high strength at 900°C and creep resistance. Best choice?

(a) Nylon
(b) Alumina
(c) Nickel superalloy
(d) Aluminum 6061

(p. 144)

⋆⋆ Carbon-fiber-reinforced polymer (CFRP) is an example of:

(a) A metal-matrix composite
(b) A polymer-matrix composite
(c) A ceramic-matrix composite
(d) A metal alloy

(p. 144)

⋆⋆⋆ For an unknown metal, $n = 2$ atoms/cell, $A = 55.85$ g/mol, $a = 0.2866$ nm (BCC). Compute theoretical density and name the likely metal. (p. 144)

16. Flash Cards

Flash Card

Front: What does the MSE paradigm say about changing a property?

Back: You must change the *structure* (via processing); properties follow structure.

Flash Card

Front: Name the 5 material families.

Back: Metals, Ceramics, Polymers, Composites, Advanced materials.

Flash Card

Front: Alloy vs composite?

Back: Alloy = single-phase metal solid solution; composite = $\geqslant 2$ distinct phases (e.g. CFRP).

Flash Card

Front: Why are ceramics brittle?

Back: Ionic/covalent bonds are directional and lack slip systems; cracks propagate easily.

Flash Card

Front: Formula for theoretical density?

Back: $\rho = nA/(V_c N_A)$.

Flash Card

Front: What made Liberty ships fail?

Back: Ductile-to-brittle transition of steel below $\sim 10°C$ + weld flaws $\rightarrow$ brittle fracture.

17. Cheat Sheet

Cheat Sheet

MSE chain: Processing → Structure → Properties → Performance.
Families & bonding:

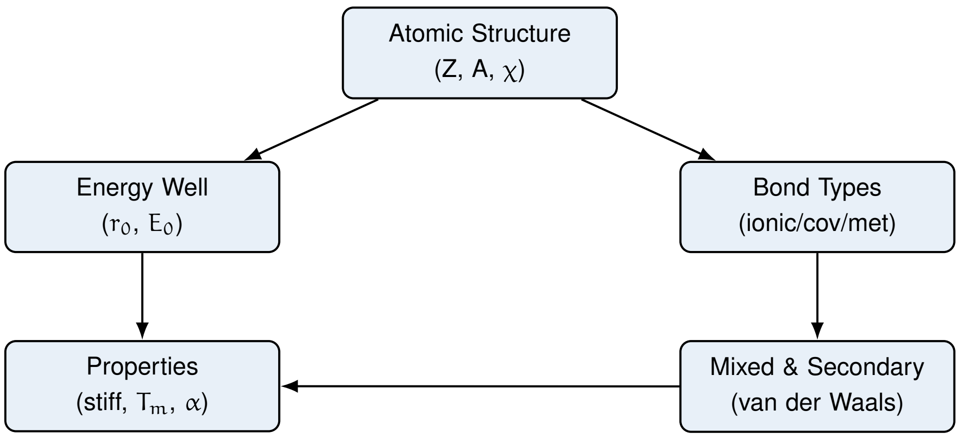
- Metals — electron sea: conductive, ductile.
- Ceramics — ionic/covalent: hard, brittle, insulator.
- Polymers — covalent chains: light, flexible, low T_{max}.
- Composites — $\geqslant 2$ phases: tailored, anisotropic.
- Advanced — semiconductors, biomaterials, nano, smart.

Density: $\rho = nA/(V_c N_A)$ (g/cm^3).
Failure lesson: most disasters are materials failures.

Chapter 2

Atomic Structure and Interatomic Bonding

1. Roadmap



This chapter connects the atom (number of protons, mass, electronegativity) to the energy well that forms between atoms, then to the three primary bond types, mixed/secondary bonding, and finally to the bulk properties engineers actually measure.

2. Learning Objectives

1. **Define** the key atomic parameters Z, A, N_A, and electronegativity χ.
2. **Predict** the dominant bond type from the Pauling electronegativity difference $\Delta\chi$.
3. **Sketch** the interatomic energy–separation curve and label r_0, E_0, and the properties derived from it.
4. **Contrast** ionic, covalent, metallic, and secondary bonds by mechanism and property signature.
5. **Estimate** ionic character from $\Delta\chi$ using Pauling's exponential relation.
6. **Compute** theoretical density $\rho = nA/(V_c N_A)$ from atomic structure.

3. Why This Matters

Bonding is the root cause of every mechanical, thermal, and electrical property you will study for the rest of the course. Roughly 10–15% of exam points test bond-type prediction and the energy-well interpretation directly, and these ideas feed crystal structure, defects, phase diagrams, and mechanical behavior. In engineering, the choice between a covalent ceramic cutting tool, a metallic wire, and a polymeric gasket is decided entirely by what holds the atoms together.

4. Intuition First

Intuition First

Imagine two magnets floating near each other. Pull them together and they snap, push them too close and they shove back. Atoms behave the same way: an attractive pull (opposite charges) and a repulsive push (crowded electron clouds) compete. At one special distance they settle into a stable hug, the *equilibrium spacing* r_0. The depth of that hug is the *bond energy* E_0 — how hard you must pull to tear the atoms apart. A deep, narrow hug means a stiff, high-melting material; a loose, wide one means something soft that melts easily.

Now picture *how* the hug forms. One atom can **hand over** an electron (ionic), **share** one (covalent), or throw all valence electrons into a common **sea** everyone swims in (metallic). The hand-over and share are directional and rigid; the sea lets atoms slide past each other, which is why metals bend instead of shattering.

5. Visual (one figure per key concept)

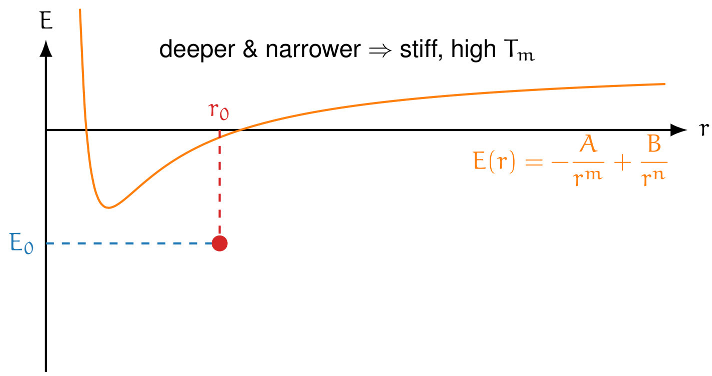


Figure 2.1. Figure 2.1 — The interatomic energy–separation curve. Attraction ($-A/r^m$) and repulsion ($+B/r^n$) sum to a well with minimum at (r_0, E_0).

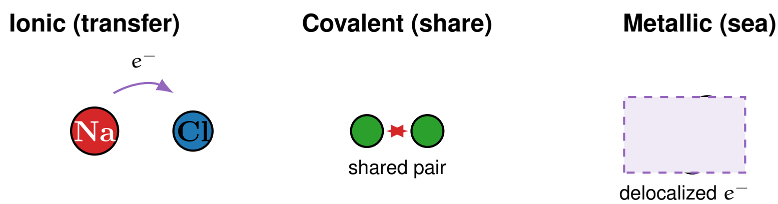


Figure 2.2. Figure 2.2 — Three primary bonding mechanisms: electron transfer (ionic), electron sharing (covalent), and a shared electron sea (metallic).

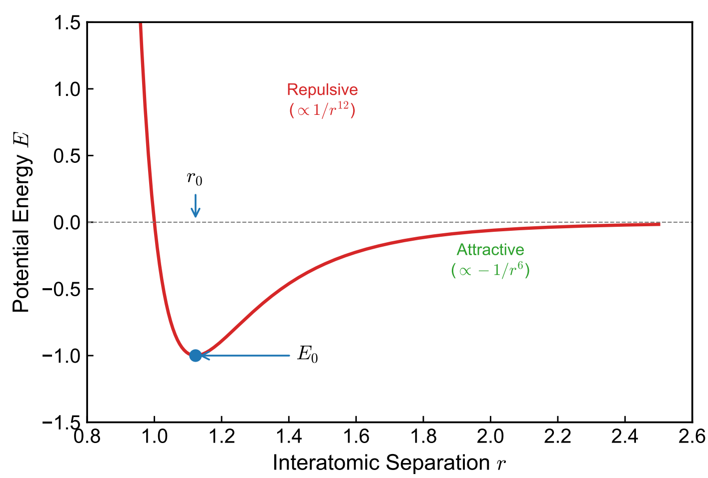


Figure 2.3. Figure 2.3 — The Lennard-Jones interatomic potential: net energy (orange) is the sum of attractive ($-1/r^6$, green) and repulsive ($+1/r^{12}$, red) terms, with minimum at (r_0, E_0).

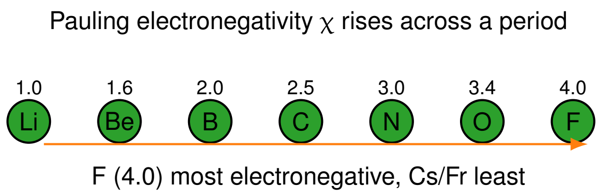


Figure 2.4. Figure 2.3 — Pauling electronegativity increases left-to-right across a period and decreases down a group; the gap $\Delta\chi$ decides bond type.

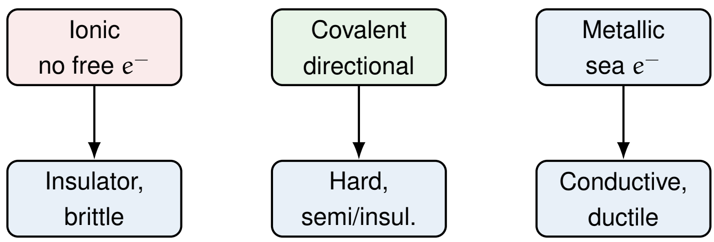


Figure 2.5. Figure 2.4 — Bond mechanism (top) maps directly to the property signature engineers select on (bottom).

6. Memory Box

Memory Box

- **Atomic number Z:** number of protons (defines element).
- **Atomic mass A:** g/mol; charge number Z of protons.
- **Avogadro N_A:** 6.022×10^{23} atoms/mol.
- **Electronegativity χ:** Pauling pull on bonding electrons.
- **Bond-rule thresholds:** $\Delta\chi > 1.7$ ionic; 0.4–1.7 polar covalent; < 0.4 nonpolar covalent/metallic.
- r_0**:** equilibrium spacing; E_0**:** bond energy at r_0.
- **Well depth** $\propto |E_0| \to$ melting point; **curvature at** $r_0 \to$ elastic modulus; **asymmetry** $\to$ thermal expansion.
- **Primary bonds:** 100–1000 kJ/mol; **secondary:** 0.1–40 kJ/mol.

7. Compare Box

Ionic vs Covalent vs Metallic vs Secondary

Ionic	Covalent	Metallic	Secondary
Electron transfer	Electron sharing	Delocalized sea	Dipole attraction
Non-directional	Highly directional	Non-directional	Non-directional
Insulator (solid)	Insulator/semicon.	Conductor	Insulator
Hard, brittle	Very hard	Ductile	Weak, soft
NaCl, MgO	diamond, Si	Fe, Cu, Al	H-bond, London
100–1500 kJ/mol	300–800 kJ/mol	100–800 kJ/mol	0.1–40 kJ/mol

8. Common Mistakes

Common Mistakes

- **"Ionic compounds conduct electricity as solids." Why wrong:** the ions are locked in the lattice; conduction only occurs when molten or dissolved in water.
- **Confusing bond energy with elastic modulus. Why wrong:** both come from the $E(r)$ well, but modulus is the *curvature* at r_0, not the well depth E_0.
- **Forgetting metallic bonding still "shares" electrons. Why wrong:** only the electrons are delocalized into a sea; the attraction is still Coulombic.
- **Using $\Delta\chi$ alone to call a ceramic "purely ionic." Why wrong:** most ceramics are mixed ionic–covalent; use $F_{\text{ionic}} = 1 - e^{-(\Delta\chi/2)^2}$ for the actual fraction.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$E_N = -\frac{A}{r^m} + \frac{B}{r^n}$	Net interatomic potential	J or eV	A, B constants; $m \approx 1$, $n \approx 6$–12	Pair potential, isolated atoms	No many-body/ environmental effects	Shape of the well sets properties
$r_0 = \sqrt[m-n]{\frac{nB}{mA}}$	Equilibrium separation	m or nm	from $dE_N/dr = 0$	Same as above	No temperature, no defects	Atomic spacing in a solid
$F = -k_0 \frac{(Z_1 e)(Z_2 e)}{r^2}$	Coulombic attraction	N	k_0 const, Z valence, e charge	Point charges, vacuum	Screened in real solids	Source of bond strength
$F_{ionic} = 1 - e^{-(\Delta\chi/2)^2}$	Ionic character fraction	–	$\Delta\chi$ Pauling gap	Pauling approximation	Empirical only	Predicts ceramic behavior
$\rho = \frac{nA}{V_c N_A}$	Theoretical density	g/cm^3	n atoms/cell, A at. wt, V_c cell vol	Perfect crystal, no voids	Ignores defects/-porosity	Mass screening of parts

10. Worked Examples

Basic

Example

Problem: For Na ($\chi = 0.93$) and Cl ($\chi = 3.16$), predict the dominant bond type.
Thinking: Compare $\Delta\chi$ to the thresholds in the memory box.
Solution:

$$\Delta\chi = 3.16 - 0.93 = 2.23$$
$$\Delta\chi > 1.7 \quad \Rightarrow \quad \textbf{ionic bond (NaCl).}$$

Intermediate

Example

Problem: MgO: $\chi_{\text{Mg}} = 1.31$, $\chi_{\text{O}} = 3.44$. Find the ionic-character fraction.
Thinking: Use Pauling's relation, then read the property signature.
Solution:

$$\begin{aligned} \Delta\chi &= 3.44 - 1.31 = 2.13 \\ F_{\text{ionic}} &= 1 - e^{-(2.13/2)^2} = 1 - e^{-1.134} = 1 - 0.322 = 0.678 \approx 68\%. \end{aligned}$$

MgO is strongly ionic (hard, high-T_m insulator) — consistent with the 63–67% table values.

Advanced

Example

Problem: Copper is FCC: $n = 4$ atoms/cell, $A = 63.55$ g/mol, $a = 0.3615$ nm. Compute theoretical ρ.
Thinking: $V_c = a^3$; convert nm^3 to cm^3 using $(10^{-7}\ \text{cm/nm})^3$.
Solution:

$$\begin{aligned} V_c &= (0.3615 \times 10^{-7}\ \text{cm})^3 = 4.724 \times 10^{-23}\ \text{cm}^3 \\ \rho &= \frac{nA}{V_c N_A} = \frac{4(63.55)}{(4.724 \times 10^{-23})(6.022 \times 10^{23})} \\ &= \frac{254.2}{28.45} = 8.94\ \text{g/cm}^3. \end{aligned}$$

Matches the handbook value (8.96 g/cm^3), confirming FCC and the metallic bond's close packing.

11. Engineering Insight

Engineering Insight

Bonding is the first lever in material selection. To raise stiffness or melting point you need deeper wells (stronger bonds) — which is why cutting tools use covalent diamond or ionic Al_2O_3 rather than metals. To get formability you need non-directional bonding, so you pick a metal. Thermal-expansion mismatch between a covalent ceramic and a metallic lead frame cracks electronics packages; understanding the well's asymmetry explains *why* ceramics expand less. Mixed ionic–covalent character also controls dielectric constant and corrosion resistance in ceramics.

12. Industrial Applications

Industrial Applications

- **Cutting tools:** covalent diamond / cubic-BN or ionic–covalent Al_2O_3, SiC for extreme hardness.
- **Electrical wiring:** metallic Cu, Al (ductile + conductive).
- **Thermal barrier coatings:** ionic ZrO_2 / YSZ (low k, high T_m).
- **Transparent windows:** ionic sapphire (Al_2O_3 single crystal, optical + refractory).
- **Semiconductors:** covalent Si, GaAs (band gap tuned by doping).
- **Polymer chains & water:** secondary H-bonds set T_g and boiling point.

13. Summary

Atoms attract and repel to form an energy well with equilibrium spacing r_0 and bond energy E_0; the well shape dictates stiffness, melting point, and thermal expansion. The three primary bonds — ionic (transfer), covalent (share), metallic (sea) — give distinct property signatures, while secondary van der Waals and hydrogen bonds dominate polymers and water. Most real ceramics are mixed ionic–covalent, with ionic character set by the Pauling gap $\Delta\chi$. The single most useful equation this chapter is $\rho = nA/(V_c N_A)$, linking atomic structure to a measurable bulk property.

14. Quick Review

Quick Review

1. The bond-energy minimum occurs at separation ____________.
2. The elastic modulus comes from the ____________ of the $E(r)$ well at r_0.
3. True / False: A larger $\Delta\chi$ means a more ionic bond. __________
4. A ductile, conductive material is held by ____________ bonds.
5. Pauling's ionic fraction: $F_{\text{ionic}} =$ ______________.

15. Practice Problems

⋆ Which bond type is characterized by directionality and very high stiffness?

(a) Ionic
(b) Covalent
(c) Metallic
(d) Secondary

(p. 144)

⋆ Given $\chi_{\text{Na}} = 0.93$ and $\chi_{\text{Cl}} = 3.16$, the Na–Cl bond is:

(a) Ionic
(b) Covalent
(c) Metallic
(d) Secondary

(p. 144)

⋆⋆ Which material property is determined by the *curvature* of the $E(r)$ well at r_0?

(a) Elastic modulus
(b) Melting point
(c) Thermal expansion
(d) Conductivity

(p. 144)

⋆⋆ A material with very high melting point, no electrical conductivity, and brittle fracture is most likely bonded by:

(a) Metallic

(b) Ionic

(c) Secondary

(d) Hydrogen

(p. 144)

⋆⋆⋆ Copper is ductile because its bonding is:

(a) Non-directional (metallic)

(b) Directional covalent

(c) Ionic with repulsion

(d) Secondary only

(p. 144)

⋆⋆⋆ For SiC, $\chi_{Si} = 1.90$, $\chi_{C} = 2.55$. Compute $\Delta\chi$ and the ionic-character fraction, then name the dominant bond character. (p. 144)

16. Flash Cards

Flash Card

Front: What does a deep, narrow energy well imply?

Back: Stiff material (high modulus) and high melting point.

Flash Card

Front: Threshold $\Delta\chi$ for an ionic bond?

Back: $\Delta\chi > 1.7$ is predominantly ionic.

Flash Card

Front: Why are metals ductile?

Back: Non-directional metallic (electron-sea) bonds let atoms slide without breaking.

Flash Card

Front: Modulus vs bond energy from the well?

Back: Modulus = curvature at r_0; bond energy = well depth E_0.

Flash Card

Front: Pauling ionic-character formula?

Back: $F_{\text{ionic}} = 1 - e^{-(\Delta\chi/2)^2}$.

Flash Card

Front: Theoretical density formula?

Back: $\rho = nA/(V_c N_A)$ g/cm^3.

Flash Card

Front: Why do ionic solids not conduct?

Back: Ions are fixed in the lattice; only molten/dissolved ions carry current.

17. Cheat Sheet

Cheat Sheet

Bond-rule: $\Delta\chi > 1.7$ ionic; 0.4–1.7 polar cov.; < 0.4 nonpolar cov./metallic.

Well: r_0 = spacing, E_0 = bond energy; depth $\to T_m$, curvature $\to E$, asymmetry $\to \alpha$.

Bonds:

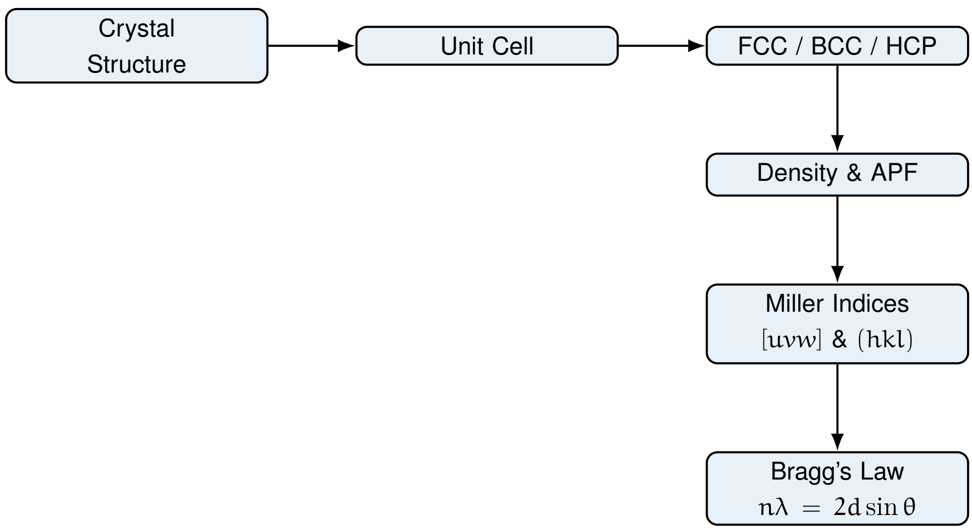
- Ionic: transfer, insulator, brittle, hard (NaCl, MgO).
- Covalent: share, directional, hard (diamond, Si).
- Metallic: sea, conductive, ductile (Fe, Cu, Al).
- Secondary: dipole, weak 0.1–40 kJ/mol (H-bond, London).

Key formulas: $E_N = -A/r^m + B/r^n$; $F_{\text{ionic}} = 1 - e^{-(\Delta\chi/2)^2}$; $\rho = nA/(V_c N_A)$.

Chapter 3

The Structure of Crystalline Solids

1. Roadmap



2. Learning Objectives

1. Sketch the BCC, FCC, and HCP unit cells and state where atoms sit.
2. Compute atoms per cell, coordination number, and atomic packing factor (APF).
3. Calculate theoretical density from crystal-structure data.
4. Index crystallographic directions $[uvw]$ and planes (hkl).
5. Use the cubic interplanar spacing d_{hkl} and Bragg's law for X-ray diffraction.
6. Distinguish FCC and BCC reflection (selection) rules for XRD.

3. Why This Matters

Crystal structure is the foundation of every mechanical, electrical, and thermal property you will meet later in the course. Whether a metal is ductile or brittle, how tightly atoms pack, and whether a semiconductor wafer will diffract a beam all trace back to the arrangement of atoms in the unit cell. On exams this chapter is a guaranteed source of calculation problems: density, APF, Miller indices, and Bragg analysis appear every year. In engineering it decides alloy choice, heat-treatment response, and failure analysis of load-bearing parts. Master the geometry here and the rest of materials science becomes far easier to learn and to apply.

4. Intuition First

Intuition First

Imagine packing oranges in a crate. If you place each layer so that every orange nestles into the hollows of the layer below, you fill more of the box than if you line them up in a simple square grid. That single habit of "nestling tightly" is the whole origin of crystal structures. A *crystal* is just a 3D repeating pattern of atoms; the smallest brick that repeats to build the whole pattern is the *unit cell*. The three metal structures differ only in how the layers offset: FCC and HCP both nestle perfectly (same efficiency), while BCC leaves a little more empty space. Density is simply "how much atom mass sits inside one brick divided by the brick's volume."

5. Visual (one figure per key concept)

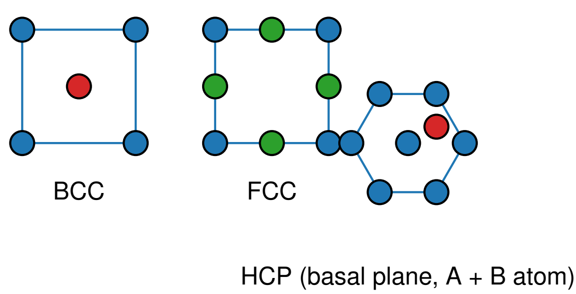


Figure 3.1. Figure 3.1 — 2D projections of the BCC, FCC and HCP unit cells. Blue = corner atoms; red = body-center; green = face-center atoms.

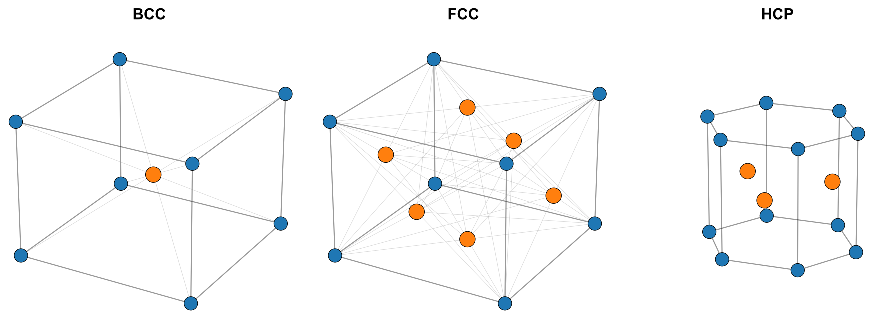
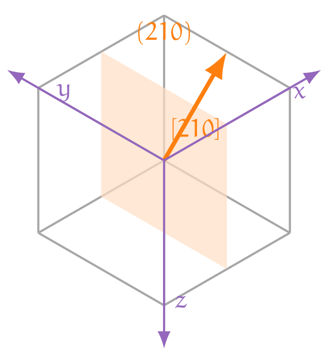


Figure 3.2. Figure 3.2 — Three-dimensional unit cells (3D, per the crystal-cell override of FIGURE_GUIDE): BCC (body-centered atom in orange), FCC (face-centered atoms in orange), and HCP.



Miller direction [210] (arrow) and plane (210) (face) in a 3D unit cell.

Figure 3.3. Figure 3.3 — A Miller direction [210] (orange arrow) and a Miller plane (210) (orange face) shown inside a 3D unit cell. In cubic crystals the direction is perpendicular to its like-indexed plane.

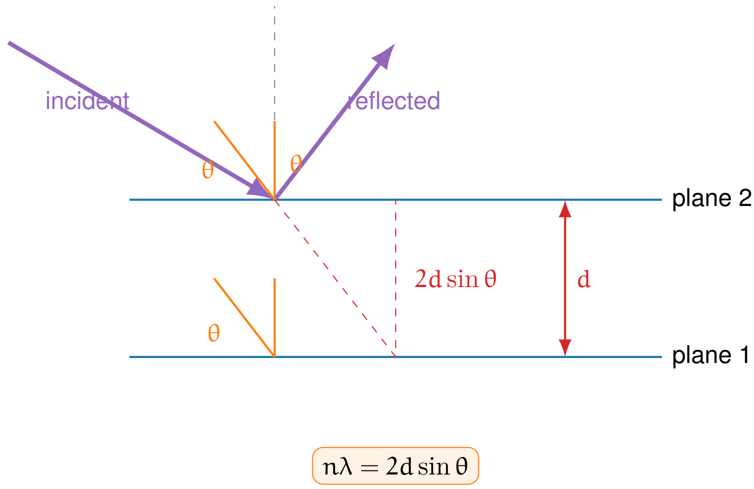


Figure 3.4. Figure 3.4 — Bragg diffraction geometry. An X-ray wave reflects from successive lattice planes (spacing d) at a glancing angle θ; the extra path travelled between planes is $2d\sin\theta$, and constructive interference occurs when $n\lambda = 2d\sin\theta$.

6. Memory Box

Memory Box

- **FCC**: 4 atoms/cell; CN 12; $a = 2\sqrt{2}R$; APF 0.74; Al, Cu, Ni, Pt, Au.
- **BCC**: 2 atoms/cell; CN 8; $a = 4R/\sqrt{3}$; APF 0.68; α-Fe, Cr, W, Mo.
- **HCP**: 6 atoms/cell (hex. prism); CN 12; ideal $c/a = \sqrt{8/3} \approx 1.633$; APF 0.74; Mg, Zn, Ti, Co.
- **Density**: $\rho = nA/(V_C N_A)$ with $N_A = 6.022 \times 10^{23}$ atoms/mol.
- **APF**: fraction of cell volume filled by hard spheres; FCC = HCP > BCC > SC.
- **Bragg**: $n\lambda = 2d\sin\theta$; FCC reflects when h, k, l all even or all odd; BCC when $h + k + l$ even.

7. Compare Box

FCC vs BCC vs HCP

FCC	BCC	HCP
4 atoms/cell	2 atoms/cell	6 atoms/cell
CN = 12	CN = 8	CN = 12
APF 0.74	APF 0.68	APF 0.74
$a = 2\sqrt{2}R$	$a = 4R/\sqrt{3}$	$a = 2R$, $c/a \approx 1.633$
Cubic	Cubic	Hexagonal
Close-packed	Not close-packed	Close-packed

8. Common Mistakes

Common Mistakes

- **Unit error in density:** forgetting $1\ \text{nm} = 10^{-7}$ cm (not 10^{-9}). Volume must be in cm^3 when ρ is in g/cm^3. **Why:** this single power-of-ten error ruins the answer.
- **Confusing direction and plane:** writing $[210]$ for a plane or (210) for a direction. **Why:** square brackets are directions, parentheses are planes.
- **Same reflection rule for FCC and BCC. Why:** FCC needs all even or all odd (hkl); BCC needs $h + k + l$ even. Mixing them mis-indexes the whole XRD pattern.
- **Mismatched units in Bragg's law. Why:** λ and d must share the same unit; θ is the glancing angle, not 2θ.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$n_{FCC} = 4$, $n_{BCC} = 2$, $n_{HCP} = 6$	Atoms per unit cell	–	n atoms	Hard-sphere atoms	–	Counts for density
$APF = \frac{n(4/3)\pi R^3}{V_C}$	Packing efficiency	–	R radius, V_C cell vol.	Spheres touch along close-packed dir.	Ideal spheres only	Higher APF = denser packing
$\rho = \frac{nA}{V_C N_A}$	Theoretical density	g/cm^3	A atomic wt., N_A Avogadro	Cubic cell, no defects	Vacancies lower real ρ	Alloy density control
$a = 2\sqrt{2}R$ (FCC), $a = 4R/\sqrt{3}$ (BCC)	Lattice param. vs radius	nm	a lattice param.	Touching along body/face diagonal	Single-element only	Sizing from R
$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$	Interplanar spacing	nm	a, Miller indices	Cubic crystals	Not for non-cubic	Indexes XRD peaks
$n\lambda = 2d\sin\theta$	Bragg's law	nm	λ wavelength, θ glancing angle	Elastic, ordered planes	Ignore disorder	XRD structure ID

10. Worked Examples

Basic

Atoms in an FCC Cell

Problem: How many atoms are in one FCC unit cell?
Thinking: Count contributions. Each of the 8 corners is shared by 8 cells; each of the 6 face centers is shared by 2 cells.
Solution:

$$n = 8\left(\frac{1}{8}\right) + 6\left(\frac{1}{2}\right) = 1 + 3 = \mathbf{4}.$$

Intermediate

Density of Copper

Problem: Cu is FCC with $R = 0.128$ nm and $A = 63.55$ g/mol. Find ρ.
Thinking: Get a from the FCC relation, cube it, convert to cm^3, then use $\rho = nA/(V_C N_A)$ with $n = 4$.
Solution:

$$\begin{aligned} a &= 2\sqrt{2}R = 2\sqrt{2}(0.128) = 0.362 \text{ nm}, \\ V_C &= a^3 = (0.362 \times 10^{-7} \text{ cm})^3 = 4.75 \times 10^{-23} \text{ cm}^3, \\ \rho &= \frac{4(63.55)}{(4.75 \times 10^{-23})(6.022 \times 10^{23})} = \mathbf{8.89 \text{ g/cm}^3}. \end{aligned}$$

(Measured 8.96 g/cm^3; small gap from vacancies and thermal expansion.)

Advanced

Indexing an Unknown XRD Pattern (BCC Fe)

Problem: A cubic metal gives $2\theta = 44.8°, 65.2°, 82.6°$ with $\lambda = 0.1542$ nm. Identify it.
Thinking: Compute $\sin^2\theta$ for each peak; their ratios reveal $h^2 + k^2 + l^2$. BCC allows only even $h + k + l$, giving $2, 4, 6, \ldots$ (ratio $1 : 2 : 3$).
Solution:

$$\begin{aligned} \sin^2(22.4°) : \sin^2(32.6°) : \sin^2(41.3°) &= 0.145 : 0.290 : 0.435 = 1 : 2 : 3, \\ d_{110} &= \frac{0.1542}{2\sin 22.4°} = 0.2027 \text{ nm}, \\ a &= d_{110}\sqrt{2} = \mathbf{0.2866} \text{ nm} \;\Rightarrow\; \textbf{BCC Fe}. \end{aligned}$$

11. Engineering Insight

Engineering Insight

Atomic packing decides how a metal deforms. FCC metals (Cu, Al, Ni, Au) have 12 nearest neighbors and many close-packed slip systems, so they are ductile and form easily. BCC metals (α-Fe, Cr, W) have a less symmetric slip set, giving higher strength but more brittle behavior at low temperature. HCP metals (Mg, Zn, Ti) slip on few systems and are often hard to forge. Engineers exploit this: they pick FCC copper for wire, BCC steel for strength, and titanium alloys for aerospace where weight and strength both matter. The same geometry that we count in a unit cell ultimately limits a bridge, an engine block, or a turbine blade.

12. Industrial Applications

Industrial Applications

- **X-ray diffraction (XRD):** phase identification and quality control in metallurgy and semiconductor fabs.
- **Alloy design:** density prediction for automotive and aerospace weight reduction.
- **Additive manufacturing:** crystal-structure checks on printed Ti-6Al-4V parts.
- **Thin-film electronics:** orienting FCC Cu interconnects and Al metallization.
- **Materials verification:** matching measured density and XRD peaks to a known structure in failure analysis.

13. Summary

Crystalline solids repeat a unit cell, and the three common metal cells differ only in how atoms nestle. FCC and HCP are the most efficient (APF 0.74, CN 12), while BCC is slightly looser (APF 0.68, CN 8). You can turn geometry into numbers: atoms per cell give density $\rho = nA/(V_C N_A)$, and the cubic interplanar spacing $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$ feeds Bragg's law $n\lambda = 2d \sin\theta$ for X-ray identification. FCC and BCC show different reflection rules, which is the key to reading a diffraction pattern. The single most important equation of the chapter is Bragg's law, because it links measurable diffraction angles back to the atomic lattice.

Key Formula

$n\lambda = 2d \sin\theta$

14. Quick Review

Quick Review

1. An FCC unit cell contains ________ atoms and has coordination number ________.
2. True / False: HCP has the same atomic packing factor as FCC. ________
3. The relation between lattice parameter and radius for BCC is $a =$ ________.
4. True / False: In Bragg's law, θ is the angle between the incident beam and the crystal plane. ________
5. For a BCC crystal, the sum $h + k + l$ of a reflecting plane must be ________.
6. One-sentence summary: ________________________

15. Practice Problems

⋆ How many atoms are in one BCC unit cell?

(a) 1 (b) 2 (c) 4 (d) 6

(p. 145)

⋆ The atomic packing factor of FCC is:

(a) 0.52 (b) 0.68 (c) 0.74 (d) 0.82

(p. 145)

⋆⋆ Copper is FCC with $R = 0.128$ nm. Its lattice parameter a is approximately:

(a) 0.256 nm (b) 0.362 nm (c) 0.512 nm (d) 0.296 nm

(p. 145)

★★ A peak at $2\theta = 44.8°$ for BCC Fe arises from the reflecting plane:

(a) (100) (b) (110) (c) (111) (d) (200)

(p. 145)

★★★ For FCC Al ($a = 0.405$ nm, $\lambda = 0.1542$ nm), the first three XRD peaks correspond to which planes?

(a) (100), (110), (111)

(b) (111), (200), (220)

(c) (110), (200), (211)

(d) (100), (200), (300)

(p. 145)

16. Flash Cards

Flash Card

Front: What are the atoms per cell for FCC, BCC, and HCP?

Back: FCC = 4, BCC = 2, HCP = 6 (hexagonal prism).

Flash Card

Front: What is the atomic packing factor (APF)?

Back: The fraction of unit-cell volume occupied by hard-sphere atoms: $n(4/3)\pi R^3/V_C$. FCC and HCP = 0.74; BCC = 0.68.

Flash Card

Front: Write Bragg's law and name each symbol.

Back: $n\lambda = 2d\sin\theta$; n order, λ wavelength, d interplanar spacing, θ glancing angle.

Flash Card

Front: How do you index a crystallographic direction $[uvw]$?

Back: Take vector components in lattice units, clear fractions, enclose in square brackets; use $\bar{u}$ for negative.

Flash Card

Front: What are the XRD reflection rules for FCC and BCC?

Back: FCC: h, k, l all even or all odd. BCC: $h + k + l$ is even.

Flash Card

Front: Give the cubic interplanar spacing formula.

Back: $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$, valid only for cubic crystals.

17. Cheat Sheet

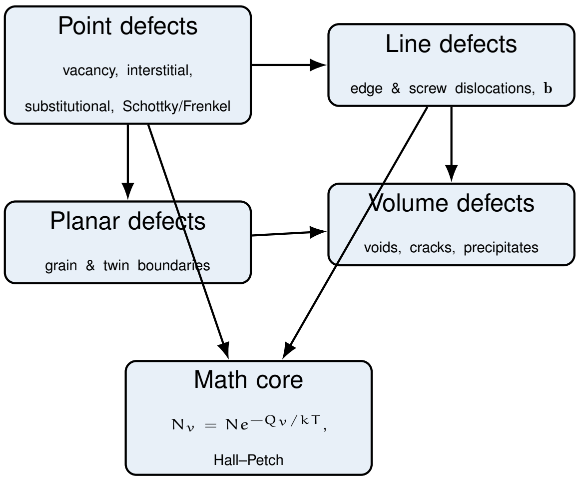
Cheat Sheet

- **FCC:** $n = 4$, CN 12, APF 0.74, $a = 2\sqrt{2}R$, Al Cu Ni Pt Au.
- **BCC:** $n = 2$, CN 8, APF 0.68, $a = 4R/\sqrt{3}$, α-Fe Cr W Mo.
- **HCP:** $n = 6$, CN 12, APF 0.74, $c/a \approx 1.633$, Mg Zn Ti Co.
- **Density:** $\rho = nA/(V_C N_A)$; use cm^3, 1 nm $= 10^{-7}$ cm.
- **APF:** $n(4/3)\pi R^3/V_C$.
- **Direction** $[uvw]$: vector comps $\to$ clear fractions $\to$ brackets.
- **Plane** (hkl): intercepts $\to$ reciprocals $\to$ clear fractions $\to$ parentheses.
- **Spacing:** $d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$ (cubic only).
- **Bragg:** $n\lambda = 2d \sin\theta$; FCC all-even/all-odd, BCC $h + k + l$ even.

Chapter 4

Imperfections in Solids

1. Roadmap



Real crystals are never perfect. Every deviation from an ideal lattice — an empty site, an extra atom, a slipped plane, a misoriented grain — is a *defect*, and these defects control diffusion, strength, conductivity, and failure. This chapter classifies defects by dimensionality (0-D, 1-D, 2-D, 3-D), gives the equation for equilibrium vacancy concentration, defines the Burgers vector that characterizes dislocations, and links grain size to strength through the Hall–Petch relation.

2. Learning Objectives

1. **Classify defects** by dimensionality: point (0-D), line (1-D), planar (2-D), and volume (3-D), with examples of each.
2. **Compute equilibrium vacancy concentration** using the Arrhenius expression $N_v = N\exp(-Q_v/kT)$ and convert temperature to Kelvin.
3. **Describe edge and screw dislocations**, locate the dislocation line, and state the direction of the Burgers vector $\mathbf{b}$ relative to it.
4. **Apply Hume-Rothery rules** to predict extensive vs. limited substitutional solid solubility.

5. **Explain Schottky and Frenkel defects** in ionic crystals and their link to diffusion and conductivity.

6. **Use the Hall–Petch relation** $\sigma_y = \sigma_0 + k_y d^{-1/2}$ to relate grain size to yield strength.

3. Why This Matters

Defects are not annoying flaws to be eliminated; they are the levers engineers pull. Vacancies make diffusion and heat treatment possible, dislocations make metals ductile instead of brittle, and grain boundaries make steel strong. On exams this chapter is high-yield: it feeds directly into diffusion, phase transformations, and mechanical behavior, and it appears as both formula questions (vacancy count, Hall–Petch) and concept questions (edge vs. screw, Schottky vs. Frenkel). In industry, controlling defects is how we make everything from turbine blades to semiconductor dopants.

4. Intuition First

Intuition First

Imagine a perfectly packed parking lot: every spot filled, every car aligned. A *vacancy* is an empty spot. A *self-interstitial* is a car parked on the grass between two spots — it does not fit, so it costs a lot of energy. A *substitutional impurity* is a different car in one spot; a *small interstitial impurity* (like a bicycle) fits in the gap.

Now picture a bookshelf. An *edge dislocation* is one extra shelf inserted halfway down, so the rows above are pushed out of line. A *screw dislocation* is the result of shearing the block with a screwdriver so the pages form a spiral ramp. Both let the whole stack deform a little at a time instead of breaking all at once.

A *grain boundary* is the messy line where two crystals meet at different angles, like two crowds drifting together. The mismatch stores energy and blocks movement — which is exactly why smaller grains make a material stronger.

5. Visual (one figure per key concept)

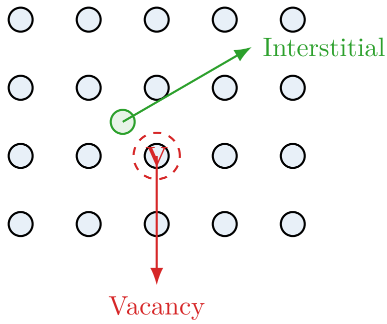


Figure 4.1. Figure 4.1 — A vacancy (missing host atom, red) and a self-interstitial (extra atom in an interstitial site, green).

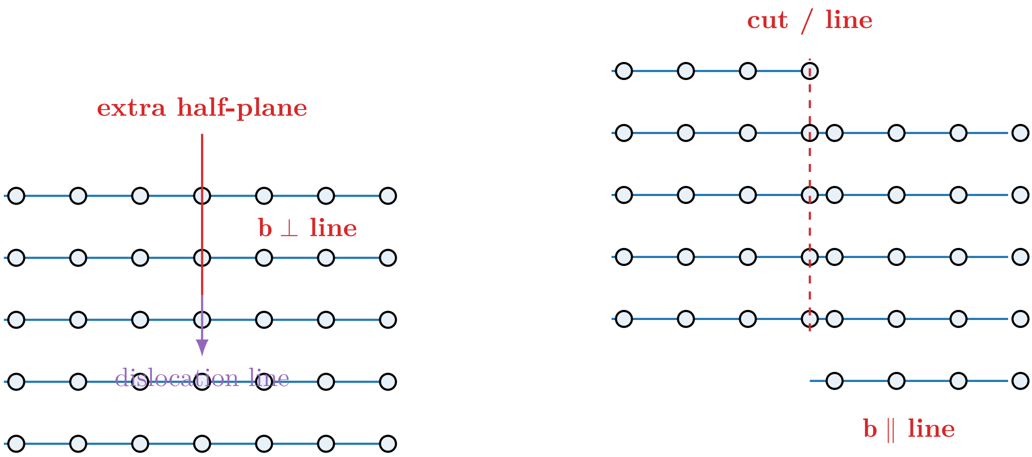


Figure 4.2. Figure 4.2 — Edge dislocation (left): an extra half-plane ends at the dislocation line, **b** perpendicular to the line. Screw dislocation (right): the lattice shears into a spiral ramp, **b** parallel to the line.

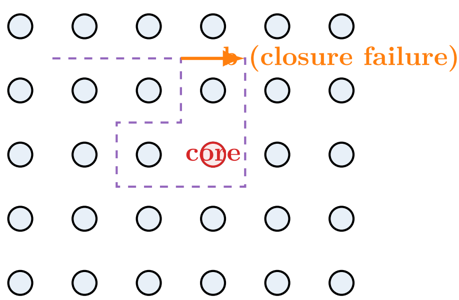


Figure 4.3. Figure 4.3 — Burgers circuit: a closed loop of lattice steps fails to close by the Burgers vector **b** when traced around a dislocation core.

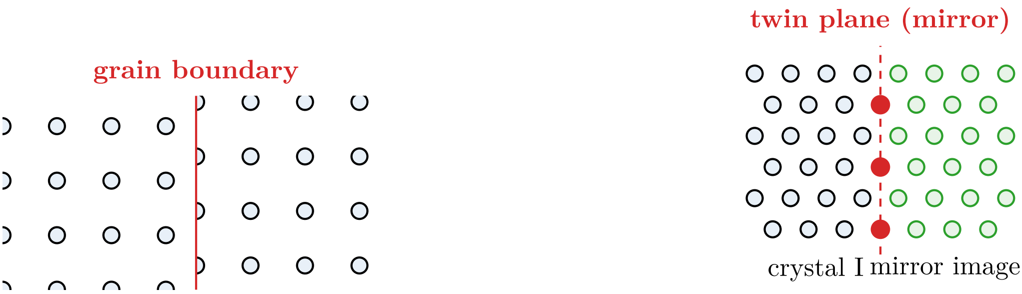


Figure 4.4. Figure 4.4 — A high-angle grain boundary (left) joins two misoriented crystals; a coherent twin boundary (right) is a low-energy *mirror-image* relationship: the lattice on one side is the exact reflection of the other across the twin (composition) plane, and the atoms on the plane are shared.

6. Memory Box

Memory Box

- **Equilibrium vacancies:** $N_v/N = \exp(-Q_v/kT)$. Always > 0 for $T > 0$; vanishes near 0 K.
- **Edge dislocation:** extra half-plane; $\mathbf{b} \perp$ dislocation line; line is the half-plane edge.
- **Screw dislocation:** spiral shear; $\mathbf{b} \parallel$ dislocation line; glide can cross-slip.
- **Burgers vector b:** lattice distortion magnitude & direction; closure failure of a Burgers circuit.
- **Dislocation density ρ:** m/m^3; annealed $\sim 10^{10} - 10^{12}$, cold-worked $\sim 10^{15} - 10^{16}$.
- **Hume-Rothery:** $<15\%$ size, same structure, similar electronegativity, same valence $\Rightarrow$ wide solubility.
- **Hall–Petch:** $\sigma_y = \sigma_0 + k_y d^{-1/2}$; smaller grains $\Rightarrow$ stronger.
- **Schottky:** paired vacancies (cation+anion), neutral. **Frenkel:** vacancy + interstitial, neutral.
- **Slip systems:** FCC $\{111\} = 12$, BCC $\{110/112/123\} = 48$, HCP basal $\{0001\} = 3$.

7. Compare Box

Edge vs. Screw Dislocation

Edge dislocation	**Screw dislocation**
Extra half-plane of atoms inserted.	Lattice sheared into a spiral ramp.
$\mathbf{b} \perp$ dislocation line.	$\mathbf{b} \parallel$ dislocation line.
Glides parallel to $\mathbf{b}$.	Glides perpendicular to $\mathbf{b}$; can cross-slip.
Climbs by vacancy absorption/emission.	No climb (no half-plane).

Vacancy vs. Interstitial

Vacancy	**Interstitial**
A lattice site left empty.	An atom in a normally empty gap site.
Low formation energy (common).	High formation energy (rare for self).
Promotes diffusion.	Small impurity atoms (C, H, N) fit well.

Schottky vs. Frenkel (ionic)

Schottky defect	**Frenkel defect**
Cation vacancy + anion vacancy.	Cation vacancy + cation interstitial.
Charge neutrality preserved.	Charge neutrality preserved.
e.g. NaCl, MgO.	e.g. AgCl, CaF_2.
$n \propto \exp(-E_s/2kT)$.	$n \propto \exp(-E_f/2kT)$.

8. Common Mistakes

Common Mistakes

- **Forgetting Kelvin:** The vacancy formula uses absolute temperature T in kelvin. Plugging in 900 (Celsius) instead of 1173 K gives a wildly wrong answer. **Why:** the exponent $-Q_v/kT$ is dimensionless and only correct in K.
- **Mixing Burgers-vector direction:** Students write $\mathbf{b} \parallel$ line for edge and $\perp$ for screw. **Why:** it is the opposite — edge is $\perp$, screw is $\parallel$.
- **Thinking dislocations make crystals *weaker*:** Dislocations let deformation happen at low stress, but high ρ *strengthens* (work hardening). **Why:** more dislocations obstruct each other; $\tau \propto Gb\sqrt{\rho}$.
- **Using R with per-atom Q_v:** If Q_v is in eV/atom, the constant is Boltzmann k; if in J/mol, use gas constant R. **Why:** unit mismatch silently breaks the exponent.
- **Assuming Hall–Petch always holds:** At very small (nanoscale) grains the inverse relation can reverse (gb sliding). **Why:** the ρ (dislocation) term weakens as grains shrink.
- **Confusing Schottky and Frenkel:** Schottky removes two atoms to surfaces; Frenkel merely moves one atom to an interstitial. **Why:** both keep neutrality but differ in atom count and density change.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$N_v = N e^{-Q_v/kT}$	Equilibrium vacancy number	m^{-3}	N sites/m^3, Q_v eV/atom	Thermal equilibrium, dilute	Not valid near melting	Vacancies drive diffusion
$N_v/N = e^{-Q_v/kT}$	Vacancy fraction	–	–	Same as above	–	Fraction grows with T
$N_v = N e^{-Q_v/RT}$	Vacancy number (molar)	m^{-3}	$R = 8.314$ J/molK	Q_v in J/mol	–	Use with molar Q_v
$n_s = N e^{-E_s/2kT}$	Schottky pairs	m^{-3}	E_s pair energy	Ionic, neutral pairs	Stoichiometric limit	Controls ceramic conductivity
$n_f = \sqrt{N N_i} e^{-E_f/2kT}$	Frenkel defects	m^{-3}	N_i interstitial sites	Ionic, one species moves	Low-T approx	AgCl, CaF_2 doping
$\sigma_y = \sigma_0 + k_y d^{-1/2}$	Hall–Petch strength	MPa	d grain dia.	Polycrystal, glide	Fails at nm grains	Smaller grain = stronger
$\tau \propto G b \sqrt{\rho}$	Dislocation strengthening	MPa	ρ density	Random forest	No dynamic recov.	Work hardening
$D = D_0 e^{-Q/RT}$	Arrhenius diffusivity	m^2/s	D_0, Q	Thermally activated	Not for short-circuit	Heat-treatment design

10. Worked Examples

Basic

Vacancy Fraction in Copper at 900°C

Problem: Given $Q_v = 0.9$ eV/atom for Cu, find N_v/N at $T = 900°C$.
Thinking: Convert Celsius to kelvin first ($T = 900 + 273 = 1173$ K), then evaluate the exponent with $k = 8.62 \times 10^{-5}$ eV/K.
Solution:

$$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{kT}\right) = \exp\left(-\frac{0.9}{(8.62 \times 10^{-5})(1173)}\right)$$
$$= \exp(-8.91) = 1.35 \times 10^{-4}.$$

So about 1 in 7400 sites is vacant at this temperature.

Intermediate

Absolute Vacancy Count in Copper

Problem: For FCC Cu, $a = 0.362$ nm, $A = 63.55$ g/mol. Find N_v (vacancies/m^3) at 1173 K with the fraction above.
Thinking: Compute atomic site density $N = n/V_C$ for FCC ($n = 4$), then multiply by the fraction.
Solution:

$$\begin{aligned} V_C &= (0.362 \times 10^{-9})^3 = 4.75 \times 10^{-29} \text{ m}^3, \\ N &= \frac{4}{4.75 \times 10^{-29}} = 8.42 \times 10^{28} \text{ atoms/m}^3, \\ N_v &= (1.35 \times 10^{-4})(8.42 \times 10^{28}) = 1.14 \times 10^{25} \text{ vacancies/m}^3. \end{aligned}$$

At 300 K the fraction is $\exp(-34.8) = 7.7 \times 10^{-16}$, effectively zero.

Advanced

Hall–Petch Grain Refinement

Problem: A steel has $\sigma_0 = 100$ MPa, $k_y = 0.8$ MPa·mm$^{1/2}$. Compare yield strength at $d = 10$ μm and $d = 1$ μm.
Thinking: Convert μm to mm (10 μm = 0.01 mm, 1 μm = 0.001 mm), then apply $\sigma_y = \sigma_0 + k_y d^{-1/2}$.
Solution:

$$\begin{aligned} d = 10 \ \mu\text{m} : &\quad \sigma_y = 100 + 0.8\,(0.01)^{-1/2} = 100 + 0.8 \times 10 = 108 \text{ MPa}. \\ d = 1 \ \mu\text{m} : &\quad \sigma_y = 100 + 0.8\,(0.001)^{-1/2} = 100 + 0.8 \times 31.6 = 125 \text{ MPa}. \end{aligned}$$

Refining the grain tenfold raises yield strength by about 16% through added boundary area.

11. Engineering Insight

Engineering Insight

Dislocations are why metals are ductile and why pure metals are soft. A perfect crystal would need $\sim G/10$ shear stress to slip; real crystals yield near $G/10^3$–$G/10^4$ because dislocations move a little at a time. Engineers exploit this two ways. First, *work hardening*: cold rolling multiplies ρ from 10^{10} to 10^{16} m/m^3, so flow stress $\tau \propto Gb\sqrt{\rho}$ rises a thousandfold. Second, *grain refinement*: every grain boundary is a wall dislocations cannot cross, so the Hall–Petch law lets us strengthen steel just by making grains smaller — the basis of high-strength low-alloy steels and nanocrystalline alloys. The same defects that "weaken" a perfect lattice are the knobs we turn to design toughness, formability, and strength.

12. Industrial Applications

Industrial Applications

- **Heat treatment & diffusion:** Vacancy concentration sets the rate of carburizing, nitriding, and annealing; higher T means more vacancies and faster case hardening.
- **Work-hardened sheet metal:** Beverage cans and auto body panels gain strength from dislocation density without adding alloy.
- **Grain-refined steels:** Controlled rolling and recrystallization tune d for pipelines and bridges (Hall–Petch).
- **Semiconductor doping:** Interstitial and substitutional impurities (P, B in Si) are placed via diffusion driven by point defects.
- **Ceramic ionic conductors:** Schottky/Frenkel defects govern oxygen-ion conduction in SOFC electrolytes (YSZ).
- **Powder metallurgy:** Voids and inclusions (volume defects) are controlled to avoid fracture-initiation sites.

13. Summary

Real solids contain defects at four scales. *Point defects* (vacancy, self-interstitial, substitutional and interstitial impurity, plus Schottky/Frenkel pairs in ionic crystals) exist in thermodynamic equilibrium; their concentration follows $N_v = N\exp(-Q_v/kT)$ and rises steeply with temperature. *Line defects* — edge and screw dislocations — carry plastic slip; the Burgers vector **b** is perpendicular to the line for edge and parallel for screw. *Planar defects* (grain and twin boundaries) store energy and block slip, giving the Hall–Petch strengthening $\sigma_y = \sigma_0 + k_y d^{-1/2}$. *Volume defects* (voids, cracks, precipitates) act as stress concentrators and fracture sites. The single equation to memorize is the equilibrium vacancy relation; it underpins diffusion, heat treatment, and conductivity.

14. Quick Review

Quick Review

1. A __________ is a lattice site left empty; a __________ is an extra atom in a gap site.
2. True / False: An edge dislocation has **b** parallel to its line. __________
3. In the Hall–Petch relation, decreasing grain size d __________ (increases / decreases) yield strength.
4. A Schottky defect in NaCl consists of a __________ vacancy plus an __________ vacancy.
5. Hume-Rothery requires atomic radii to differ by less than __________% for wide solubility.
6. FCC crystals have __________ slip systems, making them __________ (ductile / brittle).
7. One-sentence summary: Defects are the __________ that control diffusion, strength, and conductivity in real materials.

15. Practice Problems

⋆ The equilibrium concentration of vacancies in a metal:

(a) Increases exponentially with temperature.
(b) Decreases with temperature.
(c) Is independent of temperature.
(d) Is zero below the melting point.

(p. 145)

⋆ In an edge dislocation, the Burgers vector is:

(a) Parallel to the dislocation line.
(b) Perpendicular to the dislocation line.
(c) Zero.
(d) Equal to the lattice parameter.

(p. 145)

⋆⋆ According to the Hall–Petch relation, refining grain size from 10 μm to 1 μm will:

(a) Decrease strength.
(b) Increase strength.
(c) Not affect strength.
(d) Eliminate dislocations.

(p. 145)

⋆⋆ Which is **not** a Hume-Rothery rule for solid solubility?

(a) Similar atomic radii ($< 15\%$ difference).
(b) Same crystal structure.
(c) High melting point.
(d) Similar electronegativity.

(p. 145)

⋆⋆ A low-angle grain boundary consists of:

(a) Amorphous material.
(b) An array of dislocations.
(c) Precipitates.
(d) Twin boundaries.

(p. 145)

⋆⋆⋆ A Schottky defect differs from a Frenkel defect in that it:

(a) Involves only cation vacancies.
(b) Creates a vacancy-interstitial pair of the same ion.
(c) Removes a cation-anion pair from the lattice.
(d) Changes the crystal structure.

(p. 145)

16. Flash Cards

Flash Card

Front: What is a vacancy, and does it exist at 0 K?

Back: A lattice site with the atom missing. It exists in equilibrium only for $T > 0$; at 0 K the equilibrium concentration is zero.

Flash Card

Front: Edge vs. screw: direction of the Burgers vector **b**?

Back: Edge: **b** perpendicular to the dislocation line. Screw: **b** parallel to the line.

Flash Card

Front: Write the equilibrium vacancy concentration formula.

Back: $N_v = N \exp(-Q_v/kT)$, with N atomic sites/m^3, Q_v activation energy (eV/atom or J/mol), k or R the gas constant, T in kelvin.

Flash Card

Front: What does the Hall–Petch relation say?

Back: $\sigma_y = \sigma_0 + k_y d^{-1/2}$: smaller average grain diameter d gives higher yield strength.

Flash Card

Front: State the Hume-Rothery rules for wide solubility.

Back: Atomic radii within 15%, same crystal structure, similar electronegativity, same valence.

Flash Card

Front: Define Schottky and Frenkel defects (ionic crystals).

Back: Schottky: paired cation + anion vacancies (neutral). Frenkel: one cation vacancy + one cation interstitial (neutral).

Flash Card

Front: Why are FCC metals ductile but HCP often brittle?

Back: FCC has 12 slip systems; HCP has only 3 (basal). More slip systems let dislocations accommodate shape change.

Flash Card

Front: What is dislocation density ρ, and its typical range?

Back: Total dislocation length per unit volume (m/m^3). Annealed $\sim 10^{10}$–10^{12}; heavily deformed $\sim 10^{15}-10^{16}$.

17. Cheat Sheet

Cheat Sheet

Defect families (by dimension)

0-D point: vacancy, interstitial, substitutional/impurity, Schottky/Frenkel.
1-D line: edge (extra half-plane, $\mathbf{b} \perp$ line), screw (spiral, $\mathbf{b} \parallel$ line).
2-D planar: grain boundary (high energy), twin boundary (low energy, mirror), low-angle = dislocation array.
3-D volume: void, crack, precipitate, inclusion.

Key formulas

$N_v/N = \exp(-Q_v/kT)$ (use K! $k = 8.62 \times 10^{-5}$ eV/K, or R with J/mol)
Schottky: $n \propto \exp(-E_s/2kT)$; Frenkel: $n \propto \exp(-E_f/2kT)$
Hall–Petch: $\sigma_y = \sigma_0 + k_y d^{-1/2}$ (smaller $d \to$ stronger)
$\tau \propto Gb\sqrt{\rho}$ (more dislocations $\to$ stronger)

Memory numbers

Dislocation density: annealed 10^{10}–10^{12}; cold-worked 10^{15}–10^{16} m/m^3.
Slip systems: FCC 12, BCC 48, HCP 3.
Hume-Rothery size limit: $< 15\%$ radius difference.

Quick contrasts

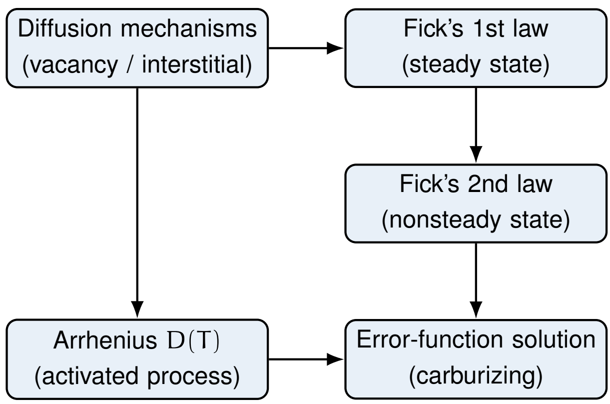
Edge glide $\parallel$ $\mathbf{b}$; screw glide $\perp$ $\mathbf{b}$ and can cross-slip.
Schottky = pair removed; Frenkel = one atom relocated to interstitial.
Climb (edge only) needs vacancies + temperature (creep).

Chapter 5

Diffusion

1. Roadmap

A concept map showing how the ideas in this chapter connect. Diffusion is the movement of atoms driven by a concentration gradient. The two laws describe the flux, the Arrhenius relation gives the temperature dependence, and the error-function solution lets us predict case depths in real treatments.



2. Learning Objectives

1. Distinguish vacancy diffusion from interstitial diffusion and state which is faster and why.
2. Apply Fick's first law to steady-state diffusion through a membrane.
3. Apply Fick's second law and the error-function solution to nonsteady-state diffusion.
4. Use the Arrhenius relation $D = D_0 \exp(-Q_d/RT)$ to find D at any temperature.
5. Solve carburizing problems for case depth, concentration, or required time.

3. Why This Matters

Diffusion appears on nearly every materials-science exam because it links atomic motion to properties we can engineer. Case-hardening of gears, doping of semiconductor wafers, and the densification of ceramics during sintering are all diffusion-controlled processes. Understanding the two Fick laws and the Arrhenius temperature dependence is the difference between predicting a heat-treatment time and guessing it in the lab.

4. Intuition First

Intuition First

Imagine a crowded room where people enter on the left and leave on the right. Nobody is told to move, yet the crowd flows from the crowded side to the empty side simply because random steps are more likely to go toward open space. That net drift is diffusion. Atoms do the same: thermal vibration makes each atom take random hops, but where the concentration is lower there are more empty sites to hop into, so the net flux points "downhill" in concentration. The negative sign in Fick's first law is just this downhill rule. If the two ends stay fixed, the profile never changes (steady state); if one end is being filled or drained, the profile moves with time (nonsteady state).

5. Visual (one figure per key concept)

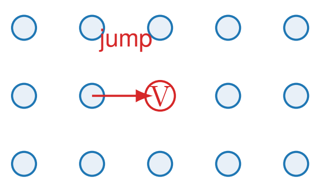


Figure 5.1. Figure 5.1 — Atomic jump mechanism: an atom hops into an adjacent vacancy, effectively moving the vacancy the other way.

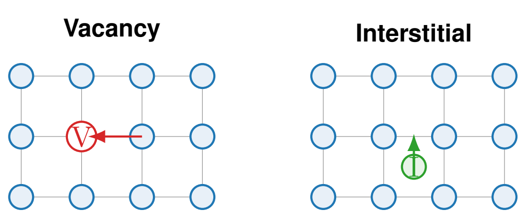


Figure 5.2. Figure 5.2 — Vacancy diffusion needs an empty lattice site; interstitial diffusion uses the gaps between host atoms, so it is faster.

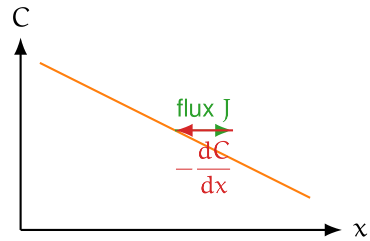


Figure 5.3. Figure 5.3 — Concentration gradient schematic. The flux points down the gradient (positive x), while the gradient dC/dx is negative.

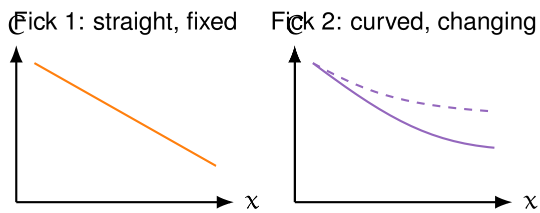


Figure 5.4. Figure 5.4 — Fick's first law: a straight profile that does not change with time. Fick's second law: a profile whose slope changes with time (dashed = earlier, solid = later).

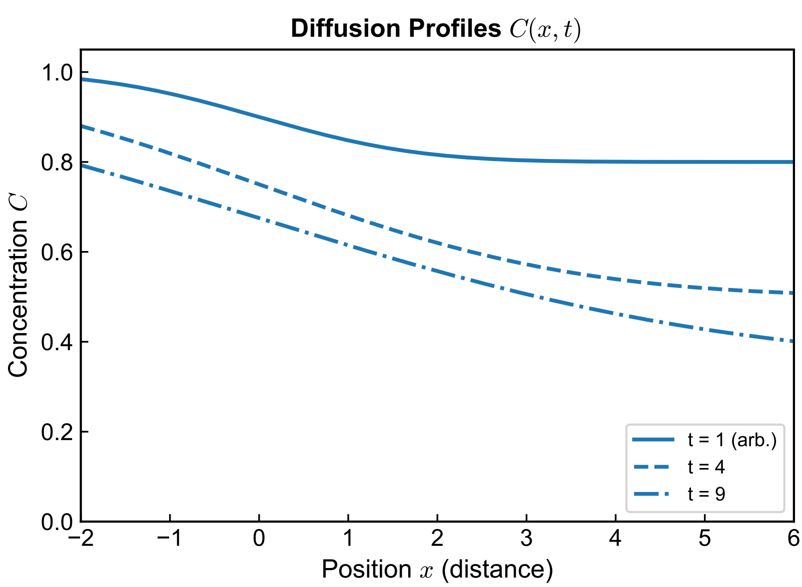


Figure 5.5. Figure 5.5 — Typical concentration profiles: steady-state linear profile and a nonsteady-state carburizing profile versus depth.

6. Memory Box

Memory Box

- **Diffusion** = mass transport by atomic motion driven by a concentration gradient.
- **Vacancy diffusion**: atom swaps with an empty site; dominant for substitutional alloys (Cu in Ni).
- **Interstitial diffusion**: small atoms (C, H, N) hop between gaps; faster, lower Q_d.
- **Fick 1**: $J = -D\, dC/dx$ (steady state, profile fixed).
- **Fick 2**: $\partial C/\partial t = D\, \partial^2 C/\partial x^2$ (transient).
- **Error fn**: $(C - C_0)/(C_S - C_0) = 1 - \mathrm{erf}(x/2\sqrt{Dt})$.
- **Arrhenius**: $D = D_0 \exp(-Q_d/RT)$; plot $\ln D$ vs $1/T$ gives slope $-Q_d/R$.
- **Diffusion length**: $\sqrt{Dt}$ is the characteristic penetration depth.

7. Compare Box

Vacancy vs Interstitial Diffusion

Vacancy diffusion	**Interstitial diffusion**
Atom moves into an adjacent empty lattice site	Small atom moves through the gaps between host atoms
Requires vacancies to be present	No vacancy needed; sites widely available
Slower; higher activation energy Q_d	Faster; lower activation energy Q_d
Dominant in substitutional alloys (e.g. Cu in Ni, self-diffusion)	Dominant for small atoms (C, H, N in Fe)

8. Common Mistakes

Common Mistakes

- **Forgetting to convert °C to Kelvin** ($T_K = T_C + 273$) in the Arrhenius equation. **Why:** the gas constant R is per Kelvin, so Celsius gives a wrong exponent by orders of magnitude.
- **Confusing the sign in Fick's first law. Why:** the minus sign means flux is opposite the gradient (downhill); dropping it predicts flow from low to high concentration.
- **Treating D as a constant when it varies with composition. Why:** the error-function solution assumes constant D; real systems need an averaged or effective value.
- **Reading $\sqrt{Dt}$ as an exact front. Why:** it is only a characteristic penetration depth, not the position of a sharp concentration step.
- **Using the steady-state law for a growing case. Why:** carburizing deepens with $\sqrt{t}$, which needs Fick's second law, not the straight-line first law.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$J = -D\,\frac{dC}{dx}$	Flux down the gradient (steady state)	$kg/(m^2 s)$	J flux, D diffusivity, C conc.	Steady state; constant D	No time dependence	Linear membrane permeation
$J = D\,\frac{C_A - C_B}{\Delta x}$	Flux through a flat membrane	$kg/(m^2 s)$	Δx thickness	Linear gradient; steady state	Neglects edge effects	Gas / carbon through a wall
$\frac{\partial C}{\partial t} = D\,\frac{\partial^2 C}{\partial x^2}$	Rate of change of concentration	$kg/(m^3 s)$	t time, x position	Constant D; 1-D	No sources/sinks	Transient profiles
$\frac{C - C_0}{C_S - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$	Semi-infinite solid, fixed surface C_S	dimensionless	C_0 initial, C_S surface	Constant C_S, D	Semi-infinite only	Carburizing case depth
$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$	Diffusivity vs temperature	m^2/s	D_0 prefactor, Q_d act. energy	Thermally activated	Single mechanism	Predict D at any T
$J = \mathscr{P}\,\frac{dC}{dx}$	Permeability form ($L = D/\Delta x$)	$kg/(m^2 s)$	$\mathscr{P} = D/\Delta x$ permeability	Steady state	Thin membrane	Throughput of a barrier

10. Worked Examples

Basic

Steady-state flux through a plate

Problem: Carbon in γ-Fe at 1000°C diffuses through a steel plate 2 mm thick. Surface concentrations are held at 1.2 wt% (high side) and 0.2 wt% (low side). Use $D = 1.4 \times 10^{-11}$ m^2/s. Find the flux J.

Thinking: This is steady state, so Fick's first law with a linear gradient applies: $J = D(C_A - C_B)/\Delta x$. Keep concentration in wt% units consistently; the wt% difference cancels with any consistent conversion.

Solution:

$$J = D\frac{C_A - C_B}{\Delta x} = (1.4 \times 10^{-11})\frac{1.2 - 0.2}{2 \times 10^{-3}}$$
$$= (1.4 \times 10^{-11})\frac{1.0}{2 \times 10^{-3}} = 7.0 \times 10^{-9} \text{ (wt\% · m/s)}.$$

Intermediate

Diffusivity at a different temperature

Problem: For C in γ-Fe, $D = 1.4 \times 10^{-11}$ m^2/s at 1000°C. Given $D_0 = 2.0 \times 10^{-5}$ m^2/s and $Q_d = 142$ kJ/mol, find D at 800°C.

Thinking: Use the Arrhenius form directly from the prefactor D_0 and activation energy Q_d. Temperature must be in Kelvin (800°C = 1073 K).

Solution:

$$\begin{aligned}\ln D &= \ln D_0 - \frac{Q_d}{R} \cdot \frac{1}{T} \\ &= \ln(2.0 \times 10^{-5}) - \frac{142000}{8.314} \cdot \frac{1}{1073} \\ &= -10.82 - 15.91 = -26.73, \\ D &= e^{-26.73} = \mathbf{2.7 \times 10^{-12}} \text{ m}^2\text{/s}.\end{aligned}$$

For comparison, at 1000°C (1273 K) $D = 1.4 \times 10^{-11}$; dropping 200°C reduces D by a factor of about 5.

Advanced

Carburizing time for a target case depth

Problem: A 0.20 wt% C steel is carburized at 1000°C in a C-rich atmosphere ($C_S = 1.2$ wt%) with $D = 1.4 \times 10^{-11}$ m^2/s. How long to reach 0.40 wt% C at a depth of 0.5 mm?

Thinking: Nonsteady state with fixed surface concentration calls for the error-function solution. Solve for $z = x/(2\sqrt{Dt})$, read erf from the table, then invert for t.

Solution:

$$\frac{C - C_0}{C_S - C_0} = \frac{0.40 - 0.20}{1.2 - 0.20} = \frac{0.20}{1.00} = 0.20,$$
$$1 - \operatorname{erf}(z) = 0.20 \;\Rightarrow\; \operatorname{erf}(z) = 0.80.$$

From the error-function table, $\operatorname{erf}(0.90) = 0.7970$ and $\operatorname{erf}(0.95) = 0.8209$; interpolating gives $z \approx 0.906$.

$$t = \frac{x^2}{4Dz^2} = \frac{(5 \times 10^{-4})^2}{4(1.4 \times 10^{-11})(0.906)^2} = \frac{2.5 \times 10^{-7}}{4.60 \times 10^{-11}} \approx 5435 \text{ s} \approx \mathbf{90} \text{ min}.$$

11. Engineering Insight

Engineering Insight

The single most useful design rule is case depth $\propto \sqrt{Dt}$. Because D rises steeply with temperature through the Arrhenius relation, a modest temperature increase slashes the time needed for the same case depth far more than simply extending the hold. In semiconductor doping, this same $\sqrt{Dt}$ controls the junction depth that sets device geometry: a few minutes at 1100°C can place a p–n junction a micron below the surface. Engineers therefore trade temperature against time, but must respect the material's microstructural limits (grain growth, distortion) when pushing T higher.

12. Industrial Applications

Industrial Applications

- **Carburizing:** carbon diffuses into low-C steel at 900–1050°C for 2–24 h, giving a hard, wear-resistant case over a tough core. Case depth $\propto \sqrt{Dt}$.
- **Nitriding:** nitrogen diffuses into steel at ~ 500°C to form hard nitrides (AlN, CrN); harder but thinner than a carburized case.
- **Semiconductor doping:** controlled diffusion of P, B, As into Si at 1000–1200°C creates p–n junctions (drive-in diffusion profile $C_S \operatorname{erfc}(x/2\sqrt{Dt})$).
- **Sintering:** a powder compact is heated below T_m; atoms diffuse across particle boundaries to close pores and densify ceramics and PM parts.
- **Homogenization:** a high-temperature hold after casting smooths dendritic microsegregation; time scales as $t \approx \lambda^2/(\pi^2 D)$ where λ is the dendrite spacing.

13. Summary

Diffusion is atomic transport down a concentration gradient. Vacancy diffusion dominates substitutional alloys and is slower; interstitial diffusion of small atoms (C, H, N) is faster. Fick's first law $J = -D\,dC/dx$ describes steady state, while Fick's second law $\partial C/\partial t = D\,\partial^2 C/\partial x^2$ describes nonsteady state, whose fixed-surface solution uses the error function. The diffusivity follows Arrhenius behavior $D = D_0 \exp(-Q_d/RT)$, so a small temperature rise hugely increases D. The most important engineering equation is the carburizing profile, where the characteristic depth is $\sqrt{Dt}$. Master these five relations and most exam problems reduce to reading a table and inverting an exponent.

14. Quick Review

Quick Review

1. Interstitial diffusion is ________ (faster / slower) than vacancy diffusion because it has ________ (higher / lower) activation energy.
2. The minus sign in $J = -D\,dC/dx$ means flux points ________ (up / down) the concentration gradient.
3. True / False: In Fick's second law the concentration profile is constant in time. ________
4. To double the case depth at fixed temperature you must increase the time by a factor of ________.
5. On an Arrhenius plot of $\ln D$ versus $1/T$, the slope equals ________.

15. Practice Problems

⋆ Interstitial diffusion is faster than vacancy diffusion because:

(a) More sites are available and the activation energy is lower
(b) Interstitial atoms are larger
(c) Vacancy diffusion requires grain boundaries
(d) Interstitial diffusion does not follow Fick's laws

(p. 146)

⋆ Fick's first law states $J = -D\,dC/dx$. The negative sign indicates:

(a) Flux is opposite to the concentration gradient direction
(b) The diffusion coefficient is negative
(c) Flow is from high to low concentration only at steady state
(d) Concentration increases with time

(p. 146)

⋆⋆ In carburizing, to double the case depth at the same temperature the time must be increased by a factor of:

(a) 2
(b) 4

(c) $\sqrt{2}$

(d) 8

(p. 146)

★★ At a given temperature, carbon diffuses faster in α-Fe (BCC) than in γ-Fe (FCC) because:

(a) BCC has a lower atomic packing factor (0.68 vs. 0.74)

(b) BCC has higher density

(c) Carbon solubility is higher in BCC

(d) BCC has no grain boundaries

(p. 146)

★★★ The activation energy Q_d for diffusion is determined from:

(a) The slope of an Arrhenius plot ($\ln D$ vs. $1/T$)

(b) The intercept of $\ln D$ vs. $\ln t$

(c) Fick's first law

(d) The error-function table

(p. 146)

16. Flash Cards

Flash Card

Front: What is diffusion?

Back: Mass transport by atomic motion driven by a concentration (chemical potential) gradient.

Flash Card

Front: Vacancy vs interstitial diffusion — which is faster and why?

Back: Interstitial is faster: small atoms (C, H, N) move through gaps between host atoms, needing no vacancy and a lower activation energy.

Flash Card

Front: State Fick's first law and when it applies.

Back: $J = -D\,dC/dx$. Applies to steady-state diffusion where the concentration profile does not change with time.

Flash Card

Front: What does the error-function solution describe?

Back: Nonsteady diffusion into a semi-infinite solid with fixed surface concentration: $(C - C_0)/(C_S - C_0) = 1 - \mathrm{erf}(x/2\sqrt{Dt})$.

Flash Card

Front: Write the Arrhenius relation for diffusivity.

Back: $D = D_0 \exp(-Q_d/RT)$. A plot of $\ln D$ vs $1/T$ gives slope $-Q_d/R$ and intercept $\ln D_0$.

Flash Card

Front: What is the diffusion length and what is it NOT?

Back: $\sqrt{Dt}$ is the characteristic penetration depth. It is NOT a sharp concentration front or exact case boundary.

17. Cheat Sheet

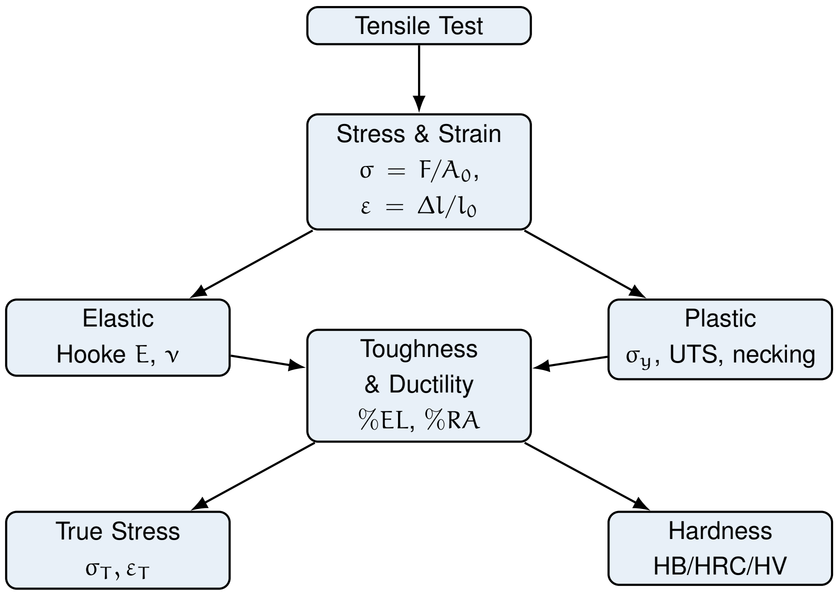
Cheat Sheet

- **Two mechanisms:** vacancy (substitutional, slower) | interstitial (C/H/N, faster).
- **Fick 1 (steady):** $J = -D\, dC/dx$; membrane: $J = D(C_A - C_B)/\Delta x$.
- **Fick 2 (transient):** $\partial C/\partial t = D\, \partial^2 C/\partial x^2$.
- **Error fn:** $\dfrac{C - C_0}{C_S - C_0} = 1 - \mathrm{erf}\Big(\dfrac{x}{2\sqrt{Dt}}\Big)$.
- **Arrhenius:** $D = D_0 \exp(-Q_d/RT)$; always use Kelvin.
- **Diffusion length:** $\sqrt{Dt}$; case depth $\propto \sqrt{t}$ at fixed T.
- **Orders of magnitude:** dropping ~ 200°C cuts D by a factor of ~ 5.
- **Typical D_0, Q_d:** C in α-Fe: 1.1×10^{-6}, 87; C in γ-Fe: 2.0×10^{-5}, 142 (kJ/mol).
- **Pitfalls:** use K not °C; keep the minus sign; assume constant D for erf.

Chapter 6

Mechanical Properties of Metals

1. Roadmap



The chapter starts with the standard tensile test, which produces the engineering stress–strain curve. From that single curve we read every major property: stiffness (E), the elastic–plastic boundary (σ_y), peak load (UTS), energy absorption (toughness), and stretch before failure (ductility). We then correct the curve for the shrinking cross-section (true stress and strain) and finish with hardness, a fast surface probe that correlates directly with tensile strength.

2. Learning Objectives

1. **Sketch and label** the engineering stress–strain curve: elastic region, yield, UTS, necking, and fracture.
2. **Compute** E, σ_y (0.2% offset), σ_{TS}, %EL, and %RA from test data.
3. **Apply** Hooke's law and Poisson's ratio to elastic deformation.
4. **Convert** between engineering and true stress/strain.
5. **Interpret** toughness and resilience as areas under the curve.

6. **Relate** Brinell/Rockwell/Vickers hardness to tensile strength.

3. Why This Matters

Mechanical properties decide whether a bracket bends, a cable snaps, or a car crumples safely. Every structural design starts by comparing the applied stress with the material's yield and tensile strength, while ductility and toughness govern how a part fails — gracefully or catastrophically. In exams this chapter is high-yield: curve-reading, offset-yield, and true-stress conversions appear almost every year. In practice, engineers quote these numbers from a datasheet before selecting any metal.

4. Intuition First

Intuition First

Think of a metal bar as a bundle of tiny springs (atomic bonds). Pull gently and the springs stretch, then snap back when released — this is *elastic* deformation, and the spring stiffness is Young's modulus E. Pull harder and atomic planes start sliding permanently: the bar stays longer even after release — this is *plastic* deformation, beginning at the yield point. Keep pulling and the bar first strengthens (atoms tangle and resist sliding), reaches a peak load, then thins locally into a "neck" and finally tears. Stiffness, strength, and stretchiness are three independent personalities: a material can be stiff but brittle, or soft but very ductile.

5. Visual (one figure per key concept)

Elastic vs. plastic regions. The elastic region is the straight line whose slope is E; beyond yield the curve bends over into permanent (plastic) deformation.

Yield by 0.2% offset. When yield is gradual, draw a line parallel to the elastic slope starting at $\varepsilon = 0.002$; its intersection with the curve defines σ_y.

Engineering vs. true stress–strain. Engineering stress uses the fixed A_0 and falls after UTS; true stress uses the instantaneous area A_i and keeps rising to fracture.

Ductility geometry (%EL and %RA). %EL compares gauge length before and after; %RA compares cross-sectional area before and at the neck.

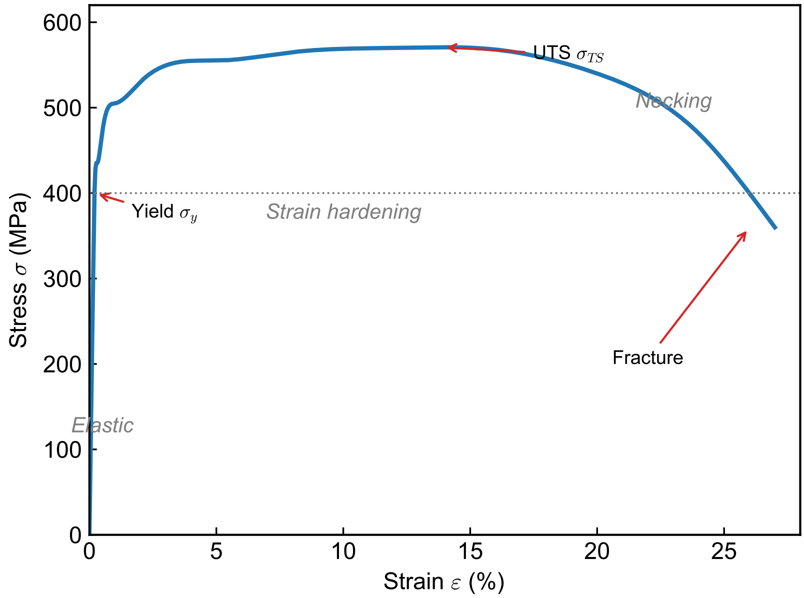


Figure 6.1. Figure 6.1 — Engineering stress–strain curve: elastic region, yield point, ultimate tensile strength, necking, and fracture. Area under the curve is toughness.

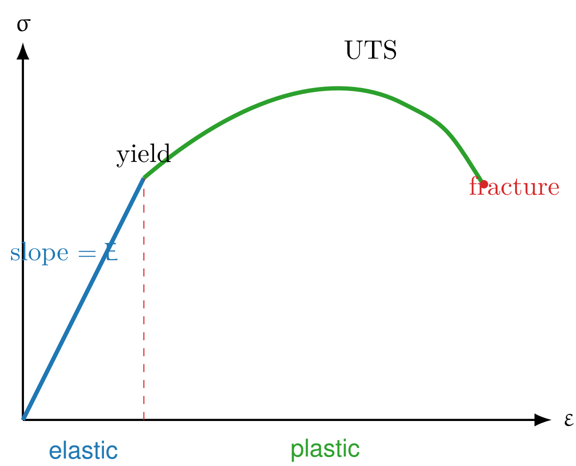


Figure 6.2. Figure 6.2 — Elastic (linear, recoverable) vs. plastic (curved, permanent) deformation.

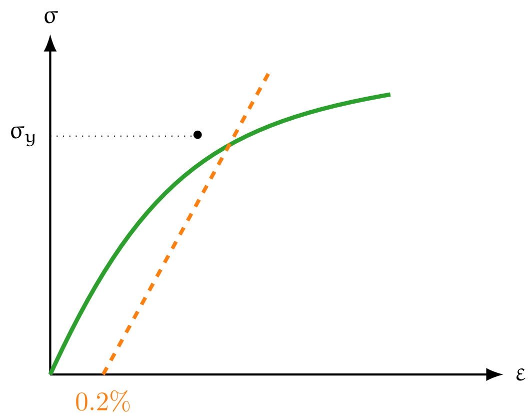


Figure 6.3. Figure 6.3 — 0.2% offset construction for yield strength.

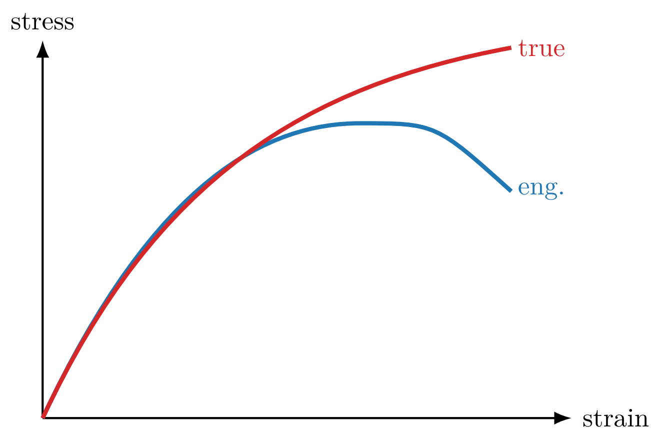


Figure 6.4. Figure 6.4 — Engineering curve (drops after necking) vs. true curve (rises to fracture).

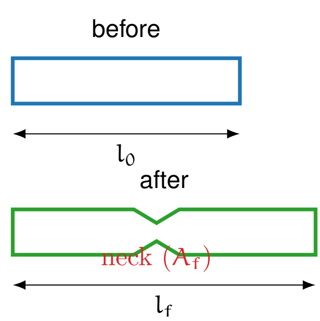


Figure 6.5. Figure 6.5 — %EL from gauge length change; %RA from cross-section reduction at the neck.

6. Memory Box

Memory Box

- E **(Young's modulus):** slope of elastic line; steels ≈ 207 GPa, Al ≈ 69 GPa, Cu ≈ 110 GPa.
- **Poisson's ratio** ν**:** metals ≈ 0.33; range 0.25–0.35.
- σ_y**:** onset of plastic flow (use 0.2% offset if gradual).
- **UTS:** peak of the engineering curve; necking starts here.
- **Toughness** = area under whole curve; **Resilience** = $\sigma_y^2/2E$ = area under elastic part.
- **Ductility:** %EL and %RA; report gauge length (50 mm standard).
- **Hardness–strength (steels):** σ_{TS} (MPa) ≈ 3.45 × HB.

7. Compare Box

Elastic vs. Plastic Deformation

Elastic	**Plastic**
Recoverable; bar returns to l_0	Permanent; bar stays deformed
Linear, $\sigma = E\varepsilon$ (Hooke)	Nonlinear, strain hardening
Bond stretching only	Atomic planes slip (dislocations)
Below σ_y	Above σ_y

Engineering vs. True Stress–Strain

Engineering	**True**
$\sigma = F/A_0$, $\varepsilon = \Delta l/l_0$	$\sigma_T = F/A_i$, $\varepsilon_T = \ln(l_i/l_0)$
Uses fixed original area	Uses instantaneous area
Curve drops after UTS	Curve rises to fracture
Design/reporting standard	Physically accurate; $\sigma_T = K\varepsilon_T^n$

8. Common Mistakes

Common Mistakes

- **Using A_i with engineering stress. Why:** engineering stress is defined with A_0; only true stress uses instantaneous area.
- **Confusing UTS with fracture strength. Why:** UTS is the peak; fracture occurs later at lower *engineering* stress after necking.
- **Forgetting the offset. Why:** many metals have no sharp yield; σ_y needs the 0.2% offset line.
- **Reporting $\%EL$ without gauge length. Why:** elongation localizes at the neck, so $\%EL$ depends on l_0.
- **Mixing up stiffness and strength. Why:** E (stiffness) and σ_y (strength) are independent; a stiff metal can be weak.
- **Applying 3.45×HB to all materials. Why:** the hardness–strength factor is calibrated for steels only.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$\sigma = F/A_0$	Engineering stress	MPa	F load, A_0 orig. area	Uniform, axial load	Ignores area change	Design stress vs. strength
$\varepsilon = \Delta l/l_0$	Engineering strain	—	Δl, l_0 gauge	Uniform strain	Only pre-necking	How much it stretches
$E = \sigma/\varepsilon$	Young's modulus (Hooke)	GPa	σ, ε	Elastic, linear	Below σ_y only	Stiffness of material
$\nu = -\varepsilon_{trans}/\varepsilon_{axial}$	Poisson's ratio	—	lateral/axial strain	Elastic, isotropic	Metals ≈ 0.33	Lateral contraction
$\sigma_T = \sigma(1+\varepsilon)$	True stress	MPa	σ, ε eng. values	Constant volume	Valid to necking	Real stress on atoms
$\varepsilon_T = \ln(1+\varepsilon)$	True strain	—	ε eng. strain	Constant volume	Valid to necking	Cumulative strain
$\%EL = \frac{\Delta l}{l_0} \times 100$	Percent elongation	%	Δl, l_0	Report l_0	Gauge dependent	Ductility measure
$\%RA = \frac{A_0 - A_f}{A_0} \times 100$	Percent area reduction	%	A_0, A_f neck	Round section	Needs neck area	Ductility (local)
$HB = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})}$	Brinell hardness	kgf/mm^2	P load, D ball, d indent	Standard ball/load	Not thin/hard parts	Fast strength proxy

10. Worked Examples

Basic

Example

Problem: A rod of $E = 200$ GPa carries $\sigma = 400$ MPa elastically. Find the strain.
Thinking: Below yield, use Hooke's law $\varepsilon = \sigma/E$; keep units consistent (MPa/MPa).
Solution:

$$\varepsilon = \frac{\sigma}{E} = \frac{400 \times 10^6}{200 \times 10^9} = 0.002.$$

Intermediate

Example

Problem: A steel bar ($A_0 = 100$ mm^2) is loaded to $F = 50$ kN at engineering strain $\varepsilon = 0.10$. Find engineering stress, true stress, and true strain.
Thinking: Engineering stress uses A_0; then convert with $\sigma_T = \sigma(1 + \varepsilon)$ and $\varepsilon_T = \ln(1 + \varepsilon)$.
Solution:

$$\sigma = \frac{50\,000}{100 \times 10^{-6}} = 500 \text{ MPa},$$
$$\sigma_T = 500(1 + 0.10) = 550 \text{ MPa},$$
$$\varepsilon_T = \ln(1.10) = 0.0953.$$

Advanced

Example

Problem: A steel has $\sigma_y = 500$ MPa and $E = 200$ GPa. Find the modulus of resilience, and estimate σ_{TS} if its Brinell hardness is HB $= 250$.
Thinking: Resilience is the elastic-triangle area $U_r = \sigma_y^2/2E$. For steels, use $\sigma_{TS} \approx 3.45\times$HB.
Solution:

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(500 \times 10^6)^2}{2(200 \times 10^9)} = 0.625 \text{ MJ/m}^3,$$

$$\sigma_{TS} \approx 3.45 \times 250 = 862 \text{ MPa}.$$

11. Engineering Insight

Engineering Insight

Designers rarely load a part to its tensile strength — they load it below yield, protected by a safety factor N ($\sigma_{\text{working}} = \sigma_y/N$, with $N \approx 1.5$–4). The choice between two metals is a trade-off: high strength buys thinner, lighter parts, but usually costs ductility and toughness, so a component may fail suddenly instead of bending as a warning. Toughness (area under the curve) is what absorbs impact in a car crash box, while stiffness (E) — not strength — controls how much a beam deflects. Understanding which property governs a given failure mode is the core engineering skill this chapter teaches.

12. Industrial Applications

Industrial Applications

- **Automotive crash structures:** high-toughness steels absorb impact energy by plastic deformation.
- **Aerospace:** high specific-strength Al and Ti alloys minimize weight while meeting yield limits.
- **Quality control:** portable hardness testers estimate σ_{TS} on the shop floor without destroying parts.
- **Metal forming:** the strain-hardening exponent n ($\sigma_T = K\varepsilon_T^n$) sets how far sheet can stretch before necking in stamping.
- **Fasteners:** bolt grades are specified directly by proof stress and σ_{TS}.

13. Summary

The tensile test converts load and elongation into an engineering stress–strain curve that encodes a material's mechanical identity. The linear elastic region gives stiffness through Hooke's law, $\sigma = E\varepsilon$, with lateral contraction set by Poisson's ratio. Yielding marks the elastic–plastic boundary, found by the 0.2% offset when gradual; the curve then rises to the ultimate tensile strength, necks, and fractures. Ductility (%EL, %RA) measures stretch before failure, while toughness is the total area absorbed and resilience the elastic part $\sigma_y^2/2E$. Because engineering values ignore the shrinking cross-section, true stress and strain correct them for accurate plastic-region work. Finally, hardness offers a quick, near-nondestructive strength estimate: for steels, $\sigma_{TS} \approx 3.45\times\text{HB}$. The single most important equation is $\sigma = E\varepsilon$.

14. Quick Review

Quick Review

1. The slope of the elastic region equals ____________.
2. True / False: necking begins at the yield point.
3. Yield strength for a gradual curve is found by the ____________% offset.
4. True / False: true stress is always greater than engineering stress in tension.
5. Toughness is the ____________ under the stress–strain curve.
6. For steels, $\sigma_{TS} \approx$ _________ $\times$ HB.

15. Practice Problems

⋆ A material with $E = 200$ GPa is stressed to $\sigma = 400$ MPa elastically. Compute the elastic strain. (p. 146)

⋆ State where necking begins on the engineering stress–strain curve and why. (p. 146)

⋆⋆ A copper rod at $\sigma = 300$ MPa, $\varepsilon = 0.05$ — find its true stress and true strain. (p. 146)

⋆⋆ A specimen ($l_0 = 50$ mm, $A_0 = 90$ mm^2) fractures at $l_f = 63$ mm, $A_f = 42$ mm^2. Compute $\%EL$ and $\%RA$. (p. 146)

⋆⋆⋆ Steel with $\sigma_y = 500$ MPa and $E = 200$ GPa: compute the modulus of resilience in MJ/m^3. (p. 146)

⋆⋆⋆ A Brinell test uses $P = 3000$ kgf, $D = 10$ mm, giving $d = 3.2$ mm. Compute HB and estimate σ_{TS} for steel. (p. 146)

16. Flash Cards

Flash Card

Front: What does Young's modulus E represent, and how is it read from the curve?

Back: Stiffness (resistance to elastic deformation); it is the slope of the linear elastic region, $E = \sigma/\varepsilon$.

Flash Card

Front: How is yield strength found when there is no sharp yield point?

Back: 0.2% offset method: draw a line parallel to the elastic slope from $\varepsilon = 0.002$; its intersection with the curve is σ_y.

Flash Card

Front: Engineering vs. true stress after UTS — which rises, which falls?

Back: Engineering stress falls (fixed A_0); true stress keeps rising because it uses the shrinking instantaneous area A_i.

Flash Card

Front: Convert engineering to true values.

Back: $\sigma_T = \sigma(1 + \varepsilon)$ and $\varepsilon_T = \ln(1 + \varepsilon)$, valid up to the onset of necking.

Flash Card

Front: Define toughness and resilience in terms of the curve.

Back: Toughness = total area under the whole curve (energy to fracture). Resilience = area under the elastic part = $\sigma_y^2/2E$.

Flash Card

Front: Give the two ductility measures and their formulas.

Back: $\%EL = \frac{l_f - l_0}{l_0} \times 100$ and $\%RA = \frac{A_0 - A_f}{A_0} \times 100$.

Flash Card

Front: Hardness–strength rule for steels?

Back: σ_{TS} (MPa) $\approx 3.45 \times$ HB — calibrated for steels only, not all metals.

Flash Card

Front: What is Poisson's ratio and its typical metal value?

Back: $\nu = -\varepsilon_{\text{trans}}/\varepsilon_{\text{axial}}$, the ratio of lateral contraction to axial extension; ≈ 0.33 for metals.

17. Cheat Sheet

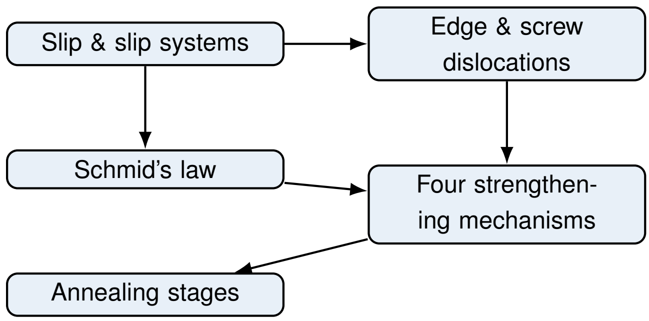
Cheat Sheet

- **Stress/strain:** $\sigma = F/A_0$, $\varepsilon = \Delta l/l_0$.
- **Elastic:** $\sigma = E\varepsilon$; shear $\tau = G\gamma$; $E = 2G(1+\nu)$; $\nu \approx 0.33$.
- **Yield:** σ_y at 0.2% offset. **UTS:** peak; necking starts here.
- **True:** $\sigma_T = \sigma(1+\varepsilon)$, $\varepsilon_T = \ln(1+\varepsilon)$; plastic $\sigma_T = K\varepsilon_T^n$.
- **Ductility:** $\%EL = \frac{\Delta l}{l_0} \times 100$, $\%RA = \frac{A_0 - A_f}{A_0} \times 100$.
- **Energy:** toughness $= \int \sigma \, d\varepsilon$; resilience $= \sigma_y^2/2E$.
- **Hardness:** $\text{HB} = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})}$; steels $\sigma_{TS} \approx 3.45\,\text{HB}$.
- **Typical** E**:** steel 207, Cu 110, Al 69 GPa.

Chapter 7

Dislocations and Strengthening Mechanisms

1. Roadmap



Plastic deformation is carried by dislocations. We first learn how slip proceeds on close-packed planes and directions, then quantify the resolved shear stress with Schmid's law. Four mechanisms (grain-size, solid-solution, strain hardening, and precipitation) all strengthen a metal by impeding dislocation motion. Finally we examine how heating a cold-worked metal reverses the damage through recovery, recrystallization, and grain growth.

2. Learning Objectives

1. **Slip systems** — Count the number of slip systems for FCC, BCC, and HCP and state their plane/direction families.
2. **Schmid's law** — Compute the resolved shear stress τ_R and the tensile yield stress σ_y for a given crystal orientation.
3. **Dislocation types** — Distinguish edge, screw, and mixed dislocations by their Burgers vector geometry.
4. **Four mechanisms** — Explain grain-size, solid-solution, strain-hardening, and precipitation strengthening and write their governing relations.
5. **Hall–Petch** — Use $\sigma_y = \sigma_0 + k_y d^{-1/2}$ to predict the grain-size effect.
6. **Annealing** — Describe recovery, recrystallization, and grain growth and the factors controlling T_{rx}.

3. Why This Matters

This chapter is the heart of mechanical behavior and carries heavy weight on undergraduate finals, the FE/PE exams, and graduate entrance tests. Every structural metal you rely on – steel in a bridge, aluminium in an aircraft skin, titanium in an implant – owes its strength to deliberately engineered dislocation obstacles. Understanding these mechanisms lets you choose a material and a heat treatment instead of guessing, and it explains the universal strength versus ductility trade-off that governs formability and failure.

4. Intuition First

Intuition First

Imagine a carpet that you want to slide across a floor. Pushing the whole carpet at once is hard. But if you bunch a small ripple into it and then walk that ripple across the carpet, the carpet moves one section at a time with very little force. A dislocation is that ripple: a line defect that lets a crystal deform piece by piece rather than all at once.

Strength, then, is simply how hard it is to move that ripple. Throw in walls (grain boundaries), obstacles (solute atoms, second-phase particles), or tangle the ripples together (more dislocations), and the ripple gets stuck – the metal gets stronger. The price you pay is that a stuck ripple also makes the metal less able to deform further, so it becomes less ductile.

5. Visual (one figure per key concept)

Slip plane and slip direction. A slip system is the combination of a slip plane and a slip direction lying in that plane. Shear translates the top of the crystal along the slip direction.

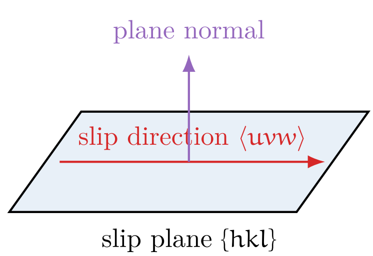


Figure 7.1. Figure 7.1 — A slip system: a close-packed plane and a close-packed direction within it.

Edge dislocation motion under shear. An edge dislocation is an extra half-plane of atoms. Shear stress pushes the half-plane sideways; as it advances one lattice spacing the crystal has slipped.

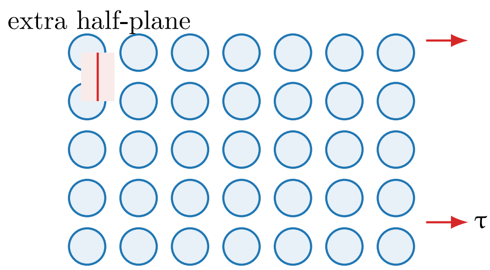


Figure 7.2. Figure 7.2 — An edge dislocation (extra half-plane, red) moves under applied shear, producing slip.

Orowan bypass of hard particles. When a particle is too strong to cut, the dislocation bows around it and leaves a dislocation loop behind. The required stress grows as particle spacing shrinks.

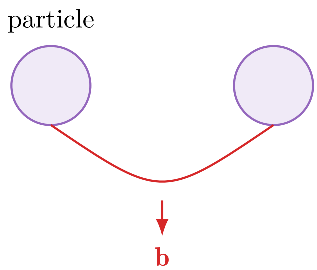


Figure 7.3. Figure 7.3 — Orowan looping: the dislocation bows between closely spaced hard precipitates.

Solid-solution distortion. A solute atom of different size distorts the surrounding lattice, creating a strain field that interacts with passing dislocations.

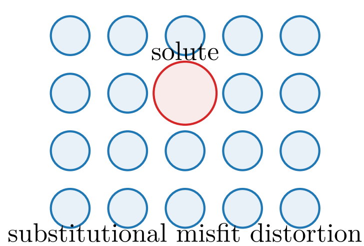


Figure 7.4. Figure 7.4 — A larger solute atom strains the lattice and pins dislocations.

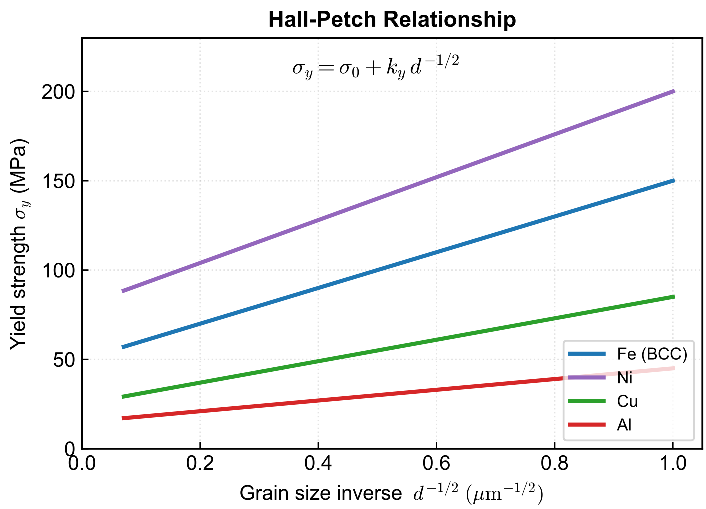


Figure 7.5. Figure 7.5 — Yield strength rises as grain size d decreases, following $\sigma_y \propto d^{-1/2}$.

Hall–Petch effect.

6. Memory Box

Memory Box

- **FCC:** 4 $\{111\}$ planes $\times$ 3 $\langle 110 \rangle$ directions $=$ **12** slip systems.
- **BCC:** $\{110\}, \{112\}, \{123\}$ with $\langle 111 \rangle$ $=$ **48** slip systems.
- **HCP:** only $\{0001\}$ basal with $\langle 11\bar{2}0 \rangle$ $=$ **3** slip systems (limited ductility).
- **Edge dislocation:** **b** perpendicular to the dislocation line.
- **Screw dislocation:** **b** parallel to the dislocation line.
- **Schmid factor:** $m = \cos\phi \cos\lambda$; max $m = 0.5$ at $\phi = \lambda = 45°$.
- **All four mechanisms** strengthen by *impeding dislocation motion.*
- **Recovery** $\sim 0.3T_m$, **recrystallization** $\sim 0.4T_m$, **grain growth** $> 0.5T_m$.

7. Compare Box

Edge vs Screw vs Mixed Dislocations

Relation of b to line	**Motion / character**
Edge: $\mathbf{b} \perp$ line (extra half-plane)	Glides only in its slip plane; climb needs diffusion.
Screw: $\mathbf{b} \parallel$ line (helical ramp)	Glides on any plane containing the line; enables cross-slip.
Mixed: angle between $\mathbf{b}$ and line	Combination; most real dislocations are mixed.

Cutting vs Orowan Bypass of Particles

Shear / cutting regime	**Orowan bypass regime**
Small, coherent particles	Large, incoherent particles
Dislocation cuts through the particle	Dislocation bows around, leaves loops
Strength rises with particle size	Strength falls as particles coarsen

8. Common Mistakes

Common Mistakes

- **Mistake 1:** Measuring ϕ and λ from each other. **Why:** Both angles are measured from the *tensile axis*; they lie in different planes.
- **Mistake 2:** Assuming HCP is as ductile as FCC. **Why:** HCP has only 3 slip systems, so it deforms on few planes and is brittle at room temperature.
- **Mistake 3:** Believing smaller grains always help. **Why:** Below ~ 1 µm grain-boundary sliding dominates and Hall–Petch can invert.
- **Mistake 4:** Treating recrystallization temperature as a fixed constant. **Why:** It depends on cold-work amount, purity, and time ($T_{rx} \approx 0.4T_m$ is only a rule of thumb).
- **Mistake 5:** Forgetting cold work trades strength for ductility. **Why:** Strain hardening raises σ_y but lowers the uniform elongation.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$\tau_R = \sigma \cos\phi \cos\lambda$	Resolved shear stress	MPa	σ, ϕ, λ	Single slip system, uniaxial tension	One active system only	Drives slip in a crystal
$\sigma_y = \tau_{CRSS}/m$	Tensile yield of crystal	MPa	$m = \cos\phi \cos\lambda$	Schmid factor known	No work hardening	Orientation sets yield
$\sigma_y = \sigma_0 + k_y d^{-1/2}$	Hall–Petch relation	MPa	d grain size	Polycrystal, d uniform	Breaks for $d < 1$ µm	Finer grains = stronger
$\sigma_T = K\varepsilon_T^n$	Hollomon (strain hardening)	MPa	n exponent, K coeff.	Isotropic, monotonic tension	Pre-necking only	Higher n = more formable
$\tau = Gb/L$	Orowan bypass stress	MPa	L particle spacing, b Burgers	Hard, non-shearable particles	Not for cutting regime	Closer particles = stronger

10. Worked Examples

Basic

Example

Problem: A single crystal has $\phi = 50°$ and $\lambda = 40°$, with $\tau_{CRSS} = 1.0$ MPa. Find the Schmid factor and the tensile yield stress.
Thinking: Schmid's law separates the orientation (through m) from the material resistance (τ_{CRSS}). Compute m first, then invert.
Solution:

$$m = \cos 50° \cos 40° = 0.643 \times 0.766 = 0.493,$$
$$\sigma_y = \frac{\tau_{CRSS}}{m} = \frac{1.0}{0.493} = 2.03 \text{ MPa}.$$

Intermediate

Example

Problem: Steel: $\sigma_0 = 100$ MPa, $k_y = 0.74$ MPa $\cdot$ m$^{1/2}$. Compare yield strength at $d = 50$ µm and $d = 5$ µm.
Thinking: Convert d to metres, then apply Hall–Petch. Smaller d means larger $d^{-1/2}$, hence higher strength.
Solution:

$$d = 50 \text{ µm} : \sigma_y = 100 + 0.74/(50 \times 10^{-6})^{1/2} = 100 + 105 = 205 \text{ MPa},$$
$$d = 5 \text{ µm} : \sigma_y = 100 + 0.74/(5 \times 10^{-6})^{1/2} = 100 + 331 = 431 \text{ MPa}.$$

Halving the grain size does not double strength; the square-root law gives a diminishing return.

Advanced

Example

Problem: A cold-worked metal follows $\sigma_T = K\varepsilon_T^n$ with $n = 0.4$ and $K = 800$ MPa. Compare true stress at $\varepsilon_T = 0.1$ and $\varepsilon_T = 0.3$. Which is more formable, this or a metal with $n = 0.15$?
Thinking: Hollomon is a power law: higher n raises stress more slowly with strain, so the material can be stretched further before necking.
Solution:

$$\varepsilon_T = 0.1 : \sigma_T = 800(0.1)^{0.4} = 800 \times 0.398 = 318 \text{ MPa},$$
$$\varepsilon_T = 0.3 : \sigma_T = 800(0.3)^{0.4} = 800 \times 0.618 = 494 \text{ MPa}.$$

The $n = 0.4$ metal (FCC-like) is far more formable than the $n = 0.15$ (BCC-like) one; deep drawing favours high n.

11. Engineering Insight

Engineering Insight

Hall–Petch explains why nanocrystalline alloys can reach strengths above 1 GPa, yet also why they can fail by grain-boundary sliding rather than dislocation glide. Engineers exploit the square-root law with diminishing returns: cutting grain size from 50 to 5 µm nearly doubles yield strength, but another decade of refinement costs far more processing for little gain and risks brittleness. That is why real alloys *stack* mechanisms – fine grains plus precipitation plus solid solution – rather than pushing any single one to its limit.

12. Industrial Applications

Industrial Applications

- **Al–Cu (2xxx series) aerospace alloy:** solution-treat at 550°C, quench, age at 160–190°C so coherent θ'' and θ' precipitates give peak strength; overaging to incoherent θ ($CuAl_2$) weakens it.
- **Steel:** combine grain refinement (normalizing), C/Mn solid solution, carbide precipitation, and cold rolling to reach ~ 800 MPa total strengthening.
- **Brass:** Zn substitutionally dissolved in Cu for solid-solution strengthening of cartridge cases.
- **Sheet forming:** high-n FCC metals (Cu, Al, austenitic stainless) are chosen for deep drawing because they strain-harden without necking.
- **Annealing of cold-rolled sheet:** recrystallization relieves hardness and restores ductility before further forming.

13. Summary

Plastic deformation proceeds by dislocation glide on slip systems: FCC (12), BCC (48), and HCP (3). Schmid's law, $\tau_R = \sigma \cos\phi \cos\lambda$, predicts when a given orientation yields. All four strengthening mechanisms – grain-size reduction (Hall–Petch, $\sigma_y = \sigma_0 + k_y d^{-1/2}$), solid-solution strengthening (strain-field pinning), strain hardening ($\sigma_T = K\varepsilon_T^n$), and precipitation hardening (shearing or Orowan bypass) – raise strength by blocking dislocation motion, at the cost of ductility. Heating a cold-worked metal reverses the damage through recovery ($\sim 0.3T_m$), recrystallization ($\sim 0.4T_m$), and grain growth ($> 0.5T_m$). The single most important equation is the Hall–Petch relation: **finer grains are stronger**.

14. Quick Review

Quick Review

1. FCC crystals have ________ slip systems; HCP have ________.
2. True / False: The maximum Schmid factor is 0.5 when $\phi = \lambda = 45°$.
3. Fill in: Hall–Petch is $\sigma_y =$ ________ $+ k_y d^{-1/2}$.
4. True / False: Recrystallization fully restores the original strength and ductility.
5. Order the annealing stages from lowest to highest temperature: ____________.
6. True / False: Interstitial carbon strengthens Fe about 100× more per atom than substitutional Mn.

15. Practice Problems

⋆ According to Schmid's law, the resolved shear stress $\tau_R = \sigma \cos\phi \cos\lambda$ is maximized for a fixed σ when:

(a) $\phi = 90°,\ \lambda = 0°$
(b) $\phi = 45°,\ \lambda = 45°$
(c) $\phi = 0°,\ \lambda = 90°$
(d) $\phi = 30°,\ \lambda = 60°$

(p. 147)

⋆ How many slip systems does an FCC metal possess?

(a) 3
(b) 12
(c) 48
(d) 4

(p. 147)

⋆⋆ Using Hall–Petch with $\sigma_0 = 100$ MPa and $k_y = 0.74$ MPa $\cdot$ m$^{1/2}$, decreasing grain size from 10 µm to 2.5 µm increases yield strength by approximately what factor?

(a) $\sqrt{2}$

(b) 2
(c) 4
(d) 10

(p. 147)

★★ During annealing of a cold-worked metal, which stage causes complete restoration of original strength and ductility?

(a) Recovery
(b) Recrystallization
(c) Grain growth
(d) All three equally

(p. 147)

★★★ Precipitation hardening in Al–Cu is most effective when particles are:

(a) Very large and widely spaced
(b) Small, coherent, and closely spaced
(c) Fully dissolved in the matrix
(d) Located only at grain boundaries

(p. 147)

★★★ Which material is expected to have the highest strain-hardening exponent n?

(a) Fully annealed copper (FCC)
(b) Tungsten (BCC)
(c) Magnesium (HCP)
(d) Glass

(p. 147)

16. Flash Cards

Flash Card

Front: What is a slip system?

Back: A slip plane combined with a slip direction within that plane; the combination on which dislocation glide occurs.

Flash Card

Front: State Schmid's law and define m.

Back: $\tau_R = \sigma \cos\phi \cos\lambda = \sigma m$, where m (the Schmid factor) is $\cos\phi\cos\lambda$, the orientation factor for a given slip system.

Flash Card

Front: What does Hall–Petch say?

Back: $\sigma_y = \sigma_0 + k_y d^{-1/2}$: yield strength increases as grain size d decreases.

Flash Card

Front: Edge vs screw dislocation?

Back: Edge: **b** is perpendicular to the line (extra half-plane). Screw: **b** is parallel to the line (helical ramp).

Flash Card

Front: What is Orowan bypass?

Back: When precipitates are too hard to cut, the dislocation bows around them, leaving loops; the stress $\tau = Gb/L$ rises as particle spacing L shrinks.

Flash Card

Front: Recovery vs recrystallization vs grain growth?

Back: Recovery ($\sim 0.3T_m$): dislocation rearrangement. Recrystallization ($\sim 0.4T_m$): new strain-free grains, full softening. Grain growth ($> 0.5T_m$): grains coarsen, slight weakening.

17. Cheat Sheet

Cheat Sheet

Slip systems: FCC $12 = \{111\}\langle 110\rangle$; BCC $48 = \{110/112/123\}\langle 111\rangle$; HCP $3 = \{0001\}\langle 11\bar{2}0\rangle$.

Schmid: $\tau_R = \sigma\cos\phi\cos\lambda = \sigma m$; $\sigma_y = \tau_{CRSS}/m$; max $m = 0.5$ at $45°/45°$.

Hall–Petch: $\sigma_y = \sigma_0 + k_y d^{-1/2}$. Finer grains → stronger (saturates $d < 1$ µm).

Solid solution: solute strain field pins dislocations; $\Delta\sigma \propto G\varepsilon^{3/2}\sqrt{c}$; interstitial C ~ $100\times$ Mn per atom.

Strain hardening: $\sigma_T = K\varepsilon_T^n$; FCC $n = 0.3$–0.5, BCC 0.15–0.25, HCP < 0.1.

Precipitation: cutting (coherent, small) vs Orowan (incoherent, large); Al–Cu peak at θ''/θ', overage to θ.

Annealing: recovery ~ $0.3T_m$ → recrystallization ~ $0.4T_m$ → grain growth $> 0.5T_m$.

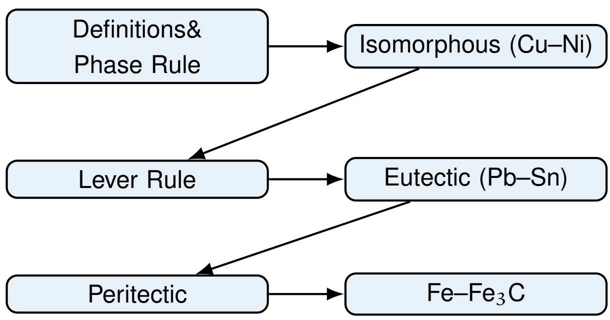
Big idea: every mechanism blocks dislocation motion → stronger but less ductile.

Chapter 8

Phase Diagrams

1. Roadmap

A phase diagram is the map that tells an engineer what phases exist, what they are made of, and how much of each is present at any temperature and overall composition. This chapter builds from the simplest possible case upward to the iron-carbon system that underpins all steel.



2. Learning Objectives

1. **Define** the terms phase, component, solubility limit, and microstructure, and state the Gibbs phase rule $F = C - P + 2$ (or $C - P + 1$ at fixed pressure).

2. **Read** a binary isomorphous diagram (Cu–Ni): locate liquidus, solidus, and the two-phase field, and name the phases present.

3. **Apply** the lever rule to compute the mass fractions of each phase from a tie line.

4. **Recognize** the eutectic reaction ($L \to \alpha + \beta$) in Pb–Sn and the peritectic reaction ($L + \delta \to \gamma$).

5. **Navigate** the Fe–Fe_3C diagram: identify the peritectic, eutectic, and eutectoid invariant points and the resulting microconstituents (ferrite, austenite, cementite, pearlite, ledeburite).

6. **Compute** phase fractions and microconstituent fractions in hypoeutectoid and hypereutectoid steels.

3. Why This Matters

Phase diagrams are the single most frequently tested topic on undergraduate materials exams and on FE/PE metallurgy sections, and they are the foundation for heat-treatment and alloy-design problems. In real engineering, every casting, weld, braze, and heat treatment follows a path across one or more phase diagrams. A steel that is quenched, a solder joint that is melted, and a turbine blade that is directionally solidified are all governed by the phase boundaries you learn here. If you can read a diagram and run the lever rule, you can predict microstructure, and microstructure predicts properties. Mastery here repays itself in nearly every later chapter on mechanical behavior, phase transformations, and processing.

4. Intuition First

Intuition First

Imagine a crowded dance floor (the liquid) where everyone is free to move. As the room cools, dancers pair off and lock into a tidy grid (the solid). In an *isomorphous* system the two atoms get along so well that the grid can hold any mixture—copper and nickel dissolve in each other at every ratio. In a *eutectic* system they do not get along: at one special cool temperature the liquid breaks up into two separate, neatly layered solids at once, like oil and water freezing side by side.

The *lever rule* is just a seesaw. Picture a balance beam on a fulcrum placed at the overall composition C_0. The ends of the beam are the two phase compositions C_α and C_β. The heavier the load you want at the right end, the closer the fulcrum must sit to that end. The fraction of each phase is simply how much beam sits on the opposite side of the fulcrum.

5. Visual (one figure per key concept)

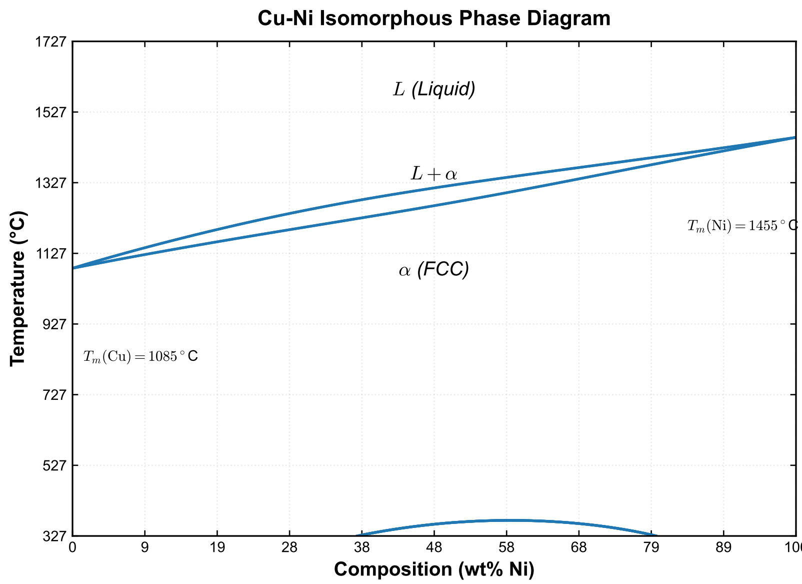


Figure 8.1. Figure 8.1 — Cu–Ni isomorphous diagram. Above the liquidus only liquid L exists; below the solidus only solid α (FCC) exists; between them L + α. Melting points: Cu 1085°C, Ni 1455°C; x-axis is wt% Ni.

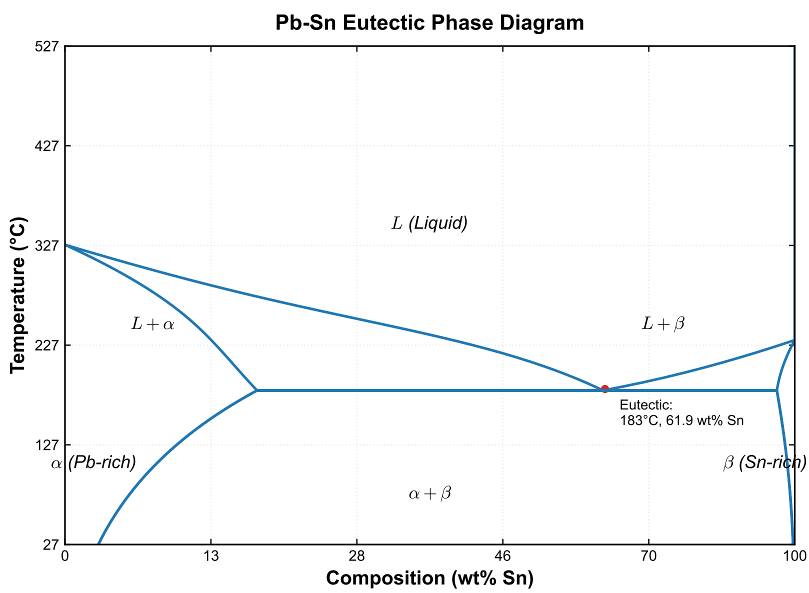


Figure 8.2. Figure 8.2 — Pb–Sn eutectic diagram. The eutectic point at 61.9 wt% Sn and 183°C is where L → α + β occurs isothermally. Solvus lines bound the limited solid solubility of each phase.

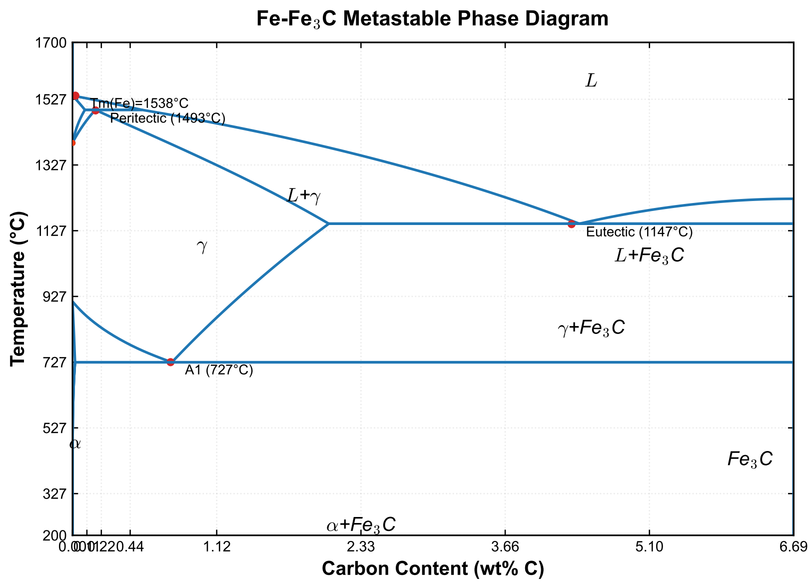


Figure 8.3. Figure 8.3 — Fe–Fe_3C metastable diagram with three invariant reactions: peritectic (P, 1493°C), eutectic (C, 1147°C), and eutectoid (S, 727°C). Cementite Fe_3C contains 6.67 wt% C.

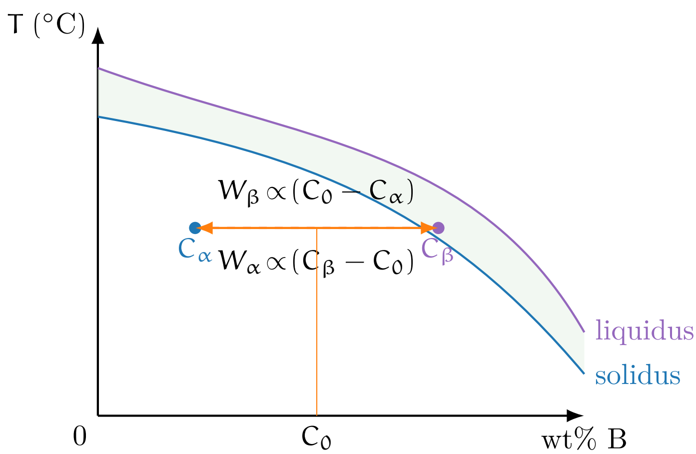


Figure 8.4. Figure 8.4 — Lever rule construction. At fixed T the tie line (dashed) joins the equilibrium phase compositions C_α and C_β; the overall composition C_0 divides the lever into arms whose lengths are inversely proportional to the opposite phase mass fraction.

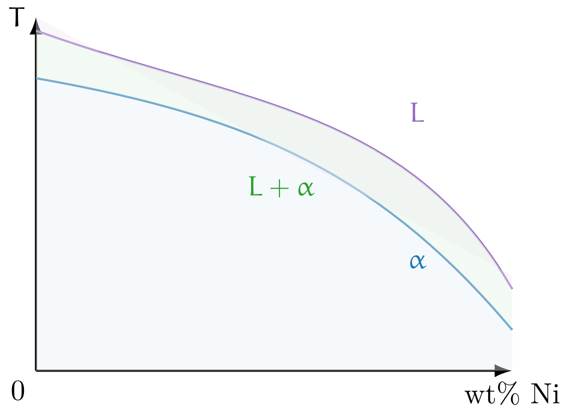


Figure 8.5. Figure 8.5 — Isomorphous regions. Above the liquidus: single-phase liquid L. Below the solidus: single-phase solid α. Between them: two-phase $\mathsf{L} + \alpha$ where a tie line gives the equilibrium compositions.

6. Memory Box

Memory Box

- **Phase rule:** $F = C - P + 2$; at fixed P, $F = C - P + 1$.
- **Isomorphous:** complete solid solubility (Cu–Ni). Liquidus = all liquid above; solidus = all solid below.
- **Lever rule:** read C_α and C_β from the *ends* of the tie line.
- **Eutectic:** $L \to \alpha + \beta$ at fixed T (Pb–Sn: 61.9 wt% Sn, 183°C).
- **Peritectic:** $L + \alpha \to \beta$ (Fe–Fe_3C: 1493°C).
- **Eutectoid:** $\gamma \to \alpha + Fe_3C$ at 727°C gives *pearlite*.
- **Fe_3C (cementite):** 6.67 wt% C; metastable but the practical diagram.
- **Hypoeutectoid** (< 0.76% C): proeutectoid ferrite + pearlite. **Hypereutectoid** (> 0.76% C): proeutectoid cementite + pearlite.

7. Compare Box

Invariant Reactions

Eutectic	**Peritectic**
$L \rightarrow \alpha + \beta$ Occurs from a *liquid* only Example: Pb–Sn at 61.9 wt% Sn Microstructure: lamellar $\alpha + \beta$	$L + \alpha \rightarrow \beta$ Consumes a *liquid* + *solid* to make a new solid Example: Fe–Fe_3C at 1493°C ($L + \delta \rightarrow \gamma$) New phase envelopes the existing phase
Eutectoid	**Solidus vs Solvus**
$\gamma \rightarrow \alpha + Fe_3C$ (solid state) Example: steel at 727°C, 0.76 wt% C Product: pearlite (no liquid involved)	**Solidus**: boundary of liquid/solid field **Solvus**: limit of solid solubility in a solid phase Found inside single-solid regions (e.g. α in Pb–Sn)

8. Common Mistakes

Common Mistakes

- **Mistake 1: Reading compositions from the overall C_0, not the tie-line ends. Why:** The lever rule uses the phase-boundary compositions C_α and C_β at the tie-line intersections, never the alloy's overall composition.
- **Mistake 2: Forgetting the phase rule's "2" becomes "1" at fixed pressure. Why:** Materials problems almost always fix P, so $F = C - P + 1$; using $+2$ gives the wrong degrees of freedom.
- **Mistake 3: Confusing eutectic, eutectoid, and peritectic. Why:** Eutectic starts from liquid; eutectoid occurs entirely in the solid; peritectic consumes a solid and liquid together.
- **Mistake 4: Treating Fe_3C as the true equilibrium phase. Why:** Graphite is the stable phase; the Fe–Fe_3C diagram is metastable but used because cementite forms faster.
- **Mistake 5: Using Celsius in thermodynamic formulas that need Kelvin. Why:** Arrhenius-type relations require absolute temperature; always convert before computing.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$F = C - P + 2$	Degrees of freedom	—	C components, P phases	Equilibrium; T, P variable	Fixed P needs +1 form	How many variables you may set
$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$	Mass fraction of α	dimensionless	Tie-line ends C_α, C_β; overall C_0	Two-phase equilibrium	Only in a 2-phase field	Fraction of one solid phase
$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha}$	Mass fraction of β	dimensionless	same as above	Two-phase equilibrium	Only in a 2-phase field	Fraction of second phase
$W_\alpha + W_\beta = 1$	Closure check	dimensionless	phase fractions	mass conservation	—	Always verify the sum
$L \to \alpha + \beta$	Eutectic reaction	—	phases at eutectic T	Invariant, $F = 0$ at fixed P	Only at the eutectic point	Lamellar two-solid product
$L + \alpha \to \beta$	Peritectic reaction	—	phases at peritectic T	Invariant, $F = 0$ at fixed P	Only at the peritectic point	New solid envelopes old
$W_p = \frac{0.76 - C_0}{0.76 - 0.022}$	Pearlite fraction (hypoeutectoid)	dimensionless	C_0 overall wt% C; 0.022 = max ferrite solubility	Slow cool, lever rule on α/Fe_3C	Below eutectoid only	Microconstituent amount

10. Worked Examples

Basic

Example

Problem: State the Gibbs phase rule value F for a binary (two-component) alloy in a two-phase region at fixed pressure.

Thinking: Use $F = C - P + 1$ with $C = 2$ and $P = 2$. One variable (temperature) can still be chosen; the phase compositions are then fixed by the boundaries.

Solution:

$$F = C - P + 1 = 2 - 2 + 1 = 1.$$

So $F = 1$: you may vary T, but then the phase compositions are locked.

Intermediate

Example

Problem: For a Cu–Ni alloy with overall composition $C_0 = 35$ wt% Ni at $T = 1250°C$, the tie line gives $C_L = 32$ wt% Ni and $C_S = 43$ wt% Ni. Find the mass fractions of liquid and solid α.
Thinking: Apply the lever rule using the tie-line ends. The solid arm runs from C_0 to C_L; the liquid arm runs from C_S to C_0.
Solution:

$$W_L = \frac{C_0 - C_S}{C_L - C_S} = \frac{35 - 43}{32 - 43} = \frac{-8}{-11} = 0.727,$$
$$W_S = \frac{C_L - C_0}{C_L - C_S} = \frac{32 - 35}{32 - 43} = \frac{-3}{-11} = 0.273.$$

Check: $W_L + W_S = 1.000$. Liquid is 72.7%, solid 27.3%.

Advanced

Example

Problem: A hypoeutectoid steel has $C_0 = 0.40$ wt% C. Estimate the mass fractions of proeutectoid ferrite (α) and pearlite after slow cooling below the eutectoid. (Use 0.022 wt% C as the maximum solubility of C in ferrite at 727°C, and 0.76 wt% C for pearlite.)
Thinking: Treat pearlite as a mixture at the eutectoid composition (0.76% C) and ferrite as the phase at 0.022% C; apply the lever rule across that two-phase region.
Solution:

$$W_p = \frac{C_0 - 0.022}{0.76 - 0.022} = \frac{0.40 - 0.022}{0.738} = \frac{0.378}{0.738} = 0.512,$$
$$W_\alpha = 1 - W_p = 0.488.$$

Thus 51.2% pearlite and 48.8% proeutectoid ferrite.

11. Engineering Insight

Engineering Insight

The lever rule is the engineer's tool for turning a diagram into a number that controls properties. In casting, the fraction of solid in the mushy $L + \alpha$ zone decides whether a part feeds cleanly or shrinks and cracks; the same two-phase region in welding sets the size of the heat-affected zone. In steel, the pearlite fraction set by the lever rule fixes hardness and tensile strength—more pearlite means a stronger but less ductile part. Heat-treatment recipes (quench, anneal, normalize) are simply chosen paths that stop the cooling trajectory at a desired microconstituent fraction. Reading the diagram correctly is therefore the difference between a part that meets spec and one that fails in service.

12. Industrial Applications

Industrial Applications

- **Soldering and brazing (Pb–Sn, SAC):** The eutectic composition melts at the lowest temperature, letting joints form without fully melting the base metal.
- **Steelmaking and heat treatment:** The Fe–Fe_3C diagram guides annealing, normalizing, and quench paths to reach target pearlite/ferrite/cementite fractions.
- **Alloy design (Cu–Ni):** Complete solubility gives monotonic property tuning for condenser tubes and coinage.
- **Single-crystal turbine blades:** Directional solidification follows the liquidus/solidus path to avoid grain boundaries.
- **Cast iron production:** The eutectic/ledeburite region controls fluidity and the final graphite or cementite microstructure.

13. Summary

Phase diagrams summarize equilibrium between phases as a function of temperature and overall composition, and the Gibbs phase rule $F = C - P + 1$ (at fixed pressure) tells you how many variables remain free. Isomorphous systems such as Cu–Ni show complete liquid and solid solubility bounded by a liquidus and solidus. Inside any two-phase field the lever rule converts a horizontal tie line into exact mass fractions: $W_\alpha = (C_\beta - C_0)/(C_\beta - C_\alpha)$ and $W_\beta = (C_0 - C_\alpha)/(C_\beta - C_\alpha)$. Eutectic systems (Pb–Sn) solidify isothermally into lamellar mixtures, while peritectic and eutectoid reactions ($L + \delta \to \gamma$ and $\gamma \to \alpha + Fe_3C$) shape the $Fe–Fe_3C$ diagram. The single most important equation of the chapter is the lever rule, because everything else—microconstituent fractions, heat-treatment selection, and property prediction—follows from it.

14. Quick Review

Quick Review

1. The Gibbs phase rule at fixed pressure is written $F =$ __________.
2. For a binary two-phase region, the number of degrees of freedom is ________.
3. True / False: The eutectic reaction occurs entirely in the solid state. __________
4. The phases that coexist at the eutectic point of a binary system number ________.
5. The eutectoid reaction in steel is $\gamma \to$ ______________ and occurs at __________ °C.
6. True / False: In the lever rule, C_α and C_β are read from the ends of the tie line. __________
7. Hypoeutectoid steels contain proeutectoid __________ plus pearlite.

15. Practice Problems

- ⋆ State the Gibbs phase rule for a binary system at fixed pressure and identify what each symbol means. (p. 147)
- ⋆⋆ For a Pb–Sn alloy with overall composition $C_0 = 40$ wt% Sn held at 200°C, a tie line gives $C_L = 46$ wt% Sn and $C_\alpha = 18$ wt% Sn. Compute the mass fraction of liquid using the lever rule. (p. 147)
- ⋆⋆ A Cu–Ni alloy contains $C_0 = 50$ wt% Ni at 1300°C, where the tie line reads $C_L = 44$ wt% Ni and $C_S = 56$ wt% Ni. Find the mass fractions of liquid and solid α, and verify they sum to 1. (p. 147)

⋆⋆⋆ A steel of composition $C_0 = 0.60$ wt% C is slowly cooled below the eutectoid temperature. Using 0.022 wt% C for maximum ferrite solubility and 0.76 wt% C for pearlite, calculate the mass fractions of proeutectoid ferrite and pearlite. (p. 147)

⋆⋆⋆ At the eutectic point of a binary system at fixed pressure, how many phases coexist and what is the value of F? Explain why this is called an invariant reaction. (p. 147)

16. Flash Cards

Flash Card

Front: What does the Gibbs phase rule $F = C - P + 1$ tell you?

Back: F is the number of independent intensive variables (e.g. temperature or composition) you may change while keeping the number of phases fixed, at constant pressure.

Flash Card

Front: Define the liquidus and solidus lines.

Back: Liquidus: the boundary above which only liquid exists. Solidus: the boundary below which only solid exists. Between them the alloy is a two-phase liquid + solid mixture.

Flash Card

Front: Write the lever rule for W_α and name its parts.

Back: $W_\alpha = (C_\beta - C_0)/(C_\beta - C_\alpha)$, where C_α and C_β are the tie-line-end phase compositions and C_0 is the overall composition.

Flash Card

Front: What is a eutectic reaction? Give the Pb–Sn example.

Back: A eutectic is $L \to \alpha + \beta$ at a fixed temperature. In Pb–Sn it occurs at 61.9 wt% Sn and 183°C, producing a lamellar $(\alpha + \beta)$ microstructure.

Flash Card

Front: What microconstituents form in a hypoeutectoid steel?

Back: Proeutectoid ferrite (α) plus pearlite. Hypereutectoid steels instead form proeutectoid cementite plus pearlite.

Flash Card

Front: What is the peritectic reaction in Fe–Fe_3C?

Back: At 1493°C: $L + \delta \to \gamma$ (austenite). A new solid phase forms by reaction of the liquid with an existing solid phase.

17. Cheat Sheet

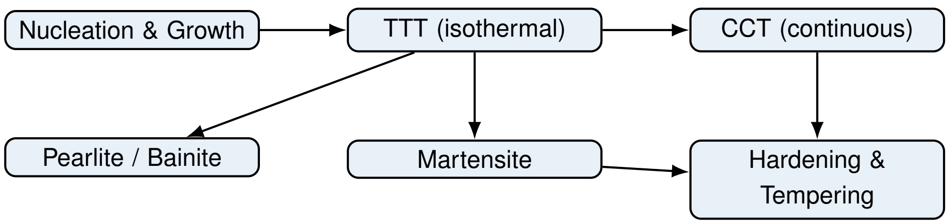
Cheat Sheet

- **Phase rule:** $F = C - P + 2$ (or $+1$ at fixed P).
- **Lever rule (2-phase):** $W_\alpha = (C_\beta - C_0)/(C_\beta - C_\alpha)$, $W_\beta = (C_0 - C_\alpha)/(C_\beta - C_\alpha)$. Read C_α, C_β from tie-line *ends*.
- **Isomorphous (Cu–Ni):** liquidus/solidus bound L and α; full solubility.
- **Eutectic (Pb–Sn):** $L \to \alpha + \beta$ at 61.9 wt% Sn, 183°C.
- **Peritectic (Fe–Fe_3C):** $L + \delta \to \gamma$ at 1493°C.
- **Eutectoid (Fe–Fe_3C):** $\gamma \to \alpha + Fe_3C$ at 727°C → pearlite. Cementite = 6.67 wt% C.
- **Steel microconstituents:** hypo (< 0.76% C) = ferrite + pearlite; hyper (> 0.76% C) = cementite + pearlite.
- **Key numbers:** eutectic Pb–Sn 61.9%/183°C; eutectoid 0.76%/727°C; peritectic 0.17%/1493°C.

Chapter 9

Phase Transformations

1. Roadmap



2. Learning Objectives

1. Distinguish the nucleation and growth stages of a phase transformation.
2. Read a TTT diagram: identify incubation time, the nose, and the transformation products.
3. Separate diffusion-controlled products (pearlite, bainite) from the diffusionless martensitic transformation.
4. Compute the M_s temperature from carbon content and explain retained austenite.
5. Interpret CCT diagrams, critical cooling rate, and the meaning of hardenability.
6. Select a heat-treatment path (austenitize, quench, temper, anneal) for a given property target.

3. Why This Matters

Phase transformations decide whether a steel component is soft and ductile or hard and brittle. Roughly a third of typical exam questions on kinetics use TTT/CCT diagrams, and the same logic governs every quenched or tempered part in a car, tool, or aircraft. Getting the cooling path wrong means a cracked gear or a spring that snaps; getting it right means tailoring strength and toughness on demand. Understanding these curves is therefore both an exam skill and a daily engineering necessity.

4. Intuition First

Intuition First

Imagine cold honey poured into a freezer. If you hold it at one fixed cold temperature, it slowly crystallizes at a steady rate — that is an *isothermal* transformation. But if you instead just leave it on the counter to cool gradually, it passes through a range of temperatures while it sets — that is *continuous cooling*, and the result can be very different. Steel behaves the same way. Quench it fast enough and the atoms freeze in place before they can shuffle into an equilibrium pattern: you get martensite, a strained, super-hard trap. Cool it slowly and the carbon and iron have time to rearrange into soft layered pearlite. The transformation diagram simply maps "how cold, how long" to "what you get".

5. Visual (one figure per key concept)

Concept A — The TTT diagram (isothermal). The complete C-curve for eutectoid steel shows the pearlite region above the nose and the bainite region below it, with the M_s/M_f lines marking where martensite begins and ends.

Concept B — The CCT diagram (continuous cooling). Under continuous cooling the curves shift to longer times and lower temperatures, and bainite can be suppressed. Different cooling rates give pearlite, a mixed structure, or full martensite.

Concept C — A cooling path drawn on the TTT diagram. The solid curve cools fast enough to dodge the nose and land in the martensite zone; the dashed curve lingers in the nose and forms pearlite. This picture is the heart of hardenability.

Concept D — Microconstituent comparison. Pearlite is layered plates, bainite is fine needles, and martensite is a strained lath/plate array. Their morphology explains their strength.

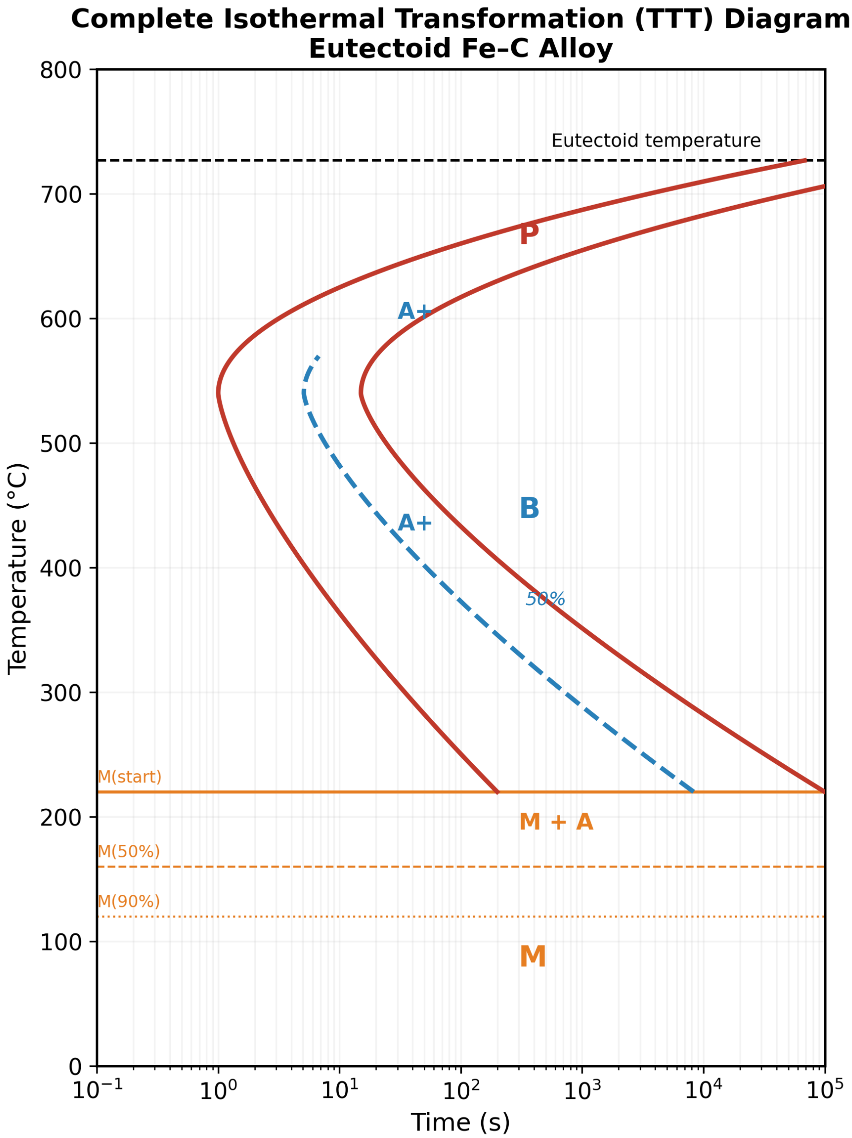


Figure 9.1. Figure 9.1 — Complete TTT diagram for a eutectoid steel: pearlite and bainite C-curves, M_s and M_f lines, and the transformation products at each isothermal hold temperature.

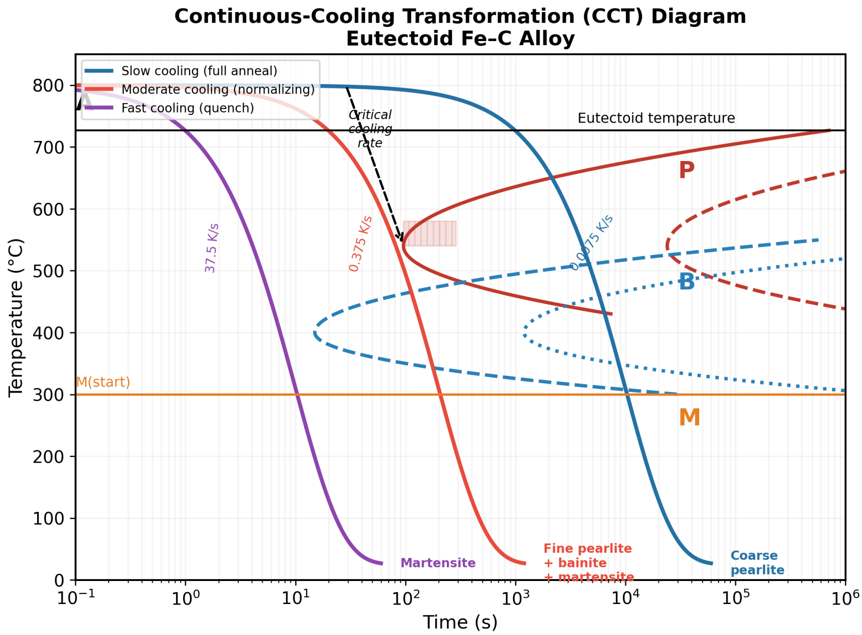


Figure 9.2. Figure 9.2 — CCT diagram with superimposed cooling curves. Slower cooling yields pearlite; faster cooling avoids the nose and forms martensite.

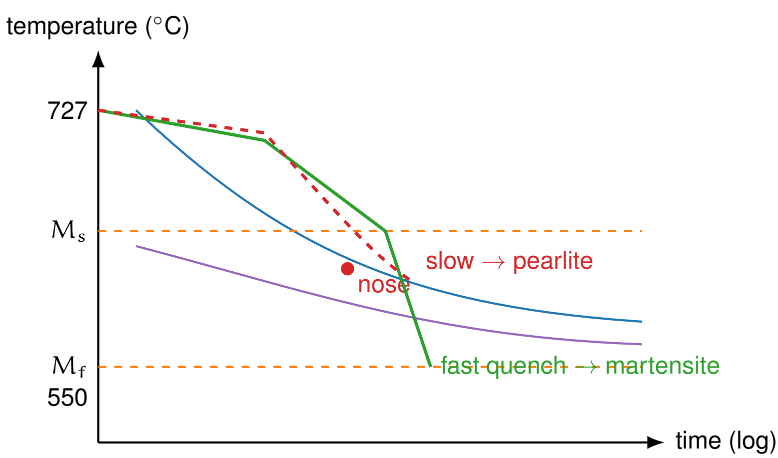


Figure 9.3. Figure 9.3 — Two cooling paths on the TTT diagram. The fast path bypasses the nose and finishes as martensite; the slow path crosses the pearlite region.

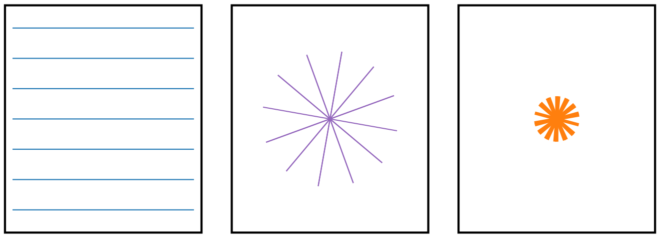


Figure 9.4. Figure 9.4 — Schematic morphologies: pearlite (layered ferrite/cementite), bainite (fine acicular), and martensite (strained laths).

6. Memory Box

Memory Box

- **TTT** = Time-Temperature-Transformation; *isothermal* (hold at fixed T).
- **CCT** = Continuous-Cooling Transformation; curves shift right and down vs. TTT.
- **Nose** = shortest time before transformation starts (compromise of driving force vs. diffusivity).
- **Pearlite**: above the nose, ~ 550–727°C, layered, medium hardness.
- **Bainite**: below the nose, ~ 250–550°C, acicular, high hardness.
- **Martensite**: below M_s, diffusionless shear, very hard and brittle.
- **Tempering**: reheat martensite (150–600°C) → tempered martensite, tougher.
- **Hardenability** = depth hardened on quench; raised by Mn, Cr, Mo, Ni.

7. Compare Box

Pearlite vs. Bainite vs. Martensite

Pearlite	**Bainite**	**Martensite**
Diffusion-controlled, layered ferrite + cementite	Diffusion-controlled, fine acicular	Diffusionless shear, supersaturated C
Forms above TTT nose (550–727°C)	Forms below nose (250–550°C)	Forms below M_s only
Medium hardness; coarse $\rightarrow$ fine toward nose	High hardness	Very hard, brittle

Isothermal (TTT) vs. Continuous Cooling (CCT)

TTT diagram	**CCT diagram**
Constant-temperature hold	Cooling through a temperature range
C-curves at shortest times	Curves shifted to longer times, lower T
Bainite always reachable	At high cooling rates bainite can be suppressed
Used to plan isothermal heat treatments	Used to plan real quenches

8. Common Mistakes

Common Mistakes

- **Mistake 1:** Reading a TTT diagram as if it were continuous cooling. **Why:** TTT curves are isothermal; real quenches follow CCT, which is shifted right and down. Use the correct diagram for the process.
- **Mistake 2:** Drawing martensite as a point on the equilibrium Fe–C phase diagram. **Why:** Martensite is metastable and diffusionless; it never appears on the equilibrium diagram. It forms only on rapid quench below M_s.
- **Mistake 3:** Treating the M_s formula as exact. **Why:** $M_s \approx 520 - 320(\text{wt\% C})$ is empirical and approximate; alloying elements beyond carbon shift it further. Treat it as a first estimate, not a precise value.
- **Mistake 4:** Confusing hardenability with hardness. **Why:** Hardenability is the *depth* that hardens on quench (a Jominy distance), not the surface hardness itself, which depends on carbon content.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$y = 1 - \exp(-kt^n)$	Fraction transformed vs. time (Avrami)	–	y fraction, k rate, n exponent (1–4)	Isothermal, constant T	Only valid at fixed T	Predicts transformation completion
$M_s \approx 520 - 320(\text{wt\% C})$	Martensite start temperature	°C	wt% C carbon mass fraction	Plain carbon steel	Ignores other alloys	Lower C → lower M_s → more retained austenite
$\dot{T}_{crit}$ (concept)	Minimum cooling rate that avoids the nose	°C/s	–	Defined by TTT nose	Not a single universal number	Sets quench severity needed
$H = H(T_{temp})$ decreasing	Tempering: hardness falls as tempering T rises	HRC	T_{temp} temper temperature	Martensitic start	Embrittlement ranges exist	Trade hardness for toughness

10. Worked Examples

Basic

Example

Problem: Find the approximate M_s for a 0.4 wt% C plain-carbon steel.
Thinking: The empirical relation uses only the carbon content; we substitute directly. Higher carbon lowers M_s, so a smaller number than 520 is expected.
Solution:

$$M_s = 520 - 320(0.4) = 520 - 128 = \mathbf{392}^{\circ}\text{C}.$$

At room temperature this steel is fully below M_s, so on a fast quench it transforms essentially completely to martensite.

Intermediate

Example

Problem: An isothermal heat treatment quenches a 1080 (eutectoid) steel to 650°C, holds 10 s, then water-quenches to room temperature. What microstructure results?
Thinking: First read the TTT nose location. At 650°C the pearlite region sits just above the nose; 10 s is long enough to start but not complete pearlite, then the quench below M_s makes martensite from the remaining austenite.
Solution: A **mixed microstructure of coarse pearlite plus martensite**. The partial pearlite softens the part; the martensite portion keeps it hard. Selection of hold time tunes the hardness-toughness balance.

Advanced

Example

Problem: A 0.4 wt% C, 50 mm diameter round bar is water-quenched, requiring ~ 55 HRC at the surface. Will it harden through?

Thinking: Connect the Fe–C diagram (hypoeutectoid, austenitize at 850°C), the TTT nose (critical rate about 100°C/s), and the CCT cooling profile. Surface and center cool at different rates.

Solution: Surface cools at ~ 150°C/s (> critical, martensite forms, hardness OK). Center cools at only ~ 30°C/s (slower than critical, pearlite forms, not hard enough). **Fix:** use a more hardenable steel (e.g. 4340) or increase quench severity (brine, agitation). The figure below shows the tempering step used after quenching to recover toughness.

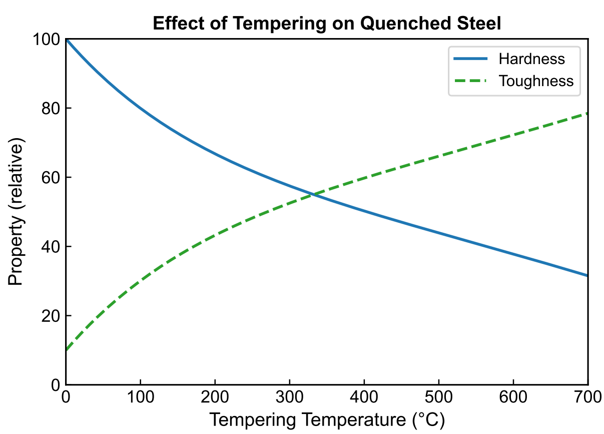


Figure 9.5. Figure 9.5 — Tempering of martensite: reheating between 150 and 600°C decomposes the metastable structure into α plus fine carbides.

11. Engineering Insight

Engineering Insight

Hardenability, not hardness, is what lets a large part be useful. A thin knife blade hardens through even in mild steel, but a thick axle needs alloying (Mn, Cr, Mo, Ni) to slow pearlite/bainite formation so the center also reaches martensite. The Jominy end-quench test turns this into a single hardness-versus-distance curve. The engineering trade is unavoidable: martensite is hard but brittle, so nearly every quenched part is tempered afterward — sacrificing some hardness to gain the toughness that prevents catastrophic failure in service.

12. Industrial Applications

Industrial Applications

- **Automotive:** Quenched-and-tempered gears, crankshafts, and springs rely on martensite plus tempering for strength with toughness.
- **Tooling:** Cutting tools use high-carbon martensite tempered at low temperature to keep an edge.
- **Aerospace:** High-hardenability low-alloy steels (e.g. 4340) give deep, uniform hardness in large forgings.
- **Annealing:** Full annealing (slow cool) softens steel for machining and forming; the figure below shows the soft lamellar/spheroidized result.

13. Summary

Phase transformations in steel are governed by two kinetic maps: the TTT diagram (isothermal) and the CCT diagram (continuous cooling). Holding above the TTT nose forms pearlite; below it forms bainite; quenching below M_s forms martensite by a diffusionless shear. Hardenability — the depth that reaches martensite — is increased by alloying and controls whether a real part hardens through. Because martensite is brittle, quenched parts are normally tempered to trade a little hardness for much-needed toughness. The single equation to remember is the Avrami law $y = 1 - \exp(-kt^n)$, describing how far a transformation has progressed at time t.

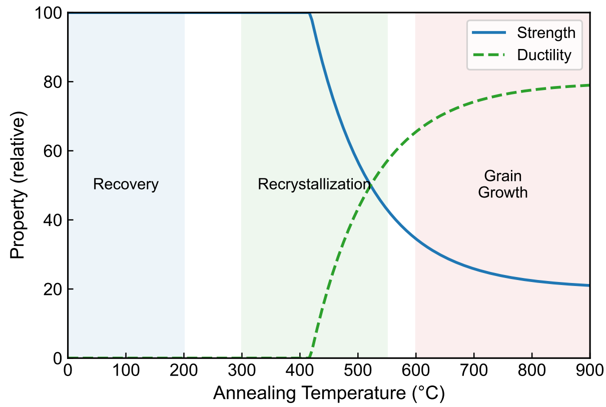


Figure 9.6. Figure 9.6 — Annealing: a slow furnace cool produces soft, machinable pearlite or spheroidized cementite, the opposite of a quench.

14. Quick Review

Quick Review

1. Pearlite forms ____________ the TTT nose, bainite forms ____________ it.
2. True / False: Martensite appears on the equilibrium Fe–C phase diagram. ____________
3. The M_s temperature ____________ (increases / decreases) as carbon content rises.
4. A faster cooling rate shifts the resulting microstructure toward ____________.
5. Tempering raises toughness by ____________ (raising / lowering) hardness.

15. Practice Problems

⋆ Which phase transformation in steel is diffusionless?

(a) Pearlite formation
(b) Martensite formation
(c) Bainite formation
(d) Spheroidite formation

(p. 148)

⋆ The nose of a TTT diagram represents:

(a) The minimum time for transformation to begin
(b) The highest temperature for any transformation
(c) The point where martensite starts
(d) The completion of transformation

(p. 148)

⋆⋆ For a 0.6 wt% C steel, the approximate M_s temperature is:

(a) 520°C
(b) 328°C
(c) 200°C
(d) 727°C

(p. 148)

⋆⋆ Tempering of martensite is performed to:

(a) Increase hardness
(b) Improve toughness while reducing hardness
(c) Form pearlite
(d) Increase the M_s temperature

(p. 148)

⋆⋆⋆ A 50 mm diameter 0.4 wt% C bar is water-quenched but the center is soft. Explain why using the CCT diagram and critical cooling rate, and propose one fix. (p. 148)

⋆⋆⋆ Which alloying element is most effective at increasing hardenability?

(a) Mo (molybdenum)
(b) Al (aluminum)
(c) Si (silicon)
(d) S (sulfur)

(p. 148)

16. Flash Cards

Flash Card

Front: What does TTT stand for, and what process does it describe?

Back: Time-Temperature-Transformation: transformation kinetics at a constant (isothermal) holding temperature.

Flash Card

Front: What is the nose of a TTT diagram?

Back: The point of shortest incubation time before transformation starts; a balance between driving force and atomic diffusivity.

Flash Card

Front: How do pearlite, bainite, and martensite differ in formation?

Back: Pearlite and bainite are diffusion-controlled (above / below the nose); martensite is a diffusionless shear below M_s.

Flash Card

Front: State the approximate M_s formula for plain carbon steel.

Back: $M_s \approx 520 - 320(\text{wt\% C})$ in °C; higher carbon gives lower M_s.

Flash Card

Front: What is hardenability, and how is it measured?

Back: The depth to which steel hardens on quenching; measured by the Jominy end-quench test (hardness vs. distance).

Flash Card

Front: Why is martensite tempered after quenching?

Back: Martensite is brittle; tempering (150–600°C) forms tempered martensite, trading hardness for toughness.

Flash Card

Front: How does a CCT diagram differ from a TTT diagram?

Back: CCT curves shift to longer times and lower temperatures; bainite can be suppressed at high cooling rates.

17. Cheat Sheet

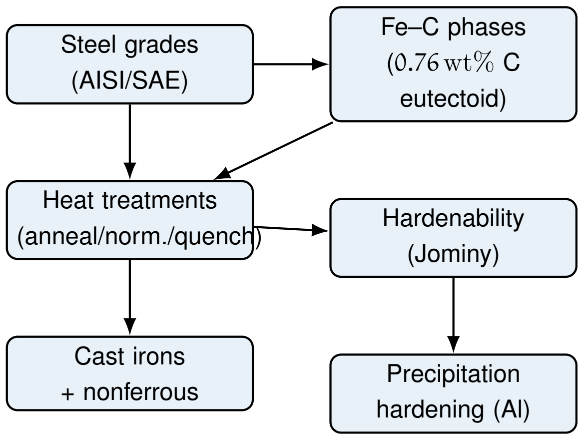
Cheat Sheet

- **TTT**: isothermal; C-curves. **CCT**: continuous; shifted right/down.
- **Nose**: fastest start of transformation.
- **Pearlite**: 550–727°C, layered, medium hardness.
- **Bainite**: 250–550°C, acicular, high hardness.
- **Martensite**: $< M_s$, diffusionless, very hard/brittle.
- $M_s \approx 520 - 320(\textbf{wt\% C})$ (°C, plain C steel).
- **Avrami**: $y = 1 - \exp(-kt^n)$.
- **Critical cooling rate**: minimum rate to dodge nose → martensite.
- **Hardenability**: depth hardened; raised by Mn, Cr, Mo, Ni.
- **Tempering**: reheat 150–600°C; hardness ↓, toughness ↑.

Chapter 10

Applications and Processing of Metal Alloys

1. Roadmap



2. Learning Objectives

1. **Classify steels** by carbon content and by AISI/SAE four-digit designation.
2. **Read the Fe–C system**: locate the eutectoid (0.76 wt% C) and explain pearlite, cementite, and martensite.
3. **Select a heat treatment** (full anneal, normalize, quench & temper, spheroidize, process anneal) for a given property target.
4. **Explain hardenability** and interpret the Jominy end-quench test.
5. **Contrast cast irons** (gray, white, ductile, malleable) by graphite morphology.
6. **Describe precipitation hardening** of aluminum alloys through its three stages.

3. Why This Matters

This chapter is the bridge between atomic structure and the parts you actually hold: gears, turbine blades, engine blocks, and body panels. On undergraduate and FE/PE exams it is a reliable source of classification, selection, and short-answer questions. For engineers the payoff is direct – the same steel chemistry can be soft enough to machine or hard enough to cut, purely by changing its heat treatment. Choosing the wrong process wastes material, energy, and sometimes lives, so the ability to match microstructure to service requirement is a core professional skill.

4. Intuition First

Intuition First

Think of a steel part as a deck of cards. Heating it to the high-temperature *austenite* state shuffles the carbon atoms into a uniform, face-centered arrangement. How you cool that deck decides the final hand. Cool it slowly in the furnace and the cards re-stack into soft, layered *pearlite*. Quench it in water and the cards freeze mid-shuffle into a tangled, very hard but brittle *martensite*. Then a gentle reheat – tempering – lets the cards relax just enough to stop being brittle. Add a pinch of other elements (Cr, Ni, Mo) and the deck stays "frozen" deeper inside, which is what we call *hardenability*.

5. Visual (one figure per key concept)

Concept 1 – The Fe–C phases and the eutectoid. The figure below shows the iron–carbon equilibrium diagram. The eutectoid point at 0.76 wt% C is where austenite splits into pearlite (alternating ferrite and cementite). Steels lie between 0.008 and 2.14 wt% C; cast irons are richer.

Concept 2 – The annealing of cold-worked metal. Three temperature-ordered stages convert a deformed, work-hardened part back toward soft, strain-free metal.

Concept 3 – The quench-and-temper cycle. Austenitize, quench to martensite, then temper to trade a little hardness for toughness.

Concept 4 – Tempering response.

Concept 5 – Precipitation hardening sequence. Solution treat, quench to a supersaturated solid solution, then age to grow fine precipitates that block dislocations.

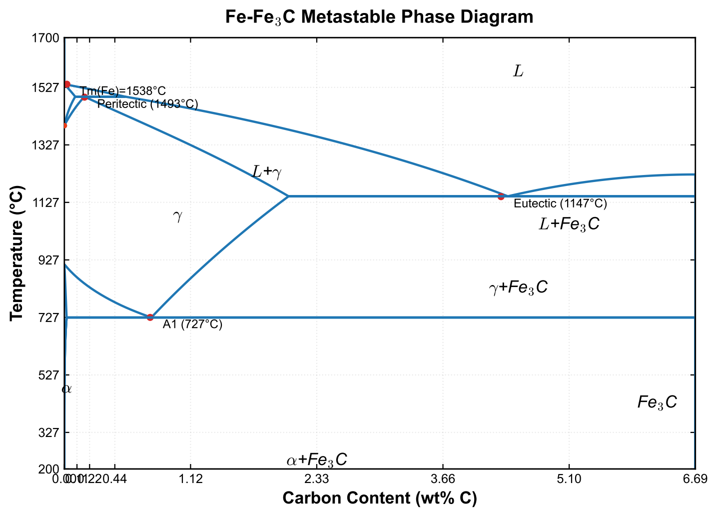


Figure 10.1. Figure 10.1 — Iron–carbon phase diagram highlighting the eutectoid composition at 0.76 wt% C.

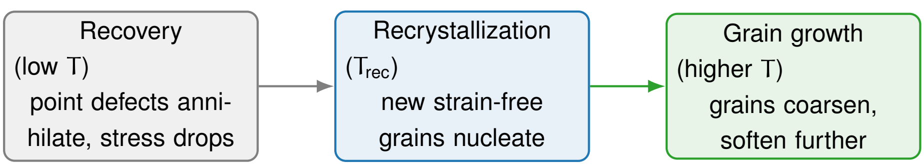


Figure 10.2. Figure 10.2 — Annealing stages ordered by increasing temperature: recovery, recrystallization, grain growth.

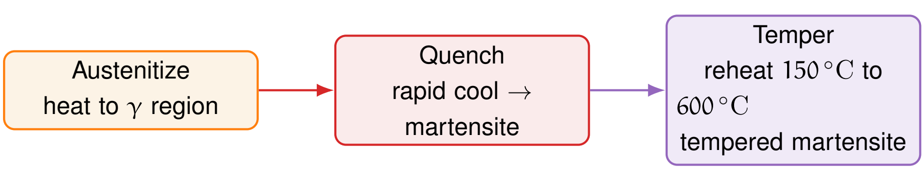


Figure 10.3. Figure 10.3 — Heat-treatment cycle: austenitize → quench → temper.

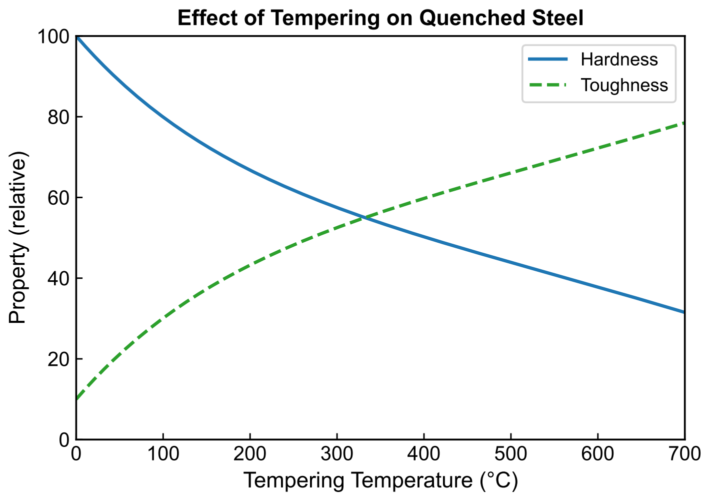


Figure 10.4. Figure 10.4 — As tempering temperature rises, hardness falls and toughness rises; the balance is chosen for service.

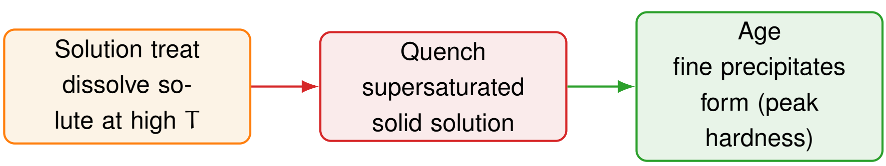


Figure 10.5. Figure 10.5 — Precipitation-hardening (age-hardening) sequence for aluminum alloys.

6. Memory Box

Memory Box

- Steels: 0.008–2.14 wt% C. Cast irons: >2.14 wt% C.
- AISI/SAE: first two digits = *alloy type*; last two = *C in hundredths of wt%*. 1045 = 0.45 % C plain carbon.
- Low-C <0.25 %: not hardenable by heat treatment. Medium-C 0.25–0.60 %: quench & temper. High-C 0.60–1.40 %: tools/dies.
- Eutectoid = 0.76 wt% C → pearlite (ferrite + cementite lamellae).
- Full anneal = furnace cool (softest). Normalize = air cool. Quench & temper = hardest + tough.
- Cast iron types depend on **graphite morphology**: flake (gray), spheroidal (ductile), temper carbon (malleable), cementite (white).
- Hardenability $\neq$ hardness: it is *depth* of martensite, measured by the Jominy end-quench test.

7. Compare Box

Annealing vs. Normalizing vs. Quench & Temper

Full anneal / normalize	**Quench & temper**
Cool from γ to soft pearlite (furnace or air).	Rapid quench to martensite, then reheat to temper.
Low-to-medium hardness, high machinability.	High strength with controlled toughness.
Used when shape is final and softness helps machining.	Used for shafts, gears, tools needing strength.
No martensite formed.	Martensite is the starting, then tempered phase.

Gray vs. Ductile vs. White vs. Malleable Cast Iron

Gray / ductile (graphite)	**White / malleable (cementite)**
Graphite as flakes (gray) or spheres (ductile).	Carbon as cementite / temper carbon – hard, brittle.
Good damping (gray), strong & ductile (ductile).	Very hard, wear-resistant, brittle.
Engine blocks, pipes, automotive parts.	Wear surfaces, annealed-to-ductile fittings.
Property set by graphite *morphology*, not just % C.	Property set by carbide formation.

8. Common Mistakes

Common Mistakes

- **Mistake 1: Reading the SAE carbon number as a percent.** SAE 1045 is 0.45 % C, not 45 % C – the last two digits are hundredths of a percent. **Why:** the scale is compressed by a factor of 100.
- **Mistake 2: Assuming all steels harden by heat treatment.** Low-C steels (<0.25 % C) form too little martensite; strengthen them by cold work instead. **Why:** martensite needs enough carbon to transform usefully.
- **Mistake 3: Confusing hardness with hardenability.** A shallow-hardening steel can still reach high surface hardness; hardenability is about how *deep* martensite forms. **Why:** the Jominy curve is about distance, not peak value.
- **Mistake 4: Treating cast irons as one material.** Their behavior is governed by graphite *morphology* (flake vs. spheroidal), not only carbon content. **Why:** ductile iron is far tougher than gray iron at similar composition.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
$C_0 = 0.76$	Eutectoid carbon content of Fe–C	wt% C	C_0: eutectoid % C	Equilibrium, slow cool	Not valid far from eutectoid	Separates hypoeutectoid/-eutectoid/hypereutectoid steels
$T_{temper} = 150\,°C$ to $600\,°C$	Tempering window after quench	°C	T_{temper}: reheat T	Martensitic start structure	Above $600\,°C$ over-softens	Higher T → softer, tougher
HRC(d)	Jominy hardness at distance d	HRC	d: distance from quenched end	Standard 1 in × 4 in bar	Relative, not absolute	Lower curve slope = higher hardenability
Recovery < Recryst. < Growth	Annealing temperature bands (ordered)	°C	stages by rising T	Cold-worked metal	Exact T alloy-specific	Pick T to control final grain size
Solution → Quench → Age	Precipitation-hardening stages	–	three-stage sequence	Supersaturated solid solution	Overaging lowers strength	Peak hardness at optimum aging time

10. Worked Examples

Basic

Example

Problem: What is the carbon content of SAE 4140 steel, and what alloying family does it belong to?
Thinking: The first two digits give the alloy type and the last two give carbon in hundredths of a percent. "41" is the Cr–Mo (chromoly) family, and "40" is 0.40 % C.
Solution: SAE 4140 is a chromium–molybdenum steel with 0.40 wt% C, suitable for quench-and-temper to high strength.

Intermediate

Example

Problem: A connecting rod must resist high cyclic loads and needs both strength and toughness. Recommend a steel and treatment.
Thinking: High strength plus fatigue resistance points to medium-carbon alloy steel given a quench-and-temper at a moderate tempering temperature.
Solution: Use SAE 4140 (medium-C, Cr–Mo) and apply quench & temper at about 400 °C. This gives tempered martensite: a strong core with enough toughness to survive fatigue.

Advanced

Example

Problem: Two bars, SAE 1040 (plain C) and SAE 4340 (Ni–Cr–Mo), are quenched from the same section. Which keeps martensite deeper, and why?
Thinking: Hardenability – not hardness – governs depth. Alloying elements shift the transformation (C) curves to the right, so pearlite/bainite start later during cooling.
Solution: SAE 4340 keeps martensite to far greater depth (still ~53 HRC near 50 mm in Jominy testing versus 1040 dropping to 35 HRC by 12 mm). Its alloy content makes slow cooling still form martensite, so large sections can be through-hardened.

11. Engineering Insight

Engineering Insight

The central engineering idea is *microstructure is a design variable.* The same 0.40 % C bar can be a soft, machinable blank (full anneal), a tough shaft (quench & temper at 400 °C), or a wear-resistant edge (quench & low temper at 200 °C) – purely by process choice. Hardenability decides whether a thick part hardens all the way through; that is why 4340, not 1040, is chosen for large fatigue-critical sections. Mistaking hardness for hardenability, or ignoring graphite morphology in cast iron, is how parts fail in service.

12. Industrial Applications

Industrial Applications

- **Automotive:** low-C 1008 panels are cold-worked and process-annealed for formability; 4140 shafts and 8620 case-carburized gears for strength with a tough core.
- **Cutting tools & dies:** high-C W1 / O1 steels are quenched and lightly tempered, or spheroidized first for machinability then hardened.
- **Engine blocks & pipes:** gray cast iron for damping and machinability; ductile iron where toughness is needed.
- **Aerospace:** age-hardened Al (2xxx, 6xxx, 7xxx) for airframes; Ti–6Al–4V and nickel superalloys (Inconel) for high-temperature turbine and implant service.

13. Summary

Steels are Fe–C alloys classified by carbon content and AISI/SAE grade; the eutectoid at 0.76 wt% C marks the boundary to pearlite formation. Heat treatment converts austenite into the desired microstructure: slow cooling gives soft pearlite, rapid quenching gives hard martensite, and tempering trades hardness for toughness. Hardenability – measured by the Jominy test – describes how deeply martensite forms and is raised by alloying elements. Cast irons are defined by graphite morphology, and aluminum alloys gain strength by precipitation hardening through solution, quench, and aging. The single most important number to remember is the eutectoid composition, 0.76 wt% C.

14. Quick Review

Quick Review

1. The eutectoid composition of the Fe–C system is ________ wt% C.
2. True / False: Hardenability and hardness mean the same thing. ________
3. SAE 1045 contains ________ % carbon.
4. List the three annealing stages in temperature order: ________________.
5. True / False: Low-carbon steels can be hardened by quench and temper. ________

15. Practice Problems

⋆ SAE 4140 steel has approximately what carbon content?

(a) 0.14 % C
(b) 0.40 % C
(c) 1.40 % C
(d) 4.0 % C

(p. 149)

⋆ Gray cast iron gets its name from:

(a) the gray appearance of graphite flakes on a fractured surface
(b) the color of cementite
(c) the heat treatment applied
(d) the nickel content

(p. 149)

⋆⋆ Which heat treatment produces the softest, most machinable condition in high-C steel?

(a) Quench and temper
(b) Spheroidize anneal

(c) Normalize

(d) Process anneal

(p. 149)

★★ A steel with 0.10 % C is cold rolled to increase strength. This strengthening method is:

(a) Precipitation hardening

(b) Strain hardening (work hardening)

(c) Quench hardening

(d) Solid-solution strengthening

(p. 149)

★★★ Why can SAE 4340 be through-hardened in a larger cross-section than SAE 1040 after the same quench? Reference hardenability and the Jominy test in your answer. (p. 149)

16. Flash Cards

Flash Card

Front: What does the last two digits of an AISI/SAE number mean?

Back: Carbon content in hundredths of wt%. 1045 = 0.45 % C.

Flash Card

Front: What is the eutectoid composition of Fe–C?

Back: 0.76 wt% C; austenite there transforms to pearlite on slow cooling.

Flash Card

Front: Hardenability vs. hardness – what is the difference?

Back: Hardness = surface strength reached. Hardenability = how deeply martensite forms (Jominy distance).

Flash Card

Front: Name the three annealing stages in order of rising temperature.

Back: Recovery (defects annihilate), recrystallization (new grains), grain growth (coarsening).

Flash Card

Front: What distinguishes gray from ductile cast iron?

Back: Graphite morphology: gray = flakes (brittle in tension); ductile = spheroids (strong, ductile).

Flash Card

Front: What are the three stages of precipitation hardening?

Back: Solution heat treatment, rapid quench to a supersaturated solid solution, then aging to form precipitates.

Flash Card

Front: Which steel class cannot be hardened by heat treatment and why?

Back: Low-carbon steel ($<0.25\,\%$ C); too little carbon to form useful martensite – use cold work instead.

17. Cheat Sheet

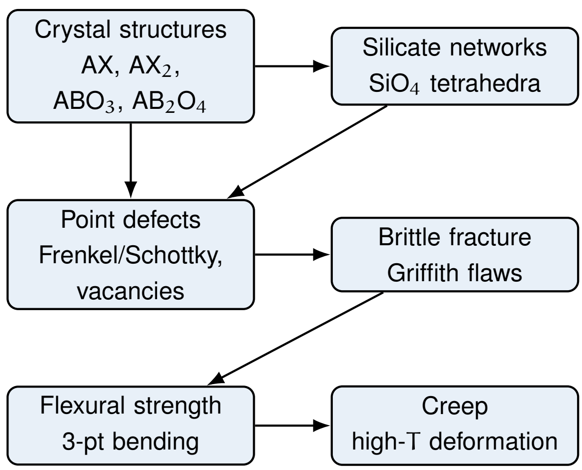
Cheat Sheet

- **Steel grade:** 1st two digits = type (10xx plain, 11xx resulfurized, 41xx Cr–Mo, 43xx Ni–Cr–Mo, 61xx Cr–V); last two = C in hundredths %.
- **Carbon classes:** Low <0.25 % (cold work), Medium 0.25–0.60 % (quench & temper), High 0.60–1.40 % (tools).
- **Eutectoid:** 0.76 wt% C → pearlite.
- **Heat treatments:** anneal (furnace cool, softest) | normalize (air cool) | quench & temper (150 °C to 600 °C) | spheroidize (below A_1) | process anneal (recovery).
- **Hardenability:** depth of martensite; Jominy end-quench; alloying shifts C-curves right.
- **Cast irons:** gray (flake), white (cementite), ductile (spheroidal), malleable (temper carbon).
- **Nonferrous:** Al age-hardenable (2xxx/6xxx/7xxx), Cu brasses/bronzes, Ti–6Al–4V, Ni superalloys (Inconel).
- **Precipitation hardening:** solution → quench → age (peak then overage).

Chapter 11

Structures and Properties of Ceramics

1. Roadmap



This chapter traces the chain from *atomic structure* (how ions pack and bond) to *engineering behaviour* (why ceramics are hard but brittle, and how they fail). Master the structure-to-property logic: radius-ratio predicts coordination, defects set the flaw population, flaws set the strength, and high temperature turns slow diffusion into creep.

2. Learning Objectives

Key Points

- Given cation and anion radii, predict the coordination number and crystal structure using the radius-ratio rule.
- Describe the AX, AX_2, ABO_3 (perovskite), and AB_2O_4 (spinel) structures and name key examples.
- Explain why silicate structures are built from SiO_4^{4-} tetrahedra and how network modifiers form glass.
- Distinguish Frenkel, Schottky, and non-stoichiometric (anion/cation vacancy) defects and their charge compensation.
- Apply the Griffith criterion to relate flaw size to fracture stress and compute flexural strength.
- Describe brittle fracture, the influence of flaws, and the mechanisms of ceramic creep.

3. Why This Matters

Ceramics dominate every field that needs hardness, heat resistance, or electrical function that metals cannot deliver: cutting tools, engine components, capacitors, piezoelectric sensors, and thermal-barrier coatings. On exams they test the radius-ratio rule, the Griffith equation, and defect terminology heavily. In practice the single biggest engineering problem with ceramics is *reliability* — strength scatters because failure starts at random flaws, so designers use statistical (Weibull) criteria instead of a single safety factor. Understanding flaws is the difference between a reliable component and a sudden catastrophic failure.

4. Intuition First

Intuition First

Imagine ceramics as a rigid 3-D jigsaw puzzle of charged balls. The big negatively charged balls (oxygen anions) form a cage; the small metal cations squeeze into the gaps. If a cation is too big for the gap it pushes the cage apart and demands more neighbours; if it is tiny it rattles in a small pocket with few neighbours. That geometric fit *is* the radius-ratio rule.

Now picture a pane of glass with a tiny scratch. Pull on it and the scratch behaves like the tip of a whip — stress piles up enormously at that one sharp point. Ceramics cannot yield to relieve that pile-up, so the crack simply runs. This is why a hairline flaw, not the bulk material, decides when a ceramic breaks. Heat it enough, though, and the balls slowly diffuse and slide past each other: that is creep.

5. Visual (one figure per key concept)

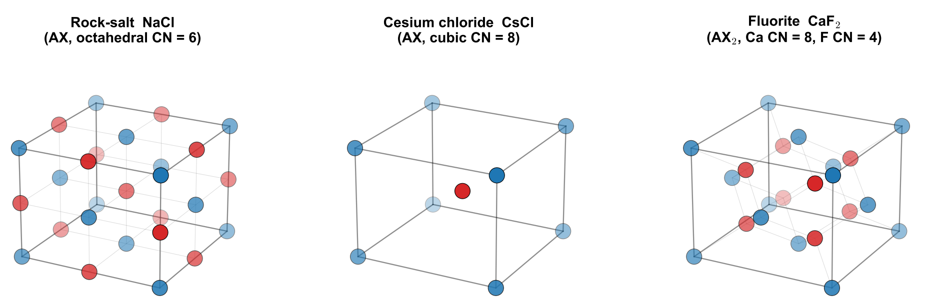


Figure 11.1. Figure 11.1 — Representative ceramic crystal cells: rock-salt NaCl (AX, octahedral CN = 6), cesium chloride CsCl (AX, cubic CN = 8), and fluorite CaF_2 (AX_2, Ca CN = 8, F CN = 4). Blue = anion, red = cation; faint grey bonds show the coordination polyhedron.

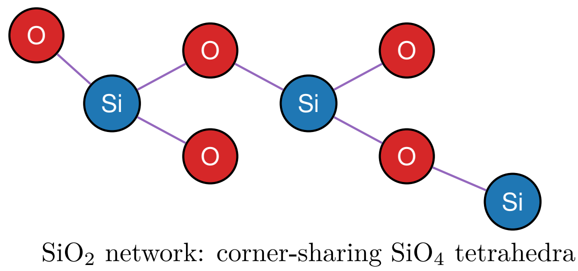


Figure 11.2. Figure 11.2 — Silicate structure built from SiO_4^{4-} tetrahedra. Each Si (blue) bonds to four O (red); adjacent tetrahedra share only corner oxygens, giving the open 3-D SiO_2 network.

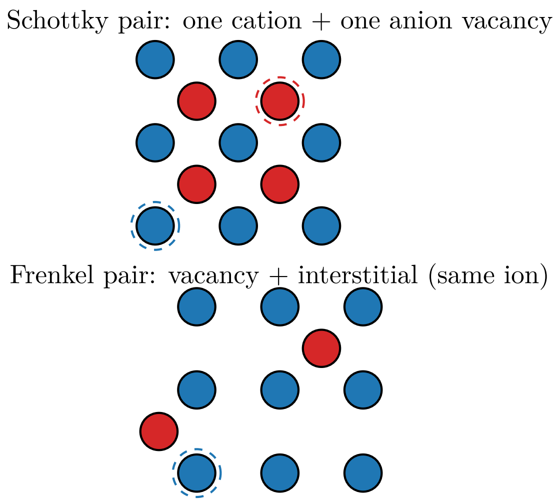


Figure 11.3. Figure 11.3 — Defects in ionic ceramics. Left: a Schottky pair (one cation and one anion vacancy) preserves stoichiometry. Right: a Frenkel defect (vacancy plus interstitial of the same ion) leaves composition unchanged.

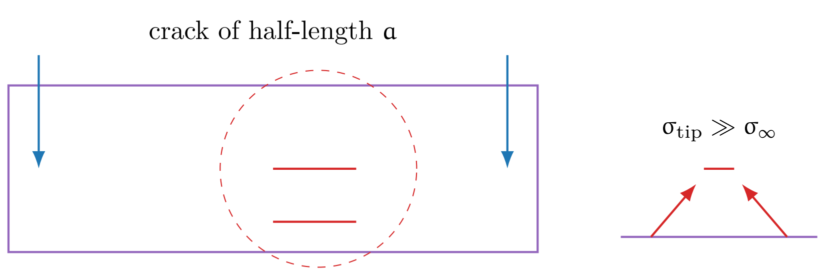


Figure 11.4. Figure 11.4 — Griffith flaw and crack-tip stress concentration. A small internal crack concentrates the remotely applied stress σ_∞ at its tip by roughly a factor of $2\sqrt{a/\rho}$; ceramics cannot relieve it plastically, so the crack runs.

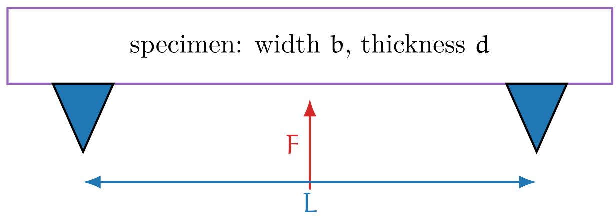


Figure 11.5. Figure 11.5 — Three-point flexural test geometry. A center load F is applied between two supports separated by span L; flexural strength is computed from the maximum load at fracture.

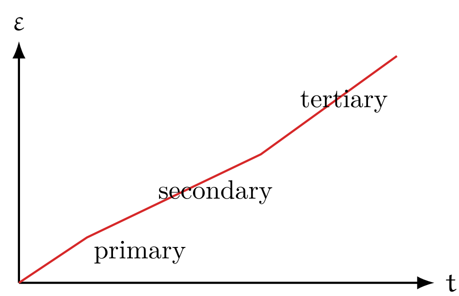


Figure 11.6. Figure 11.6 — Creep curve of a ceramic under constant load: primary (decelerating), secondary (steady-state, slope = minimum creep rate), and tertiary (accelerating to failure).

6. Memory Box

Memory Box

- **Ceramic** = compound of metal + nonmetal; ionic, covalent, or mixed bonding; inorganic, nonmetallic (includes glass).
- **Radius ratio** r_C/r_A: $> 0.732 \to$ CN 8; 0.414–$0.732 \to 6$; 0.225–$0.414 \to 4$; $0.155-0.225 \to 3$.
- **AX**: NaCl (6:6), CsCl (8:8), ZnS (4:4). **AX$_2$**: CaF_2 (8:4), TiO_2 rutile (6:3).
- **ABO$_3$ perovskite**: $CaTiO_3$, $BaTiO_3$, PZT. **AB$_2$O$_4$ spinel**: $MgAl_2O_4$, Fe_3O_4.
- **Silicates**: built from SiO_4^{4-} tetrahedra; modifiers (Na_2O, CaO) break Si–O–Si bridges → glass.
- **Defects**: Frenkel = vacancy + interstitial (same ion); Schottky = paired vacancies (keeps stoichiometry).
- **Fracture**: $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$; compressive strength $5-10\times$ tensile.
- **Creep** in ceramics is diffusion-controlled and dominates at high T / fine grains.

7. Compare Box

Defect families in ceramics

Frenkel defect	**Schottky defect**
A vacancy paired with an interstitial of the *same* ion. Composition (atom ratios) unchanged; density barely changes. Common for small cations (e.g. Ag^+, cations in close-packed arrays).	A cation vacancy paired with an anion vacancy. Composition unchanged; density *decreases* (mass loss, same volume). Common in strongly ionic compounds (e.g. NaCl, MgO).

Strength vs. toughness

Strength (flexural / tensile)	Toughness (K_{Ic})
Depends on the largest flaw present; scatter is large.	A material property; small for ceramics (0.5–5 MPa$\sqrt{\text{m}}$).
Raised by removing flaws (fine powders, polishing).	Raised by toughening (transformation, fibers, cracks bridging).
High strength **does not** imply high toughness.	A tough ceramic can absorb more energy before failure.

8. Common Mistakes

Common Mistakes

- **Mistake 1:** Thinking all ceramics are crystalline. **Why:** soda-lime glass and many silicates are amorphous; "ceramic" means inorganic and nonmetallic, glass included.
- **Mistake 2:** Confusing the two Griffith forms $\sigma_f = \sqrt{2E\gamma_s/(\pi a)}$ and $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$. **Why:** the first uses surface energy (ideal brittle solid); the second uses fracture toughness and is the practical engineering form. Use the toughness form unless told otherwise.
- **Mistake 3:** Assuming higher strength means higher toughness. **Why:** strength depends on flaw size, while K_{Ic} is intrinsic; glass fibers can be very strong yet very brittle.
- **Mistake 4:** Forgetting charge neutrality in defect chemistry. **Why:** a single cation vacancy leaves net negative charge; it must be compensated (another vacancy or a lower-valence substitution).
- **Mistake 5:** Treating compressive and tensile strength as equal. **Why:** flaws are closed (not opened) under compression, so compressive strength is 5–$10\times$ the tensile value.

9. Formula Sheet

Formula	Meaning	Units	Variables	Assumptions	Limitations	Eng. meaning
r_C/r_A rules	Predicts CN from ionic radii	–	r_C, r_A	Hard spheres, electrostatic	Covalency distorts	Choose crystal structure
$\rho = \frac{n' \sum A}{V_c N_A}$	Theoretical ceramic density	g/cm^3	n' f.u./cell, $\sum A$ atomic wt, V_c cell vol	Perfect crystal	Porosity lowers measured ρ	Compare to measured density
$\sigma_f = \sqrt{\frac{2E\gamma_s}{\pi a}}$	Griffith (surface energy)	MPa	E, γ_s, a	Perfectly brittle, through crack	Ignores plasticity	Flaw size → strength
$\sigma_f = \frac{K_{Ic}}{Y\sqrt{\pi a}}$	Griffith (toughness)	MPa	K_{Ic}, Y, a	Linear elastic	Not for ductile / R-curve	Design against fracture
$\sigma_{fs} = \frac{3FL}{2bd^2}$	Flexural strength (3-pt)	MPa	F, L, b, d	Beam, center load	4-pt uses $3F(L-l)/(2bd^2)$	Bend-test strength
$R = \frac{\sigma_f k}{\alpha E(1-\nu)}$	Thermal-shock resistance	–	k, α, E, ν	Elastic, no phase change	Approximate	Resist quenching
$P_f = 1 - e^{-(\sigma/\sigma_0)^m}$	Weibull failure prob.	–	σ_0, m	Random flaw population	Needs large data set	Reliability design

10. Worked Examples

Basic

Radius-ratio prediction for MgO

Problem: $r_{Mg^{2+}} = 0.072$ nm, $r_{O^{2-}} = 0.140$ nm. Predict the coordination number and structure.

Thinking: Compute the ratio and map it onto the radius-ratio bands.

Solution:

$$\frac{r_C}{r_A} = \frac{0.072}{0.140} = 0.514$$

This lies in 0.414–0.732, so CN = 6 (octahedral); MgO adopts the rock-salt (NaCl) structure.

Intermediate

Griffith fracture from flaw size

Problem: A glass plate has $K_{Ic} = 0.8$ MPa$\sqrt{\text{m}}$ and a surface crack $a = 0.1$ mm. Find the fracture stress (take $Y \approx 1$).
Thinking: Use the toughness form of Griffith; convert mm to m.
Solution:

$$\sigma_f = \frac{0.8}{1.0\sqrt{\pi(0.1 \times 10^{-3})}} = \frac{0.8}{\sqrt{3.14 \times 10^{-4}}} = \frac{0.8}{0.0177} = \mathbf{45.2\ MPa}$$

Reducing the crack to $a = 0.01$ mm gives $\sigma_f = 143$ MPa: smaller flaws → higher strength (the reason glass fibers beat bulk glass).

Advanced

Flexural strength from a bend test

Problem: A ceramic beam of width $b = 5$ mm, thickness $d = 3$ mm, span $L = 30$ mm fractures at center load $F = 120$ N in a 3-point test. Compute σ_{fs}.
Thinking: Convert all lengths to metres, then use the 3-point flexural formula.
Solution:

$$\sigma_{fs} = \frac{3FL}{2bd^2} = \frac{3(120)(30 \times 10^{-3})}{2(5 \times 10^{-3})(3 \times 10^{-3})^2} = \frac{10.8}{9.0 \times 10^{-8}} = \mathbf{120\ MPa}$$

This is the flexural (modulus of rupture) strength; it exceeds the true tensile strength because the stress peak is localized and ceramics fail where the largest flaw sits.

11. Engineering Insight

Engineering Insight

The governing engineering truth for ceramics is: *strength is a flaw statistic, not a material constant*. Two identically specified alumina parts can fail at very different stresses because one happens to contain a larger pore or surface scratch. This is why ceramic design uses a *survival probability* (Weibull) rather than a fixed safety factor, and why processing quality (powder purity, sintering density, surface finish) matters more than the nominal composition. Toughening strategies — transformation toughening in zirconia, fiber/whisker reinforcement, and crack bridging — all work by making a running crack spend more energy, raising the effective K_{Ic}. For high-temperature parts, the limit is creep: fine-grained ceramics creep faster (grain-boundary diffusion), so coarse grains or single crystals are preferred when dimensional stability under load is critical.

12. Industrial Applications

Industrial Applications

- **Dielectrics & capacitors:** $BaTiO_3$ perovskite has $\varepsilon_r > 1000$; PZT (lead zirconate titanate) is the standard piezoelectric for sensors, actuators, and ultrasonic transducers.
- **Cutting tools & wear parts:** Al_2O_3 and SiC retain hardness at temperature; SiC and ZnO act as varistors for surge protection.
- **Thermal management:** AlN and BeO ($k > 200$ W/m·K) are heat-sink substrates; YSZ (ZrO_2, $k \approx 2$ W/m·K) is a thermal-barrier coating on turbine blades.
- **Transparent & functional:** ITO (In_2O_3:Sn) for transparent electrodes; YBCO as a high-T_c superconductor.
- **Structural & glass:** soda-lime and borosilicate glass, porcelain, and glass-ceramics (CorningWare) for cookware and telescope mirrors.

13. Summary

Ceramic behaviour follows directly from structure. The radius-ratio rule converts ionic sizes into coordination numbers and thus into AX, AX_2, perovskite (ABO_3), or spinel (AB_2O_4) structures. Silicates are built from corner-sharing SiO_4^{4-} tetrahedra, and modifiers turn the network into glass. Point defects — Frenkel, Schottky, and non-stoichiometric vacancies — are governed by charge neutrality. Because ceramics are brittle, the largest flaw, not the bulk, sets the strength through the Griffith relation $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$; compressive strength is $5-10\times$ tensile. At elevated temperature, diffusion-driven creep limits service life. *Most important equation:* $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$ — flaw size is destiny.

14. Quick Review

Quick Review

1. A radius ratio $r_C/r_A = 0.514$ gives coordination number __________.
2. True / False: All ceramics are crystalline. __________
3. The Griffith equation shows fracture stress scales with $1/\sqrt{\_\_\_\_\_\_}$.
4. A Frenkel defect is a __________ paired with an interstitial of the same ion.
5. Flexural strength uses a __________-point bend test.
6. Fill: Ceramic compressive strength is roughly ________ times its tensile strength.
7. True / False: Higher strength always means higher fracture toughness. __________

15. Practice Problems

⋆ The radius ratio $r_C/r_A = 0.45$ predicts a coordination number of:

(a) 4
(b) 6
(c) 8
(d) 12

(p. 149)

⋆ Soda-lime glass is an example of:

(a) A crystalline ceramic
(b) An amorphous (glassy) ceramic
(c) A metal-matrix composite
(d) A polymer

(p. 149)

⋆⋆ Which factor is most responsible for the low tensile strength of ceramics?

(a) Low elastic modulus
(b) Pre-existing flaws that concentrate stress
(c) High thermal expansion
(d) Ionic bonding

(p. 149)

⋆⋆ The Griffith equation $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$ predicts that reducing flaw size from $100\ \mu m$ to $25\ \mu m$ increases strength by a factor of:

(a) 1.5
(b) 2
(c) 2.5
(d) 4

(p. 149)

⋆⋆⋆ A ceramic beam ($b = 4$ mm, $d = 3$ mm, $L = 40$ mm) fractures in a 3-point test at $F = 200$ N. Compute its flexural strength in MPa. (p. 149)

⋆⋆⋆ Distinguish a Schottky defect from a Frenkel defect and state why each preserves electroneutrality and stoichiometry. (p. 149)

16. Flash Cards

Flash Card

Front: What coordination number does $r_C/r_A = 0.60$ predict?

Back: CN $= 6$ (octahedral); it falls in the 0.414–0.732 band.

Flash Card

Front: Give the Griffith fracture equation in toughness form.

Back: $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$; a is flaw size, Y a geometry factor.

Flash Card

Front: What is a Frenkel defect?

Back: A vacancy paired with an interstitial of the SAME ion; total composition unchanged.

Flash Card

Front: What is a Schottky defect?

Back: A cation vacancy paired with an anion vacancy; keeps stoichiometry, lowers density.

Flash Card

Front: Why is ceramic compressive strength much higher than tensile?

Back: Under compression, flaws close instead of opening; stress concentration cannot drive a crack.

Flash Card

Front: What building block forms all silicate structures?

Back: The SiO_4^{4-} tetrahedron; tetrahedra share corners to build networks, sheets, or chains.

Flash Card

Front: Write the 3-point flexural strength formula.

Back: $\sigma_{fs} = 3FL/(2bd^2)$ with span L, load F, width b, thickness d.

Flash Card

Front: Name one perovskite and one spinel with formula.

Back: Perovskite ABO_3 (e.g. $BaTiO_3$); spinel AB_2O_4 (e.g. $MgAl_2O_4$).

17. Cheat Sheet

Cheat Sheet

Crystal structures: AX = NaCl(6:6), CsCl(8:8), ZnS(4:4); AX_2 = CaF_2(8:4), rutile(6:3); ABO_3 perovskite; AB_2O_4 spinel.
Radius ratio: $> 0.732 \to 8$; $0.414–0.732 \to 6$; $0.225–0.414 \to 4$; $0.155–0.225 \to 3$.
Silicates: SiO_4^{4-} tetrahedra; modifiers (Na_2O, CaO) → glass; soda-lime ≈ 72% SiO_2/14% Na_2O/10% CaO.
Defects: Frenkel = vac + interstitial (same ion); Schottky = vac pair (stoichiometric). Charge neutrality always.
Fracture: $\sigma_f = K_{Ic}/(Y\sqrt{\pi a})$; $\sigma_f \approx \sqrt{2E\gamma_s/(\pi a)}$. Compressive $\approx 5-10\times$ tensile.
Toughness: $K_{Ic} \approx 0.5-5$ MPa$\sqrt{\text{m}}$ (metals $20-200$).
Flexural (3-pt): $\sigma_{fs} = 3FL/(2bd^2)$.
Creep: diffusion-controlled; fine grains / high T → faster.
Density: $\rho = n' \sum A/(V_c N_A)$.
Key numbers: K_{Ic} glass ≈ 0.8; $BaTiO_3$ $\varepsilon_r > 1000$; Al_2O_3 $k \approx 30$; YSZ $k \approx 2$ W/m·K.

Appendix A

Master Formula Index

Scan this once before the exam. Each row: formula (left) vs. meaning + units (right).

Ch 2–3 — Atomic Structure & Crystal Structures

$\rho = \dfrac{nA}{V_c N_A}$	Theoretical density; g/cm^3. n atoms/cell, A at. wt, V_c cell vol.
$\text{APF} = \dfrac{n(4/3)\pi R^3}{V_c}$	Atomic packing factor (0–1); higher = denser packing.
$a = 2\sqrt{2}R$ (FCC), $a = 4R/\sqrt{3}$ (BCC)	Lattice parameter from atomic radius; nm.
$d_{hkl} = \dfrac{a}{\sqrt{h^2+k^2+l^2}}$	Interplanar spacing (cubic); nm.
$n\lambda = 2d\sin\theta$	Bragg's law; λ wavelength, θ glancing angle.

Ch 4 — Imperfections in Solids

$N_v = Ne^{-Q_v/kT}$	Equilibrium vacancy count; $k = 8.62 \times 10^{-5}$ eV/K.
$N_v/N = e^{-Q_v/kT}$	Vacancy fraction; grows with T.
$\sigma_y = \sigma_0 + k_y d^{-1/2}$	Hall–Petch: finer grains = stronger; MPa.

Ch 5 — Diffusion

$J = -D\,\dfrac{dC}{dx}$	Fick's 1st law (steady state); kg/(m^{2}s).
$\dfrac{\partial C}{\partial t} = D\,\dfrac{\partial^2 C}{\partial x^2}$	Fick's 2nd law (transient profile).
$D = D_0\exp(-Q_d/RT)$	Arrhenius diffusivity; m^2/s, $R = 8.314$ J/molK.
$\dfrac{C - C_0}{C_S - C_0} = 1 - \text{erf}\left(\dfrac{x}{2\sqrt{Dt}}\right)$	Carburizing / semi-infinite solid solution.

Ch 6 — Mechanical Properties

$\sigma = F/A_0,\ \ \varepsilon = \Delta l/l_0$	Engineering stress (MPa) / strain.
$E = \sigma/\varepsilon$	Young's modulus (Hooke); GPa.
$\nu = -\varepsilon_{\text{trans}}/\varepsilon_{\text{axial}}$	Poisson's ratio (metals ≈ 0.33).
$\sigma_T = \sigma(1+\varepsilon),\ \ \varepsilon_T = \ln(1+\varepsilon)$	True stress / true strain (to necking).
$\%\text{EL} = \dfrac{\Delta l}{l_0} \times 100,\ \ \%\text{RA} = \dfrac{A_0 - A_f}{A_0} \times 100$	Ductility: elongation / reduction of area.
$\text{HB} = \dfrac{2P}{\pi D(D - \sqrt{D^2 - d^2})}$	Brinell hardness; strength proxy ($\sigma_{TS} \approx 3.45\,\text{HB}$ for steel).

Ch 7 — Dislocations & Strengthening

$\tau_R = \sigma \cos\phi \cos\lambda$	Schmid: resolved shear stress; ϕ, λ angles to slip.
$\sigma_y = \tau_{CRSS}/m$	Crystal tensile yield; $m = \cos\phi\cos\lambda$.
$\sigma_T = K\varepsilon_T^n$	Hollomon strain hardening; higher n = more formable.
$\tau = Gb/L$	Orowan bypass stress; L particle spacing.

Ch 8 — Phase Diagrams

$F = C - P + 2$	Gibbs phase rule (use $+1$ at fixed P).
$W_\alpha = \dfrac{C_\beta - C_0}{C_\beta - C_\alpha}$	Lever rule mass fraction of α.
$W_\beta = \dfrac{C_0 - C_\alpha}{C_\beta - C_\alpha}$	Lever rule mass fraction of β ($W_\alpha + W_\beta = 1$).

Ch 9 — Phase Transformations

$y = 1 - \exp(-kt^n)$	Avrami fraction transformed (isothermal); $n \approx 1$–4.
$M_s \approx 520 - 320(\text{wt\% C})$	Martensite start temperature (°C), plain carbon steel.

Ch 11 — Ceramics & Fracture

r_C/r_A rules	Radius-ratio → coordination number / structure.
$\sigma_f = \sqrt{\dfrac{2E\gamma_s}{\pi a}}$	Griffith fracture (brittle); MPa.
$\sigma_f = \dfrac{K_{Ic}}{Y\sqrt{\pi a}}$	Griffith / toughness form; K_{Ic} fracture toughness.
$\sigma_{fs} = \dfrac{3FL}{2bd^2}$	Flexural strength, 3-pt bend; MPa.
$P_f = 1 - e^{-(\sigma/\sigma_0)^m}$	Weibull failure probability; m shape, σ_0 scale.

Appendix B

Answer Key

Answers — Chapter 01

- ⋆ **C. Semiconductors.** The five primary families are Metals, Ceramics, Polymers, Composites, and Advanced materials. Semiconductors are an *advanced*-material subclass (e.g. Si), not a primary classification on their own.
- ⋆ **B. Structure.** In the MSE paradigm, Processing creates the Structure; Structure is what directly determines Properties, which in turn enable Performance.
- ⋆⋆ **C. Nickel superalloy.** At 900°C (above the useful range of Al, ceramics are brittle, and Nylon melts), nickel superalloys retain strength and resist creep; they are the standard jet-turbine-blade material.
- ⋆⋆ **B. A polymer-matrix composite.** CFRP = carbon-fiber reinforcement in a polymer (epoxy) matrix; the matrix is polymer, so it is a PMC, not a metal or ceramic composite.
- ⋆⋆⋆ Theoretical density and likely metal.

$$V_c = (0.2866 \times 10^{-7}\ \mathrm{cm})^3 = 2.355 \times 10^{-23}\ \mathrm{cm}^3,$$
$$\rho = \frac{nA}{V_c N_A} = \frac{2(55.85)}{(2.355 \times 10^{-23})(6.022 \times 10^{23})} = \frac{111.7}{14.18} \approx 7.88\ \mathrm{g/cm}^3.$$

  With $n = 2$ (BCC) and $A = 55.85$ g/mol the metal is **iron** (α-Fe, handbook ≈ 7.87 g/cm^3).

Answers — Chapter 02

- ⋆ **B. Covalent.** Covalent bonds are highly directional (localized shared-electron pairs) and give very high stiffness/hardness (e.g. diamond).
- ⋆ **A. Ionic.** $\Delta\chi = 3.16 - 0.93 = 2.23 > 1.7$, so the Na–Cl bond is predominantly ionic (NaCl).
- ⋆⋆ **A. Elastic modulus.** The modulus is the *curvature* (second derivative) of the $E(r)$ well at r_0; well depth E_0 gives the melting point, asymmetry gives thermal expansion.
- ⋆⋆ **B. Ionic.** Very high T_m, electrical insulating behavior, and brittle fracture are the signature property set of ionic bonding (e.g. MgO, Al_2O_3).

⋆⋆⋆ **A. Non-directional (metallic).** The delocalized electron sea lets atomic planes slide past one another without breaking bonds, giving metals their ductility.

⋆⋆⋆ $\Delta\chi = 2.55 - 1.90 = 0.65$.

$F_{\text{ionic}} = 1 - e^{-(\Delta\chi/2)^2} = 1 - e^{-(0.65/2)^2} = 1 - e^{-0.1056} = 1 - 0.900 \approx 0.10$ (about 10%).

SiC is therefore *mostly covalent* (with only ~ 10% ionic character); this mixed character is why SiC is a hard, covalent-ceramic semiconductor.

Answers — Chapter 03

⋆ **b. 2.** A BCC cell has 8 corners $\times 1/8$ + 1 body center $= 1 + 1 = 2$ atoms.

⋆ **c. 0.74.** FCC (and HCP) are close-packed: APF $= 4(4/3)\pi R^3/(2\sqrt{2}R)^3 \approx 0.74$. BCC is 0.68.

⋆⋆ **b. 0.362 nm.** For FCC, $a = 2\sqrt{2}R = 2\sqrt{2}(0.128) \approx 0.362$ nm.

⋆⋆ **b. (110).** The first BCC reflection (smallest $h + k + l$ even) is (110) at $2\theta \approx 44.8°$ ($d_{110} = a/\sqrt{2} = 0.2027$ nm).

⋆⋆⋆ **b. (111), (200), (220).** FCC allows only all-even or all-odd (hkl). The allowed $h^2 + k^2 + l^2$ sequence is $3, 4, 8, \ldots$ giving planes (111), (200), (220) as the first three peaks.

Answers — Chapter 04

⋆ **A. Increases exponentially with temperature.** $N_v/N = \exp(-Q_v/kT)$; as T rises the exponent becomes less negative, so vacancy fraction grows steeply (use kelvin).

⋆ **B. Perpendicular to the dislocation line.** For an edge dislocation the Burgers vector **b** is perpendicular to the line (the extra half-plane edge); screw is the parallel case.

⋆⋆ **B. Increase strength.** Hall–Petch: $\sigma_y = \sigma_0 + k_y d^{-1/2}$; smaller d gives a larger $d^{-1/2}$, so yield strength rises.

⋆⋆ **C. High melting point.** Hume-Rothery rules are: atomic radii within ~ 15%, same crystal structure, similar electronegativity, same valence. Melting point is not a rule.

- ⋆⋆ **B. An array of dislocations.** A low-angle (tilt) grain boundary is modeled as a wall of edge dislocations; high-angle boundaries are misoriented crystal lattices.

- ⋆⋆⋆ **C. Removes a cation-anion pair from the lattice.** A Schottky defect is a paired cation + anion vacancy (neutral, lowers density); a Frenkel defect keeps the ion inside the lattice as a vacancy-interstitial pair.

Answers — Chapter 05

- ⋆ **A. More sites are available and the activation energy is lower.** Interstitial atoms hop through the numerous gaps between host atoms, needing no vacancy, so Q_d is lower and diffusion is faster than vacancy (substitutional) diffusion.

- ⋆ **A. Flux is opposite to the concentration gradient direction.** The minus sign in $J = -D\,dC/dx$ means atoms flow "downhill" from high to low concentration.

- ⋆⋆ **b. 4.** Case depth $\propto \sqrt{Dt}$; to double the depth at fixed T you need $(2)^2 = 4\times$ the time.

- ⋆⋆ **A. BCC has a lower atomic packing factor (0.68 vs 0.74).** The more open BCC lattice gives carbon more and larger interstitial sites, so C diffuses faster in α-Fe (BCC) than in γ-Fe (FCC).

- ⋆⋆⋆ **A. The slope of an Arrhenius plot ($\ln D$ vs. $1/T$).** $D = D_0 \exp(-Q_d/RT)$; a plot of $\ln D$ vs $1/T$ is a straight line of slope $-Q_d/R$, so $Q_d = -(\text{slope})R$.

Answers — Chapter 06

- ⋆ Elastic strain $\varepsilon = \sigma/E = 400/200000 = \mathbf{0.002}$ (0.2%). Hooke's law applies below yield.

- ⋆ Necking begins at the **ultimate tensile strength (UTS)**, the peak of the engineering stress–strain curve. Beyond UTS the cross-section locally narrows faster than strain hardening can compensate, so the load drops and a neck forms.

- ⋆⋆ True stress $\sigma_T = \sigma(1+\varepsilon) = 300(1+0.05) = \mathbf{315\ MPa}$; true strain $\varepsilon_T = \ln(1+\varepsilon) = \ln(1.05) = \mathbf{0.0488}$.

- ⋆⋆ $\%EL = (l_f - l_0)/l_0 \times 100 = (63-50)/50 \times 100 = \mathbf{26\%}$. $\%RA = (A_0 - A_f)/A_0 \times 100 = (90-42)/90 \times 100 = \mathbf{53.3\%}$.

★★★ Modulus of resilience $U_r = \sigma_y^2/(2E) = (500 \times 10^6)^2/(2 \cdot 200 \times 10^9) = 2.5 \times 10^{17}/4.0 \times 10^{11} = \mathbf{0.625\ MJ/m^3}$.

★★★ Brinell hardness:

$$HB = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})} = \frac{2(3000)}{\pi(10)(10 - \sqrt{100 - 10.24})} = \frac{6000}{\pi(10)(10 - 9.474)}$$
$$= \frac{6000}{16.52} \approx \mathbf{363}.$$

Estimated tensile strength (steels): $\sigma_{TS} \approx 3.45\,\mathrm{HB} \approx 3.45(363) = \mathbf{1252\ MPa}$.

Answers — Chapter 07

★ **B. $\phi = 45°$, $\lambda = 45°$.** Schmid factor $m = \cos\phi\cos\lambda$ is maximized at $m = 0.5$ when both angles are 45°, giving the largest resolved shear stress for a given σ.

★ **b. 12.** FCC has 4 $\{111\}$ planes × 3 $\langle 110 \rangle$ directions = 12 slip systems (BCC 48, HCP 3).

★★ **b. 2 (approximately).** With $\sigma_0 = 100$ MPa, $k_y = 0.74$ MPa·m$^{1/2}$: $d = 10\ \mu m \Rightarrow \sigma_y = 100 + 0.74/(10^{-5})^{1/2} = 100 + 234 = 334$ MPa; $d = 2.5\ \mu m \Rightarrow \sigma_y = 100 + 0.74/(2.5 \times 10^{-6})^{1/2} = 100 + 468 = 568$ MPa. The $d^{-1/2}$ term doubles (grain size cut by 4), giving a total factor of about 1.7; the nearest listed choice is **2**.

★★ **b. Recrystallization.** Recovery lowers stored energy but only partially softens; recrystallization forms new strain-free grains and *fully* restores the original softness and ductility. Grain growth then slightly coarsens/softens further.

★★★ **B. Small, coherent, and closely spaced.** Coherent, fine precipitates are cut/sheared (or Orowan-bypassed at very small spacing), maximizing the obstacle density to dislocation motion; this is the peak-aged condition. Large/widely spaced particles are weaker.

★★★ **A. Fully annealed copper (FCC).** FCC metals have the highest strain-hardening exponent ($n \approx 0.3$–0.5); BCC (~ 0.15–0.25) and HCP (< 0.1) are lower, glass essentially none.

Answers — Chapter 08

★ Gibbs phase rule at fixed pressure: $\mathbf{F = C - P + 1}$, where C = number of components, P = number of phases, and F = degrees of freedom (independent intensive variables you may set). At ordinary (variable-T, P) conditions use $+2$.

⋆⋆ Lever rule, liquid fraction:

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{40-18}{46-18} = \frac{22}{28} \approx \mathbf{0.786}\ (78.6\%).$$

(Solid α fraction $= 0.214$; check sums to 1.)

⋆⋆ Cu–Ni: $C_0 = 50$, $C_L = 44$, $C_S = 56$ wt% Ni.

$$W_L = \frac{C_0 - C_S}{C_L - C_S} = \frac{50-56}{44-56} = \frac{-6}{-12} = 0.50,$$
$$W_S = \frac{C_L - C_0}{C_L - C_S} = \frac{44-50}{44-56} = \frac{-6}{-12} = 0.50.$$

50% liquid and **50**% solid α; $W_L + W_S = 1.00$ confirmed.

⋆⋆⋆ Hypoeutectoid ($C_0 = 0.60$ wt% C):

$$W_p = \frac{C_0 - 0.022}{0.76 - 0.022} = \frac{0.60-0.022}{0.738} = \frac{0.578}{0.738} \approx \mathbf{0.783}\ (78.3\%\ \text{pearlite}),$$
$$W_\alpha = 1 - W_p \approx \mathbf{0.217}\ (21.7\%\ \text{proeutectoid ferrite}).$$

⋆⋆⋆ At the eutectic point (fixed pressure) **three phases coexist** ($L + \alpha + \beta$) and $F = 0$. It is called *invariant* because with $P = 3$, $C = 2$ and fixed P, $F = C - P + 1 = 0$: temperature and all phase compositions are locked, and the reaction proceeds isothermally.

Answers — Chapter 09

⋆ **B. Martensite formation.** Martensite is a diffusionless shear transformation that occurs on rapid quench below M_s; pearlite and bainite are diffusion-controlled, spheroidite forms by coarsening.

⋆ **A. The minimum time for transformation to begin.** The nose is the C-curve's closest approach to the time axis (shortest incubation); it is the easiest point to avoid (or hit) during cooling.

⋆⋆ **B. 328°C.** $M_s \approx 520 - 320(\text{wt\% C}) = 520 - 320(0.6) = 520 - 192 = 328°\text{C}$ for a plain-carbon steel.

⋆⋆ **B. Improve toughness while reducing hardness.** Tempering reheats martensite (150–600°C), decomposing it to tempered martensite; hardness falls but toughness rises, averting brittle failure.

⋆⋆⋆ The bar's **center cools more slowly than the critical cooling rate**, so its path on the CCT diagram crosses the pearlite/bainite region instead of dodging the nose → it forms soft pearlite, not martensite. *Fix:* use a higher-hardenability steel (e.g. 4340, with Ni–Cr–Mo shifting the C-curves right) or increase quench severity (brine, agitation).

⋆⋆⋆ **A. Mo (molybdenum).** Molybdenum is among the strongest hardenability boosters (along with Cr, Mn, and B); Al, Si, and S are much weaker for this purpose.

Answers — Chapter 10

⋆ **B. 0.40% C.** In AISI/SAE, the last two digits are carbon in hundredths of a wt%; "40" ⇒ 0.40 wt% C (the "41" prefix = Cr–Mo family).

⋆ **A. The gray appearance of graphite flakes on a fractured surface.** Gray cast iron's fracture surface looks gray because graphite exists as flakes; the name refers to that appearance, not to cementite or heat treatment.

⋆⋆ **B. Spheroidize anneal.** Spheroidizing (just below A_1) turns cementite into rounded globules, giving the softest, most machinable condition for high-C steel. Full anneal is next; normalize and quench-and-temper are harder.

⋆⋆ **B. Strain hardening (work hardening).** Cold rolling multiplies dislocation density, raising strength via $\tau \propto Gb\sqrt{\rho}$; it is not precipitation, quench, or solid-solution hardening.

⋆⋆⋆ SAE 4340 has much higher **hardenability** than 1040 because its Ni–Cr–Mo alloying shifts the transformation (C) curves to the right, lowering the critical cooling rate. Even the center of a large section then cools faster than the critical rate and forms martensite. The Jominy end-quench test shows 4340 holding high hardness to large distance (deep), whereas 1040's hardness drops sharply a few mm from the quenched end.

Answers — Chapter 11

⋆ **b. 6.** Radius ratio 0.45 lies in the 0.414–0.732 band, predicting octahedral coordination (CN = 6), e.g. rock-salt structure.

⋆ **b. An amorphous (glassy) ceramic.** Soda-lime glass is non-crystalline; "ceramic" means inorganic and nonmetallic, which includes glass, not only crystals.

⋆⋆ **B. Pre-existing flaws that concentrate stress.** Ceramics cannot yield to relax stress, so a sharp flaw raises the local stress by $\sim 2\sqrt{a/\rho}$ and triggers brittle crack growth; this flaw sensitivity, not bonding or modulus, governs the low tensile strength.

⋆⋆ **B. 2.** Griffith: $\sigma_f \propto 1/\sqrt{a}$. Reducing a from 100 µm to 25 µm is a 4× smaller flaw, so strength rises by $\sqrt{4} = 2\times$.

⋆⋆⋆ 3-point flexural strength:

$$\sigma_{fs} = \frac{3FL}{2bd^2} = \frac{3(200)(40 \times 10^{-3})}{2(4 \times 10^{-3})(3 \times 10^{-3})^2} = \frac{24}{7.2 \times 10^{-8}} \approx \mathbf{3.33 \times 10^8\ Pa = 333\ MPa.}$$

⋆⋆⋆ **Schottky defect:** a paired cation vacancy + anion vacancy removed from the lattice; it stays charge-neutral (equal ± removed) and stoichiometric, and lowers density (mass leaves, volume roughly same). **Frenkel defect:** a vacancy plus an interstitial of the *same* ion (ion relocated into a gap); also neutral and stoichiometric, with density essentially unchanged. Both preserve electroneutrality and composition because the vacancy/interstitial charges balance.